TOFME: picoDst QA and mixed-event setup for TPC–BTOF matching

Run19 Au+Au $\sqrt{s_{NN}} = 19.6$ GeV, production P24iy (test sample)

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Abstract

This note documents the TOFME project, whose goal is a mixed-event estimate of random (fake) matches between TPC tracks and Barrel Time-of-Flight (BTOF) hits in STAR. The input is the latest Run19 Au+Au 19.6 GeV production (P24iy, library SL24y). First, a QA analysis of the picoDst content was set up and run: event-level, TPC-track-level and TPC–BTOF-association variables, including $1/\beta$ and m^2 versus momentum, local hit coordinates, tray/module occupancy and the TOF matching efficiency. Then the mixed-event machinery was built (Sec. 7): a data-driven BTOF module geometry, projection of the picoDst helix of every track to a BTOF cell, and cell-level matching to hits of the same and of mixed events. For the production-like matching rule (projected cell or its neighbour), a random hit is present in the matching cells for 5.24% of the tracks, equal to the hit occupancy (1M-event sample, 15 runs, 423 775 selected events; stable from run to run once scaled by the occupancy; Sec. 7.4). Finally (Sec. 7.5), β and m^2 are computed for same-event and mixed-event matches in nine centrality classes. The mixed events are overlaid on the track's own event, normalized per projected track, and include hits without a stored time. Random matches are then 4.2% of the TOF-matched tracks in 0–5% and 0.2% in 70–80%, and 2.1% (π), 3.0% (K) and 1.0% (p) of the yield under the PID peaks in 0–5% at $0.8 < p < 1\,\mathrm{GeV}/c$. They explain only a small part of the yield between the proton and deuteron peaks, because the picoDst has no time for BTOF hits not matched to a track. On MuDst (Sec. 7.7), where all hits have a time, the production BTOF matching and calibration re-run on MuDst reproduce production exactly. They are then re-run on overlaid events (hits and tracks of a partner event added); this mixed event describes the same-event background between the p and d peaks (SE/ME = 1.39 in 0–5%, against 9.25 on picoDst). Random matches are 2.62% of the TOF matches in 0–5% and 1.57% in 0–80%, in agreement with an independent emulation (2.7% and 1.6%; four MuDst files of two runs). A HIJING+GEANT simulation of the 2019 detector (Sec. 7.10, 49857 events) shows that the remaining low-momentum excess is track-correlated: at 0.2–0.4 GeV/c it is mostly secondaries produced by the track's own particle in the BTOF trays, at a rate (2.7 per 10^4 tracks) flat in centrality as in the data. The simulated component over-predicts the K–p valley and is not used as a correction. **Limitations:** the QA sections (4–6) use a *test sample* of 10000 events from one picoDst file of a single run (20093036); the mixed-event results use 15 runs. No bad-tray list is applied. The time of random hits without a stored time is an assumption in the picoDst sections (Sec. 7.5); the MuDst results use 21160 events only. Statistical uncertainties only. Coloured boxes: **blue** = a choice made and its reason; **red** = caveat / not yet understood; **orange** = decision needed from the analyser.

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1 Overview

Goal of the project. Estimate the probability that a TPC track is matched to a BTOF hit that it did not produce, by pairing tracks with BTOF hits from *different* events of similar topology (mixed events). This first step establishes the data access, software environment and a QA baseline for the quantities the mixed-event study will use.

Main findings (test sample, 5 501 selected events).

- Event selection keeps 5 501 of 10 000 events (55.0%); the $|V_z| < 70\,\text{cm}$ cut dominates the loss (Table 2). 9 989 events carry the minimum-bias trigger 640051.
- Mean refMult is 102.3; mean number of BTOF hits stored per event is 296.5, i.e. 2.94 BTOF hits per TOF-matched primary track (100.8 per event). The number of matched tracks and of BTOF hits are strongly correlated (correlation factor 0.987).
- Track selection (Table 4): 960 157 primary tracks enter the QA sample (174.5 per event); 554 429 of them have a BTOF match with valid β.
- TOF matching efficiency (matched with $\beta > 0$ / TPC tracks) is 0.639 for $0.5 < p_T < 1.5\,\text{GeV}/c$, 0.467 for $0.15 < p_T < 0.3\,\text{GeV}/c$, 0.640 for $|\eta| < 0.5$ and 0.510 for $0.5 < |\eta| < 1$.
- m^2 peaks (for $0.4 < p < 1.0\,\text{GeV}/c$) are at 0.020, 0.244 and 0.877 GeV^2/c^4 for π, K, p, consistent with PDG values within $0.005\,\text{GeV}^2/c^4$.
- 18.7% of matched tracks have `btofMatchFlag`=2 (one hit, several track candidates).
- Mixed events (Sec. 7): the module geometry fitted from production matches reproduces the production projection points to 0.0029 cm. With the picoDst helix, 92.9% of production matches above $1.5\,\text{GeV}/c$ are projected to the same cell. The random-match probability is 5.24% (cell±1, production-like) and 2.34% (cell only), proportional to the number of BTOF hits in the event; it amounts to 6.3% of same-event matches for $0.5 < p_T < 2\,\text{GeV}/c$.
- 1M-event sample (15 runs, Sec. 7.4): $P_{\text{ME}} = 5.24\%$ (cell±1), $f_{\text{rand}} = 6.3\%$ for $0.5 < p_T < 2\,\text{GeV}/c$; per-run P_{ME} divided by the occupancy stays within 0.99–1.01; the geometry refitted on 1M events agrees with the test-run geometry to 0.0005 cm.
- m^2 (Sec. 7.5): with the hit time taken from the picoDst matches and the production path-length formula, the same-event m^2 reproduces production (RMS difference $0.0007\,\text{GeV}^2/c^4$). Mixed events overlaid on the track's own event, normalized per projected track and including hits without stored time, give random matches equal to 4.2% of the TOF matches in 0–5% and 0.2% in 70–80%. In 0–5% at $0.8 < p < 1\,\text{GeV}/c$ they are 2.1% (π), 3.0% (K) and 1.0% (p) of the yield under

the peaks. They explain only a small part of the SE yield between the proton and deuteron peaks (SE/ME = 7.41).

- Background origin (Sec. 7.6): the SE excess between the p and d peaks comes from neighbour-cell matches (82% of it), not from dead cells, start time or pile-up. Production picks the largest-ToT hit among the projected and neighbour cells, so random hits can replace the track's own hit. Emulating this choice in picoDst (no ToT) brackets the in-time random matches at 3.3–7.3% of the TOF matches in 0–5%. A late component flat in time, weakly dependent on centrality, has no time in the picoDst and cannot be mixed; this needs the MuDst.

- MuDst (Sec. 7.7, 21160 events of two runs): unmatched BTOF hits are 59% of all hits and much later than matched ones (9.5% vs 1.3% above 26 ns): slow, unreconstructed particles. Re-running `StBTofMatchMaker`/`StBTofCalibMaker` on MuDst reproduces all 16451 production matches of the closure test (time of flight to -14 ps median). Main result, production makers re-run on the overlay of partner hits and partner tracks: random matches 2.62% (0–5%) and 1.57% (0–80%), SE/ME between p and d 1.39 (1.06–1.39 per bin at 0.4–2 GeV/c). Cross-checks: emulation 2.7% and 1.6%; without partner tracks 5.98%; the pool method (own hits removed) 10.44%, which is not a random-match fraction.

- Large MuDst sample (Sec. 7.8, 102 files of 14 runs, 643650 events, method 1): random matches 2.8% (0–5%) and 1.7% (0–80%), SE/ME between p and d 1.32. In SE−ME the split Student-t line shape is best in 9 of 9 momentum bins (mean χ^2/ndf 21 vs 208 for Gaussians, 0–5%); raw $\bar{\mathrm{p}}/\mathrm{p}$ = 0.089–0.104 at 0.4–2.5 GeV/c. Method 2 (Sec. 7.9) gives 2.64% and 1.58%, within 9% of method 1, and yields within 1.8% at 0.4–2 GeV/c. The SE excess between p and d at low momentum is a fixed rate per track, independent of centrality and flat in time, so it is not a random match (Sec. 7.9.3).

- Full simulation (Sec. 7.10, HIJING+GEANT, 49857 events): 2.4% of the matches at 0.2–0.4 GeV/c are decay muons (they stay near the pion peak) and 2.7% secondaries, 100% of which start in the BTOF trays. In the p–d window the correlated rate is 2.7 per 10^4 tracks, flat in centrality (data/simulation 0.71–1.00); at 0.4–0.6 GeV/c the simulation is low (data/simulation 1.63–2.40; a simulated ToT closes part of it). Added to the data ME, the simulated component describes the data SE between p and d at 0.2–0.4 GeV/c (SE/(ME + sim.) 0.92–1.00) but over-predicts the K–p valley and the region above the deuteron by 2–3×, so it is not used as a correction.

- Occupancy: trays 30, 34, 102 have no hits at all; 140 of 3840 modules (3.6%) are empty. No matched track points to an empty module.

Choices made (one line each; details where used).

1. Production P24iy (latest of P21ic, P23id, P24iy in the file catalog) — Sec. 2.
2. Input read through XRootD from the `local` storage replicas; the NFS catalog replicas checked were absent on disk — Sec. 2.
3. STAR library SL24y run inside the SL7 container on the AlmaLinux 9 interactive node — Sec. 2.2.
4. Minimum-bias trigger IDs 640001/11/21/31/41/51 — Sec. 3.1.
5. Event cuts $|V_z| < 70$ cm, $V_r < 2$ cm; no $V_z^{\mathrm{TPC}} - V_z^{\mathrm{VPD}}$ cut — Sec. 3.1.
6. Track cuts: primary, $p_T > 0.15$ GeV/c, nHitsFit$\geq$ 15, nHitsFit/nHitsMax$>$ 0.52, gDCA$<$ 3 cm, $|\eta| < 1$ — Sec. 3.2.

7. TOF match: `btofMatchFlag`> 0 and $\beta > 0$; primary momentum used for m^2— Sec. 3.3.
8. Approximate TOF $n\sigma$ with a fixed $1/\beta$ resolution of 0.013 (QA only) — Sec. 3.3.
9. BTOF module geometry fitted from production matches (no `StBTofGeometry`) — Sec. 7.
10. Projection with the picoDst global helix; matching at cell level, "cell" and "cell±1" — Sec. 7.
11. Mixing within the same run, 14 V_z × 9 refMult classes, 5 partner events — Sec. 7.
12. Large sample: one job per run (≥60k events), bad runs excluded, catalog replicas deduplicated — Sec. 7.4.
13. Centrality from StRefMultCorr (2019 19.6 GeV), pile-up events excluded from m^2 — Sec. 7.5.
14. Random m^2 from overlay mixed events (partner hits added to the track's own event), normalized per projected track, all hits included (untimed hits assumed to have the timed-hit time distribution) — Sec. 7.5.

2 Dataset, software and computing

2.1 Dataset

The STAR file catalog was queried with `get_file_list.pl` for `filetype=daq_reco_picoDst, trgsetupname=production_19GeV_2019`. Three productions exist: P21ic (SL21c), P23id (SL23d) and P24iy (SL24y).

Choice

Use P24iy, the most recent production of the Run19 19.6 GeV data (library SL24y). The picoDst format is the one defined in `StRoot/StPicoEvent` of https://github.com/star-bnl/star-sw; the SL24y copy of that package is used to read it.

For P24iy the catalog lists 100 560 files on HPSS, 99 273 on `local` storage (1.42×10^9 events) and 100 958 NFS entries. For the latest runs, the NFS paths checked (first 3000 entries ordered by run number) do not exist on disk.

Choice

Read the `local` replicas through XRootD, `root://xrdstar.rcf.bnl.gov:1095//home/starlib/...` (absolute path, double slash). The full list is `filelists/P24iy_19GeV_all_xrd.list`; the catalog dump is `filelists/local_P24iy_all.txt`.

Test sample. One file from the last run in the catalog, run 20093036: `st_physics_20093036_raw_1000008.picoDst.root` (17041 events in the TTree). The first 10000 events were processed (`filelists/test.list`).

Caveat

A single file of a single run is not representative of the whole dataset (beam conditions, detector status, luminosity vary run by run). Statistics are adequate for shapes and mean values, not for run-dependent or tray-dependent conclusions.

Table 1: Software environment, from `logs/provenance.txt` (version banners on the node) and the run log.

Node	starsub01.sdcc.bnl.gov, AlmaLinux release 9.8 (Olive Jaguar)
Container	`/cvmfs/star.sdcc.bnl.gov/containers/rhic_sl7_05_26.sif` (Scientific Linux release 7.3 (Nitrogen))
STAR library	SL24y (`/star/nfs4/AFS/star/packages/SL24y`)
ROOT	5.34/38 (STAR spack build)
Compiler	gcc 4.8.5
Executable	`root4star` (SL24y, .sl73_x8664_gcc485)
Run date	Tue Nov 3 22:14:23 2026 (EST)

2.2 Software and computing

Choice

The interactive nodes run AlmaLinux 9 while the STAR libraries are SL7 builds. All builds (`cons`) and jobs run in the SL7 container via the wrapper `scripts/sl7.sh` (`apptainer exec -B /gpfs01,/gpfs,/star,/cvmfs,/direct,/sdcc ... tcsh -c "starver SL24y; ..."`).

Working directory: `/gpfs01/star/pwg/dongx/tof/TOFME`. Code:

- `StRoot/StTofMEQAMaker/StTofMEQAMaker.{h,cxx}` — QA maker (`StMaker`) behind `StPicoDstMaker` in read mode.
- `macros/runTofMEQA.C` — chain macro; sets cuts and trigger IDs.
- `scripts/run_qa.shNlistoutlog` — runs the macro in the container.
- `macros/drawTofMEQA.C`, `scripts/draw_qa.sh` — quick-look plots on the node.
- `StRoot/StTofMEMaker`, `macros/runTofME.C`, `macros/makeGeom.C`, `scripts/run_me.sh`, `scripts/me_test.sh`, `scripts/select_runs.py`, `scripts/me_v3.sub`, `scripts/merge_v3.sh` — projection, mixed-event and m^2 step (Sec. 7–7.5).
- Note figures/tables: `analysis_note/scripts/make_figures.C`, `make_numbers.C`, `make_me.C`, `make_me_runs.C`, `make_m2.C`, `parse_log.py`, run by `build.sh` on the local copy `analysis_note/data/qa_test_10k.root`.

Choice

Only the picoDst branches `Event`, `Track`, `BTofHit` and `BTofPidTraits` are enabled (`StPicoDstMaker::SetStatus`), since nothing else is used.

Performance. Processing 10000 events took 324 s wall time but only 13.7 s CPU; 99.4% of the wall time is spent in `StPicoDstMaker`, i.e. the job is limited by XRootD I/O, not by the analysis.

Caveat

At this rate (about 30 events/s per job) full-dataset processing needs batch jobs, ideally scheduled close to the data; this has not been set up yet.

3 Method and definitions

The QA maker loops over picoDst events; per event it (i) records trigger IDs and the vertex before any cut, (ii) applies the event selection, (iii) fills event-level and BTOF-hit histograms, (iv) loops over tracks applying the track selection and fills track and TOF histograms. All histograms are written to one ROOT file (`output/qa_test_10k.root`).

3.1 Event selection

Cuts are applied in the order listed; the cut flow is stored in `hEvtCut`.

1. **Trigger:** the event fires at least one of 640001, 640011, 640021, 640031, 640041, 640051 (`StPicoEvent::isTrigger`).
2. **Vertex z:** $|V_z| < 70\,\text{cm}$, with V_z the TPC primary vertex (`primaryVertex()`).
3. **Vertex radius:** $V_r = \sqrt{V_x^2 + V_y^2} < 2\,\text{cm}$, measured from the origin of the STAR frame (no beam-spot offset).
4. $|V_z - V_z^{\text{VPD}}|$: *not applied* (cut value set negative), only histogrammed.

Choice

The trigger list is taken as the Run19 19.6 GeV minimum-bias offline trigger IDs (640001–640051 in steps of 10). This list was *not* checked against the trigger database; only 640051 occurs in the test file (Table 3), so the other IDs are untested here.

Choice

$|V_z| < 70\,\text{cm}$ keeps the vertex well inside the TPC and BTOF acceptance and is a typical value for collider-mode BES-II analyses. $V_r < 2\,\text{cm}$ rejects beam-pipe interactions. Both are QA defaults, not optimised.

Caveat

The $V_z^{\text{TPC}} - V_z^{\text{VPD}}$ cut, often used against pile-up, is not applied because at 19.6 GeV the VPD efficiency and resolution are limited (Sec. 4). Pile-up rejection (e.g. refMult versus nBTOFMatch) is also not applied yet.

3.2 Track selection

Tracks are taken from the picoDst track array; for each track:

1. **Primary** track (`isPrimary()`, i.e. $|\vec{p}_{\text{prim}}| > 0$), $p_T^{\text{prim}} > 0.15\,\text{GeV}/c$. Histograms of nHitsFit, nHitsFit/nHitsMax and gDCA are filled here, *before* the next cuts.
2. **Quality:** nHitsFit ≥ 15, nHitsFit/nHitsMax > 0.52 (rejects split tracks), gDCA $< 3\,\text{cm}$ with gDCA $= |\vec{\text{origin}}_{\text{global}} - \vec{V}|$ (`gDCA(pv).Mag()`). The η histogram is filled here.
3. **Acceptance:** $|\eta| < 1$ (BTOF acceptance). All remaining track and TOF histograms use this sample, which is also the denominator of the matching efficiency.

Track kinematics (p, p_T, η, ϕ) use the *primary* momentum, and the charge q is the sign of nHitsFit as stored in the picoDst.

3.3 TPC–BTOF association variables

A track is *associated* with BTOF if it has a `StPicoBTofPidTraits` entry (`isTofTrack()`). From it the following are used:

- `btofMatchFlag`: 0 no match, 1 one-to-one, 2 one hit with several track candidates. **Matched** means flag > 0.
- `btofCellId` $= (\text{tray}-1)\cdot 192 + (\text{module}-1)\cdot 6 + (\text{cell}-1)$; tray $\in [1, 120]$, module $\in [1, 32]$, cell $\in [1, 6]$.
- y_{local}, z_{local}: local coordinates of the track projection relative to the centre of the matched cell; y is along the row of six pads in a module (nominal pad pitch 3.15 cm), z is along the beam (nominal pad length 6.1 cm). The production matching (`StBTofMatchMaker`, SL24y) requires $|z_{\text{local}}| < 3.05$ cm (`mZLocalCut`, checked in the source).
- global hit position `btofHitPos()` (used for the ϕ–z map);
- $\beta =$ `btofBeta()`. Matched tracks without valid timing have $\beta = 0$.

Derived quantities, with p the primary momentum:

$$m^2 = p^2\left(\frac{1}{\beta^2} - 1\right), \qquad \frac{1}{\beta^i_{\text{exp}}} = \frac{\sqrt{p^2 + m_i^2}}{p}, \qquad n\sigma_i^{\text{TOF}} = \frac{1/\beta - 1/\beta^i_{\text{exp}}}{0.013}, \quad i = \pi, \text{K}, \text{p}. \tag{1}$$

The **TOF-matched sample** (numerator of the efficiency) requires flag > 0 and $\beta > 0$. The matching efficiency is $\varepsilon_{\text{match}} = N_{\text{TOF}}/N_{\text{TPC}}$, with binomial (Bayesian, uniform prior, 68% interval) errors in the figures and $\sqrt{\varepsilon(1-\varepsilon)/N}$ in the tables.

Caveat

$n\sigma^{\text{TOF}}$ here uses a constant $1/\beta$ resolution of 0.013 for orientation only; it is not the calibrated $n\sigma$ stored in `StPicoBTofPidTraits`. $\varepsilon_{\text{match}}$ is the product of the BTOF geometric acceptance, the hit efficiency and the matching efficiency, and still contains random matches — separating those is the aim of the mixed-event step.

BTOF hits. For every stored `StPicoBTofHit` the tray, module and cell are decoded from `id()` with the encoding above. Note that `StPicoBTofHit::cell()` in SL24y returns `mId/6+1` rather than the cell number; the maker computes `id\%6+1` itself. The picoDst stores only the cell ID of each hit, not its position or time.

4 Event-level QA

All figures in Secs. 4–6 are produced by `analysis_note/scripts/make_figures.C` from `data/qa_test_10k.root` (copy of `output/qa_test_10k.root` on the node). Unless stated otherwise they show the 5 501 selected events; the event count printed above each frame is the number of selected events. Errors are statistical.

Trigger. 9 989 of 10 000 events carry 640051, the only minimum-bias ID present. IDs 6, 35, 36 and 37 also appear in a large fraction of events.

Caveat

The meaning of trigger IDs 6, 35, 36, 37, 640601 and 6408xx was not looked up. They do not affect the selection, which only requires one of the minimum-bias IDs.

Table 2: Event cut flow for the test sample (`hEvtCut`; generated table `tables/evtcut.tex`).

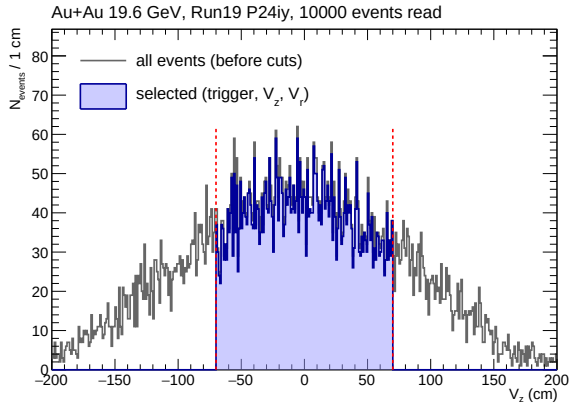
Cut (cumulative)	Events	Fraction of previous	Fraction of all
All events read	10 000	1.000	1.000
Minimum-bias trigger	9 989	0.999	0.999
$\lvert V_z\rvert < 70\,\mathrm{cm}$	5 786	0.579	0.579
$V_r < 2\,\mathrm{cm}$	5 501	0.951	0.550
$\lvert V_z - V_z^{\mathrm{VPD}}\rvert$ (not applied)	5 501	1.000	0.550

Table 3: Trigger IDs stored in the picoDst for all 10 000 events read (an event can carry several). IDs with fewer than 50 events are summed. 21 distinct IDs occur in total (full list in `logs/numbers.txt`).

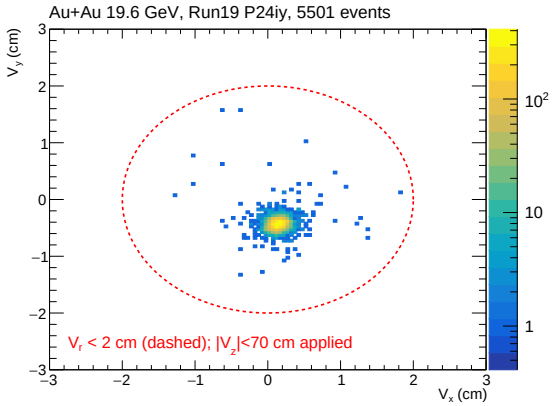
Trigger ID	Events	Fraction
640051	9 989	0.9989
6	9 371	0.9371
36	8 785	0.8785
35	8 437	0.8437
37	3 669	0.3669
640601	303	0.0303
640039	20	0.0020
Other (14 IDs, each < 50)	59	0.0059

Vertex. The V_z distribution before cuts is broad (RMS 80.9 cm, mean -8.0 cm), so the $\lvert V_z\rvert < 70\,\mathrm{cm}$ cut is the largest loss (Table 2, Fig. 1a). The transverse vertex is at $(\langle V_x\rangle, \langle V_y\rangle) = (0.149, -0.421)\,\mathrm{cm}$ with RMS 0.105 and 0.100 cm (Fig. 1b), well inside the $V_r < 2\,\mathrm{cm}$ circle centred at the origin.

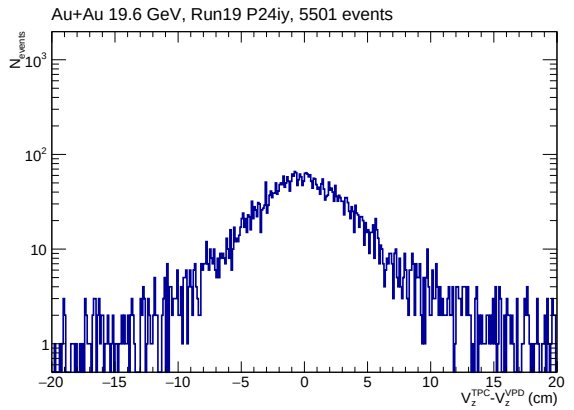
VPD vertex. For 92.9% of the selected events V_z^{VPD} lies within $\pm 200\,\mathrm{cm}$; the remainder has no valid VPD vertex. These events, together with events with $\lvert\Delta V_z\rvert > 20\,\mathrm{cm}$, populate the under-/overflow of Fig. 1c (117 / 530 events). A Gaussian fit to the core ($\lvert\Delta V_z\rvert < 3\,\mathrm{cm}$) of $\Delta V_z = V_z - V_z^{\mathrm{VPD}}$ gives mean -0.14 cm and $\sigma = 2.71\,\mathrm{cm}$; only 51.2% of the selected events have $\lvert\Delta V_z\rvert < 3\,\mathrm{cm}$.



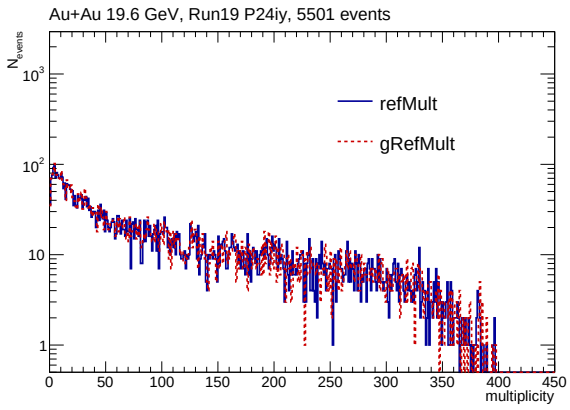
(a) V_z for all 10 000 events read (grey line) and for the selected events (blue, filled). Red dashed lines: $|V_z| < 70$ cm. 1 cm bins.



(b) V_y vs V_x for selected events; the red dashed circle is $V_r = 2$ cm. 0.05 cm bins.



(c) $V_z^{\mathrm{TPC}} - V_z^{\mathrm{VPD}}$ (0.1 cm bins, log scale). The central Gaussian core has $\sigma = 2.71$ cm; events without VPD vertex are outside the range.



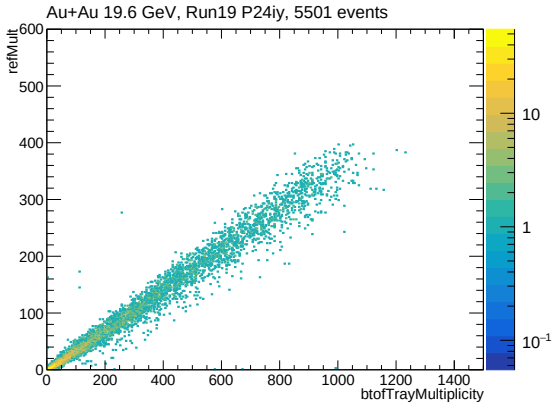
(d) refMult (blue solid) and gRefMult (red dashed), 1-unit bins, log scale. Means 102.3 and 102.4.

Figure 1: Event-level distributions of the test sample.

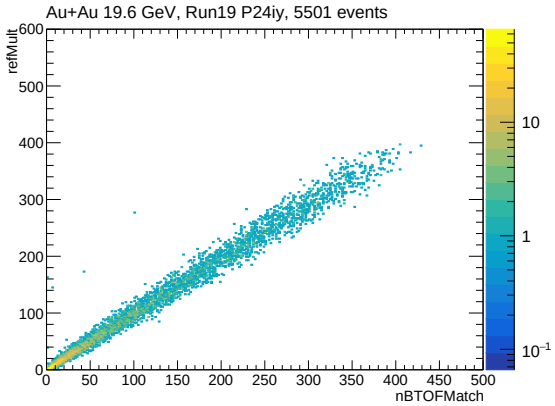
Multiplicities and BTOF hits. Per selected event the mean refMult is 102.3, the mean nBTOFMatch (`StPicoEvent::nBTOFMatch`) is 104.1 and the mean number of stored BTOF hits is 296.5 (maximum 1230). The mean of `btofTrayMultiplicity` is also 296.5. Figure 2c shows that the number of BTOF hits grows linearly with the number of matched tracks, and Fig. 2d that the number of TOF-matched *primary* QA tracks follows it closely (correlation factor 0.987). On average there are 2.94 BTOF hits per TOF-matched QA track; most hits are therefore *not* associated with a selected track — they come from tracks outside the selection, secondaries, or noise, and are the candidates for random matches.

Caveat

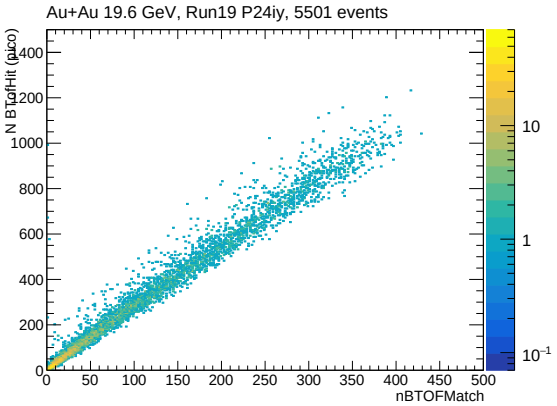
The hit counts and `btofTrayMultiplicity` have identical mean and RMS in this sample, which suggests the picoDst keeps all BTOF hits of the event. Not verified in `StPicoDstMaker`.



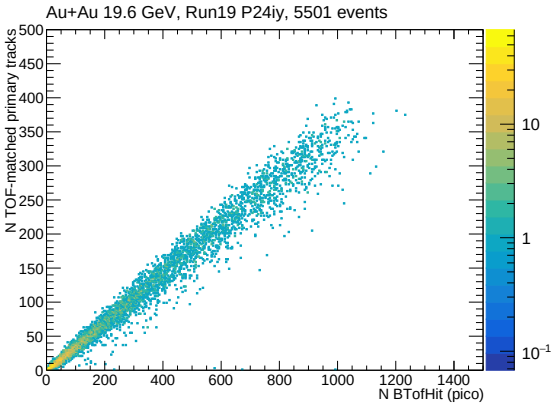
(a) refMult vs `btofTrayMultiplicity`.



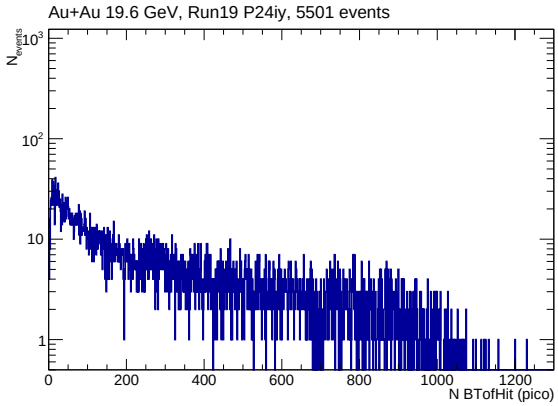
(b) refMult vs nBTOFMatch. No pile-up cut is applied; the band is narrow with few outliers in this sample.



(c) Number of stored BTOF hits vs nBTOFMatch.



(d) Number of TOF-matched primary QA tracks per event (Sec. 3.3) vs number of stored BTOF hits.



(e) Number of stored BTOF hits per selected event, log scale.

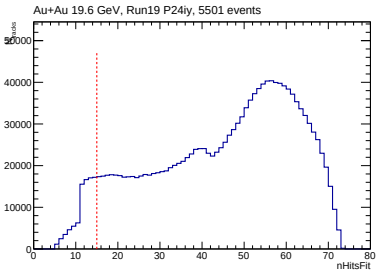
Figure 2: Event multiplicity correlations (log colour scale for 2D panels).

5 Track-level QA

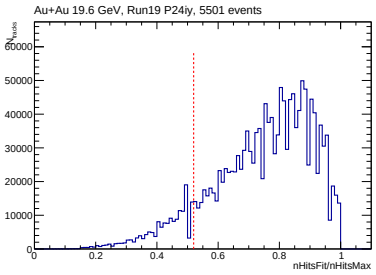
Table 4: Track cut flow, summed over the 5 501 selected events (generated table `tables/trkcut.tex`). Rows 1–3 are the TPC selection of Sec. 3.2; rows 4–6 the BTOF association of Sec. 3.3.

Selection (cumulative)	Tracks	Per event	Fraction of previous
Primary tracks, $p_T > 0.15\,\mathrm{GeV}/c$	1 542 971	280.5	–
+ nHitsFit≥ 15, nHitsFit/nHitsMax> 0.52, gDCA$< 3\,\mathrm{cm}$	1 347 381	244.9	0.873
+ $\lvert\eta\rvert < 1$ (QA track sample, denominator)	960 157	174.5	0.713
+ has BTofPidTraits	562 313	102.2	0.586
+ btofMatchFlag> 0	562 313	102.2	1.000
+ $\beta > 0$ (TOF-matched sample, numerator)	554 429	100.8	0.986

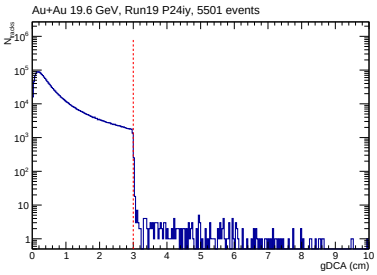
Table 4 gives the track yields. The quality cuts keep most primary tracks, and the $|\eta| < 1$ cut keeps 71.3% of those: with the inner TPC (iTPC) upgrade in Run19, primary tracks extend well beyond $|\eta| = 1$ (Fig. 3d). The QA sample has 174.5 tracks per event, mean $p_T = 0.516\,\mathrm{GeV}/c$.



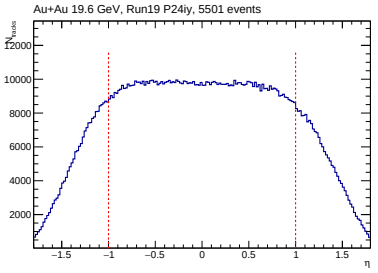
(a) nHitsFit, primary $p_T > 0.15\,\mathrm{GeV}/c$, before quality cuts. Red: cut at 15.



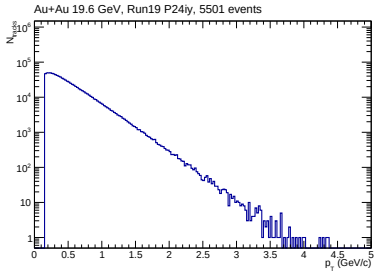
(b) nHitsFit/nHitsMax, same sample. Red: cut at 0.52. The spikes are a binning effect of the ratio of two integers.



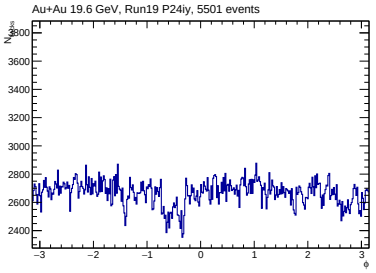
(c) gDCA, same sample, log scale. Red: cut at 3 cm.



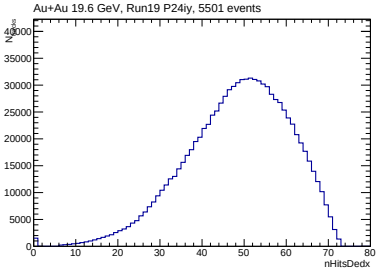
(d) η after quality cuts. Red: $|\eta| < 1$.



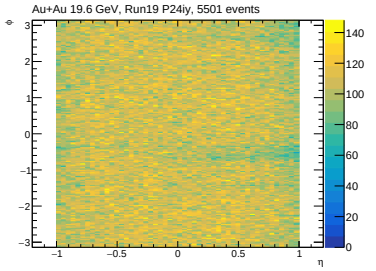
(e) p_T of QA tracks, log scale.



(f) ϕ of QA tracks (1° bins). Flat within about ±5% except a dip near $\phi \approx -0.6$ to -0.3.

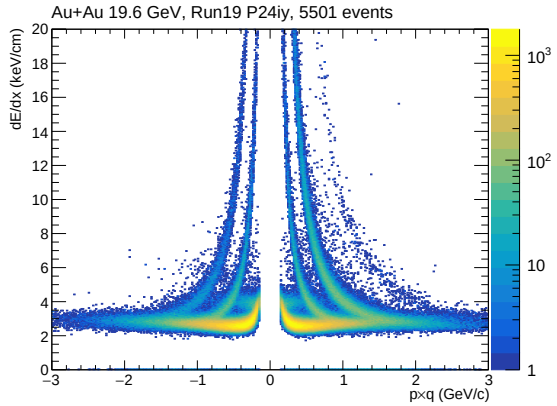


(g) nHitsDedx of QA tracks.

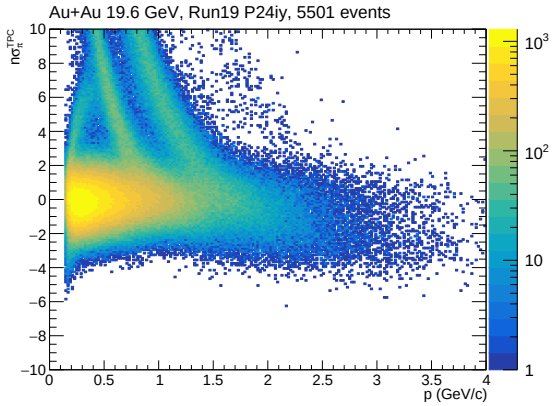


(h) ϕ vs η of QA tracks (linear colour scale). A region of lower density is visible at $\phi \approx -0.6$ for $\eta > 0$.

Figure 3: TPC track QA. Top row: distributions of the cut variables (before the cuts) and η; other panels: QA track sample (primary, $p_T > 0.15\,\mathrm{GeV}/c$, quality cuts, $|\eta| < 1$).



(a) dE/dx (keV/cm, as stored in the picoDst) vs $p \times q$ for QA tracks, log colour scale. π, K and p bands are separated at low momentum for both charges; a deuteron band is visible for positive charge.



(b) TPC $n\sigma_\pi$ vs p of QA tracks, log colour scale.

Figure 4: TPC dE/dx PID for the QA track sample.

Caveat

The reduced track density near $\phi \approx -0.6$, $\eta > 0$ (one TPC sector region of the west half, by its position) was seen by eye only and not traced to a sector or run condition. The ϕ and η–ϕ structure was not quantified.

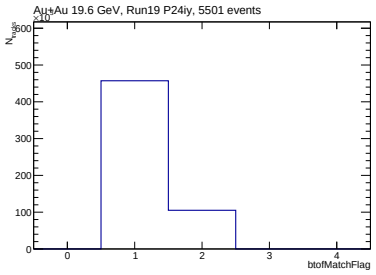
6 TPC–BTOF association QA

Match flag and β. Every QA track with a `BTofPidTraits` entry has `btofMatchFlag`> 0 (0 entries with flag 0), so the picoDst keeps only matched tracks. 457 193 (81.3%) are one-to-one matches and 105 120 (18.7%) have flag 2, i.e. the hit had more than one track candidate (Fig. 5a). 7 884 matched tracks (1.4%) have $\beta = 0$ (no valid time) and are dropped from the TOF-matched sample. The β distribution (Fig. 5b) peaks just below 1 with a tail above 1 containing 3.72% of the matched tracks; these give $m^2 < 0$.

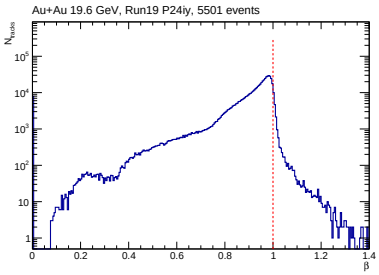
Local coordinates. z_{local} is flat with sharp edges at ± 3.05 cm, the matching window of `StBTofMatchMaker` (Fig. 5e). y_{local} (Fig. 5d) has a plateau over the pad, $|y_{\mathrm{local}}| \lesssim 1.6$ cm, falling over about 1 cm on each side, and flat tails out to ± 5 cm about a factor 50 below the plateau (read by eye from the figure); 92.4% of the entries are within $|y_{\mathrm{local}}| < 1.8$ cm, RMS 1.21 cm. The tails broaden at low momentum, where multiple scattering and the projection uncertainty are largest (Fig. 5f).

Caveat

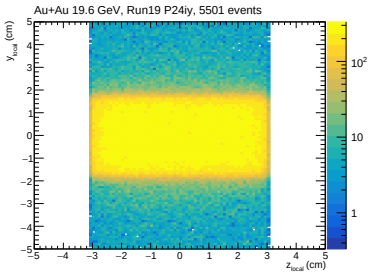
The flat y_{local} tails are the region where random matches are expected to dominate. Their size relative to the plateau is a direct handle for the mixed-event normalisation.



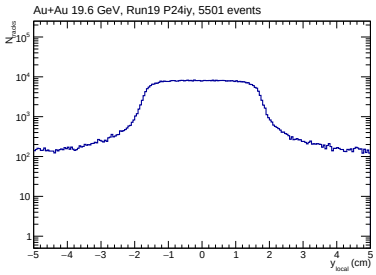
(a) `btofMatchFlag` of QA tracks with BTofPidTraits.



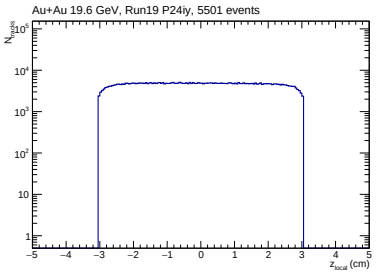
(b) β of matched tracks, log scale; red dashed line: $\beta = 1$. The $\beta = 0$ entries are in the first bin.



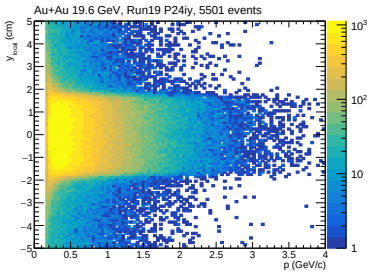
(c) y_{local} vs z_{local}, log colour scale.



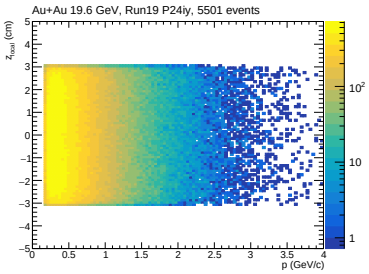
(d) y_{local} of matched tracks, 0.05 cm bins, log scale.



(e) z_{local} of matched tracks, 0.05 cm bins, log scale.



(f) y_{local} vs p, log colour scale.



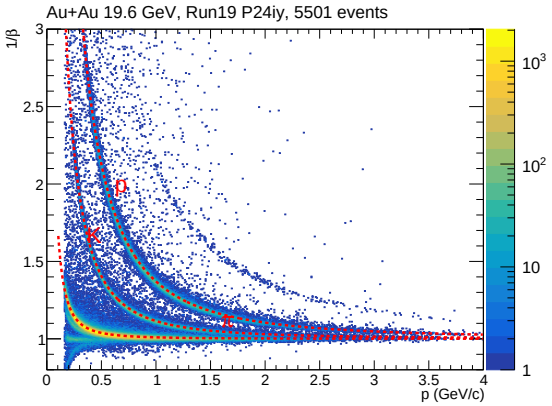
(g) z_{local} vs p, log colour scale.

Figure 5: Match flag, β and local hit coordinates for QA tracks matched to BTOF (flag>0).

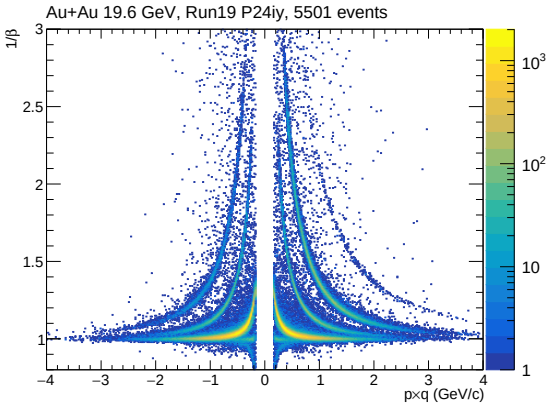
Particle identification. $1/\beta$ vs p (Fig. 6a) shows clean π, K and p bands that follow the expected curves, plus a deuteron band and a uniform background between bands. Fitting Gaussians to the m^2 projection for $0.4 < p < 1.0\,\mathrm{GeV}/c$ (Fig. 6e) gives

	mean (GeV^2/c^4)	σ (GeV^2/c^4)	PDG m^2
π	0.020	0.008	0.019
K	0.244	0.012	0.244
p	0.877	0.034	0.880
d	3.332	0.407	3.520

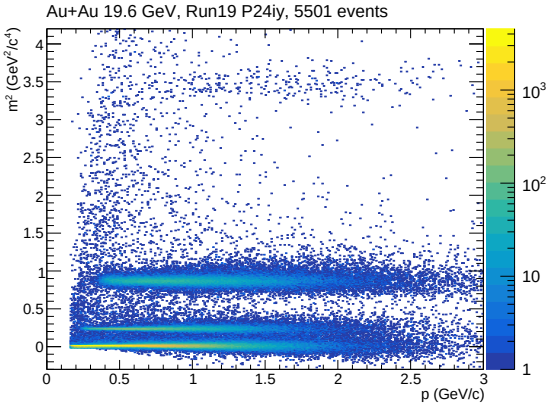
The π, K and p positions agree with PDG values within $0.005\,\mathrm{GeV}^2/c^4$, i.e. the TOF calibration in P24iy is in good shape for this run. The deuteron fit has low statistics in this momentum window and a broad, asymmetric peak; it is shown for orientation only.



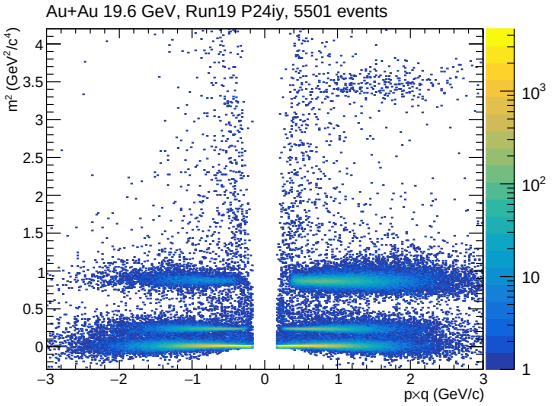
(a) $1/\beta$ vs p; red dashed curves are the expectations for π, K, p.



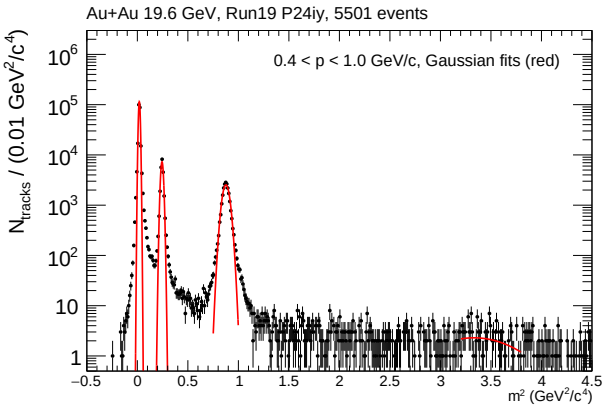
(b) $1/\beta$ vs $p \times q$.



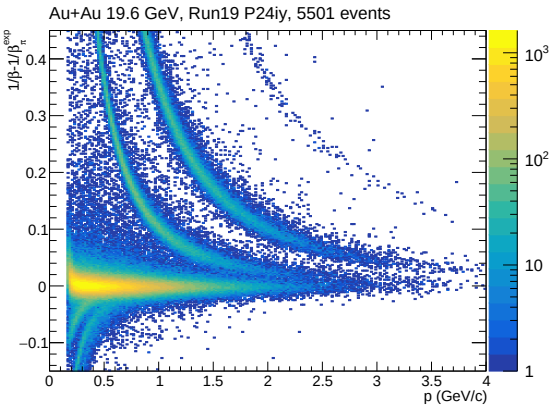
(c) m^2 vs p.



(d) m^2 vs $p \times q$; d appears only for positive charge.

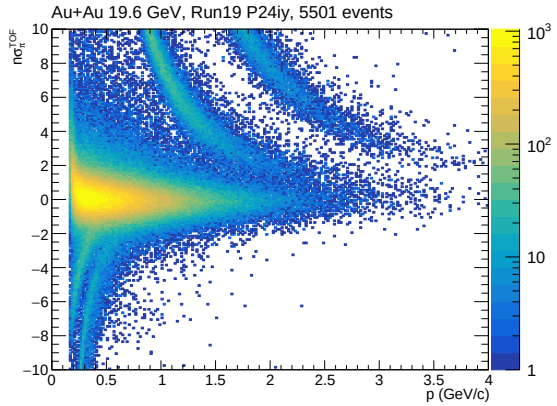


(e) m^2 for $0.4 < p < 1.0\,\mathrm{GeV}/c$ (black points), with Gaussian fits around the π, K, p and d peaks (red). Log scale; $0.01\,\mathrm{GeV}^2/c^4$ bins.

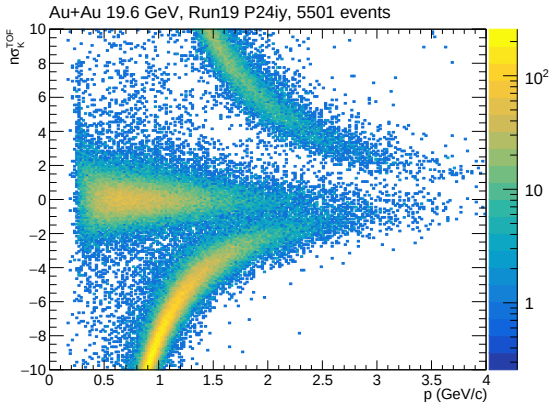


(f) $1/\beta - 1/\beta^{\pi}_{\mathrm{exp}}$ vs p.

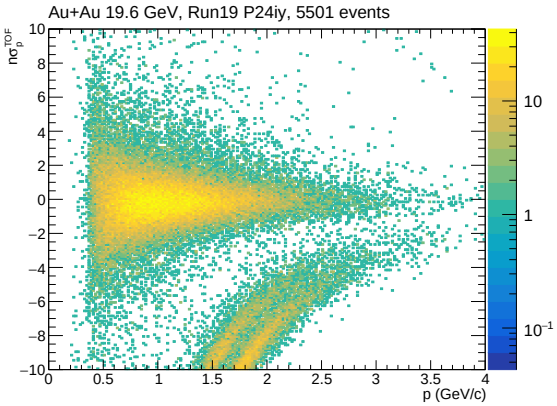
Figure 6: TOF PID for the TOF-matched sample (flag> 0, $\beta > 0$), log colour scale.



(a) $n\sigma_{\pi}^{\mathrm{TOF}}$ vs p.



(b) $n\sigma_{\mathrm{K}}^{\mathrm{TOF}}$ vs p.



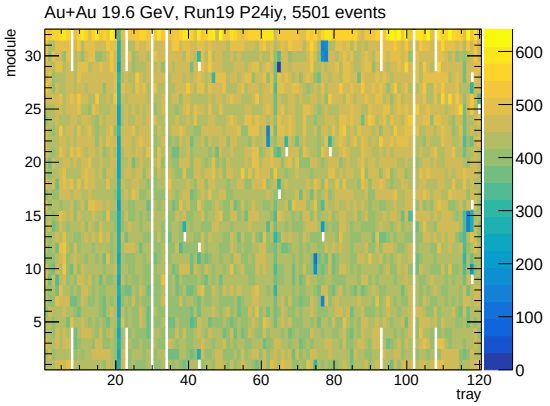
(c) $n\sigma_{\mathrm{p}}^{\mathrm{TOF}}$ vs p.

Figure 7: Approximate TOF $n\sigma$ with a fixed $1/\beta$ resolution of 0.013 (Sec. 3.3); not calibrated.

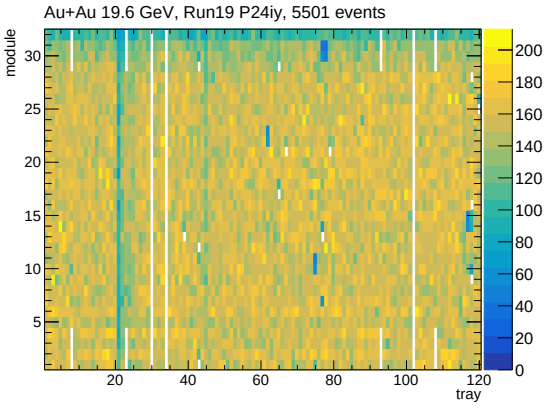
Occupancy. Figure 8a shows the tray–module occupancy of all stored hits. The median module has 440 hits in this sample. Trays 30, 34, 102 are empty; trays with empty modules, given as tray(number of empty modules), are 8 (8), 23 (8), 39 (1), 43 (3), 65 (1), 67 (1), 77 (1), 79 (1), 93 (8), 108 (8), 118 (3), 120 (1). In total 140 of 3840 modules (3.6%) are empty, and no matched track points to an empty module (0 entries), as expected. In each of the four trays with 8 empty modules, the empty modules are 1, 2, 3, 4, 29, 30, 31, 32 (same list in all four). Isolated low-occupancy modules are also visible. The global hit-position map of matched tracks (Fig. 8c) shows the same gaps as horizontal white bands at fixed ϕ, separately for $z < 0$ (east) and $z > 0$ (west). Cell occupancy (1–6) is uniform within a factor 1.224 (max/min).

Caveat

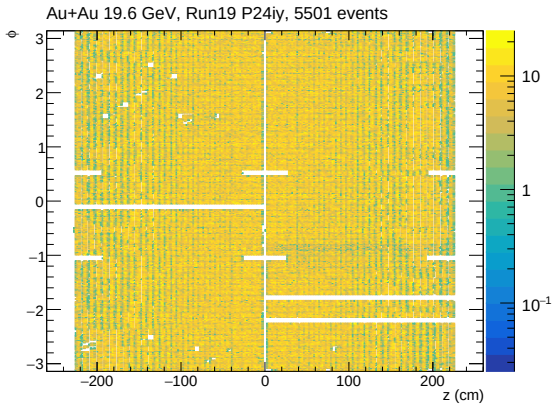
The dead-tray/module list is from one run only. A run-by-run (or time-period) BTOF status map is needed for the mixed-event step, since mixing events with different dead areas biases the random-match rate.



(a) Tray vs module for all stored BTOF hits (linear colour scale).



(b) Tray vs module of the cell matched to a QA track (flag>0), linear colour scale.



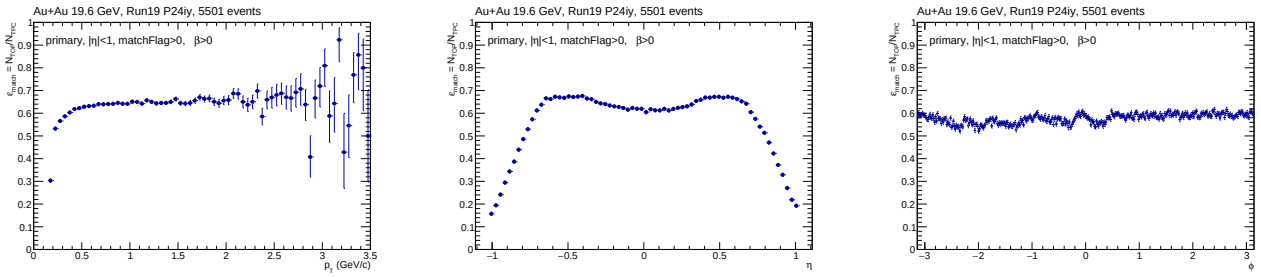
(c) Global ϕ vs z of the BTOF hit position of matched tracks, log colour scale.

Figure 8: BTOF occupancy.

Matching efficiency. Table 5 and Fig. 9a give $\varepsilon_{\text{match}}$ versus p_T. It rises from 0.467 at $0.15 < p_T < 0.3\,\text{GeV}/c$ to a plateau of 0.63–0.66 above $0.5\,\text{GeV}/c$ (Table 5). Versus η (Fig. 9b) it is 0.640 for $|\eta| < 0.5$ with a shallow dip at $\eta \approx 0$ and maxima near $|\eta| \approx 0.5$, then drops to 0.510 averaged over $0.5 < |\eta| < 1$ as the edge of the BTOF acceptance is reached. Versus ϕ (6° bins) it ranges from 0.539 (at $\phi \approx -2.36$) to 0.601.

Table 5: TOF matching efficiency vs p_T for QA tracks (generated table `tables/eff.tex`). Statistical errors.

p_T (GeV/c)	N_{TPC}	N_{TOF}	$\varepsilon_{\mathrm{match}}$
0.15–0.30	289 179	135 113	0.467 ± 0.001
0.30–0.50	289 953	175 653	0.606 ± 0.001
0.50–0.70	170 213	107 696	0.633 ± 0.001
0.70–1.00	126 415	81 087	0.641 ± 0.001
1.00–1.50	66 773	43 349	0.649 ± 0.002
1.50–2.00	14 077	9 177	0.652 ± 0.004
2.00–3.00	3 404	2 256	0.663 ± 0.008
3.00–5.00	141	98	0.695 ± 0.039



(a) $\varepsilon_{\mathrm{match}}$ vs p_T (0.05 GeV/c bins). (b) $\varepsilon_{\mathrm{match}}$ vs η. (c) $\varepsilon_{\mathrm{match}}$ vs ϕ (1° bins).

Figure 9: TOF matching efficiency $N_{\mathrm{TOF}}/N_{\mathrm{TPC}}$ (flag> 0, $\beta > 0$ over QA tracks). Error bars: 68% Bayesian binomial intervals.

7 Mixed-event setup for TPC–BTOF matching

This section describes step 2: projecting every QA track to BTOF with the helix stored in the picoDst, matching it at the cell level to the BTOF hits of the *same* event (SE) and of *other* events (mixed events, ME), and validating the projection against the production matching. Unless stated otherwise all numbers are from the same 10k-event test file as Secs. 4–6 (5 501 selected events), produced by `StRoot/StTofMEMaker` and summarised by `analysis_note/scripts/make_me.C` (`logs/me_numbers_test.txt`).

7.1 Method

BTOF geometry from data. The picoDst does not contain the BTOF geometry, and the official geometry (`StBTofGeometry`) needs the full detector description plus database alignment. Instead, the position of every module is fitted from production-matched tracks. For these, `btofHitPos()` is the production projection point in global coordinates and (y_{local}, z_{local}, cell) are its local coordinates, so for each module

$$\vec{r}_{\mathrm{hit}} = \vec{C} + y_{\mathrm{local}}\,\vec{U} + z_{\mathrm{local}}\,\vec{V} + (\mathrm{cell} - 3.5)\,\vec{W}, \tag{2}$$

which is linear in the 12 parameters $(\vec{C}, \vec{U}, \vec{V}, \vec{W})$. The maker accumulates the normal-equation sums per module (`hGeomSum`, additive over jobs) and `macros/makeGeom.C` solves them (modules with ≥ 20 matched tracks). Of 3840 modules, 3699 are fitted, with a mean residual of 0.0029 cm, i.e. the picoDst position precision, so the linear model is exact. The fit gives $|\vec{U}| = 1.00000$ (mean over modules), a

cell pitch of 3.450 cm (RMS 0.001 cm over modules), the `mWidthPad` value of `StBTofMatchMaker`, and module-centre radii of 213.54 cm (RMS 0.78 cm; Fig. 10a). The remaining 141 modules (dead in this run or with too few tracks) are copied from the same module of the nearest fitted tray in the same half (east/west), rotated about z by the tray-to-tray ϕ difference. Applying the same rotation to fitted modules (closure test) reproduces their centres to 0.297 cm on average (RMS 0.219 cm, at most 1.07 cm; Fig. 10b), small compared with the 3.45 cm pitch. Only 3.70% of the projected tracks land in a rotated module.

Choice

Data-driven module geometry instead of `StBTofGeometry`: it includes the production alignment automatically, has no database dependence, and is validated to the 0.01 cm level by the fit residuals. Geometry file: `geom/btof_geom_test.txt` (from the test run 20093036).

Projection. For each QA track (Sec. 3.2) the global helix is built with `StPicoTrack::helix(B)` (origin = DCA point, global momentum, charge, B from the event). It is intersected with a cylinder of radius 213.5 cm (the mean module radius) to select candidate modules ($|\Delta\phi| < 0.09$ to the tray centre, $|\Delta z| < 12$ cm to the module centre); then the exact intersection with each candidate module plane gives the local coordinates, the cell (pitch 3.45 cm) and z_{local}. The projection is accepted if it falls inside one of the six cells and $|z_{\mathrm{local}}| < 3.05$ cm (the production window). If several modules qualify, the one with smallest $|z_{\mathrm{local}}|$ is kept.

Matching definitions. Hits are stored only as cell IDs, so all matches are at the cell level:

- **cell**: a BTOF hit in the projected cell;
- **cell±1 (production-like)**: a hit in the projected cell or in an adjacent cell of the same module. This is the production rule: `StBTofMatchMaker` (SL24y, "match find neighbours") accepts a hit in the projected cell or its ±1 neighbour (checked in the source).

P_{SE} is the fraction of projected tracks matched to a hit of the same event. P_{ME} is the fraction of (track, partner-event) trials in which the projected cell (or cell±1) has a hit in the partner event. $f_{\mathrm{rand}} = P_{\mathrm{ME}}/P_{\mathrm{SE}}$ estimates the fraction of SE matches that are random. To first order the true matching efficiency is $\varepsilon_{\mathrm{true}} = (P_{\mathrm{SE}} - P_{\mathrm{ME}})/(1 - P_{\mathrm{ME}})$.

Mixing. Partner events come from a FIFO buffer of the previous 5 selected events with the same run number, the same V_z class (14 classes of 10 cm) and the same refMult class (boundaries 0, 10, 25, 50, 80, 120, 170, 230, 300). Buffers are cleared at each run change, so dead areas are identical in mixed pairs. 88.5% of the events have the full 5 partners.

Choice

Mixing classes in refMult (not number of BTOF hits) and V_z, depth 5, same run only. refMult is independent of BTOF, and the hit occupancy follows it closely (Fig. 2a); P_{ME} is then studied directly versus the hit multiplicity of the partner event.

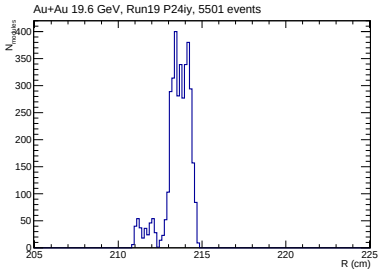
7.2 Validation of the projection

Of the 960 080 QA tracks, 76.8% have an accepted projection; the rest have no crossing with the TOF radius (1.8%), fall between modules (9.7%), outside the six cells (1.1%) or outside $|z_{\mathrm{local}}| < 3.05$ cm (10.5%) (Fig. 10c).

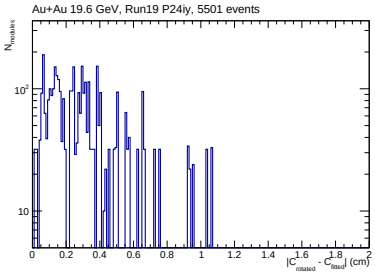
For the 562 288 production-matched QA tracks, 97.7% have an accepted projection, 92.9% are projected to the same module as production and 78.1% to the same cell (Fig. 10d). Above $p = 1.5\,\mathrm{GeV}/c$ the values are 98.2% (module) and 92.9% (cell); for $0.2 < p < 0.4\,\mathrm{GeV}/c$ the same-cell fraction is only 61.6%. Different cells in the same module are expected: 8.5% of the production matches have $|y_{\mathrm{local}}| > 1.75\,\mathrm{cm}$, i.e. production matched them to a hit in the neighbouring cell. When the cells agree, the local-coordinate differences (Figs. 10e, 10f) have RMS 0.043 cm (y) and 0.049 cm (z) for $p > 1.5\,\mathrm{GeV}/c$, growing to 0.679 cm and 0.454 cm for $p < 0.45\,\mathrm{GeV}/c$.

Caveat

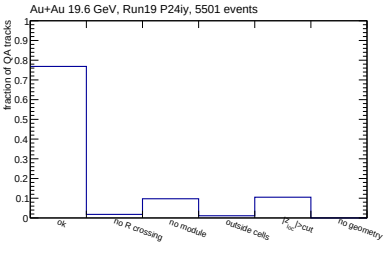
The picoDst helix starts at the DCA to the primary vertex, while production projects the outer TPC helix. Energy loss and multiple scattering between the TPC and BTOF are therefore not in this projection, which is why the agreement degrades at low momentum and P_{SE} is below the production match rate for $p_T < 0.3\,\mathrm{GeV}/c$ (Table 6). P_{ME} is not affected (Sec. 7.3).



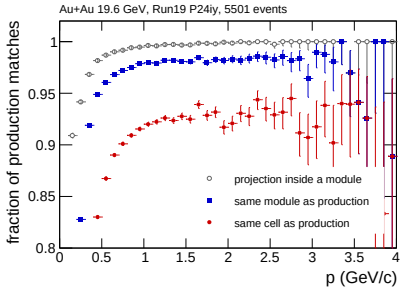
(a) Radius of the fitted module centres.



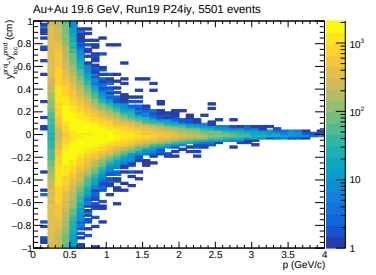
(b) Closure test of the rotated geometry: distance between rotated and fitted module centres, log scale.



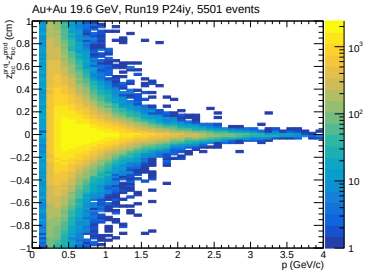
(c) Projection status of QA tracks (fractions).



(d) For production-matched tracks: fraction with an accepted projection (grey), projected to the production module (blue), and to the production cell (red), vs p.



(e) $y^{\mathrm{proj}}_{\mathrm{local}} - y^{\mathrm{prod}}_{\mathrm{local}}$ vs p when the cells agree, log colour scale.



(f) $z^{\mathrm{proj}}_{\mathrm{local}} - z^{\mathrm{prod}}_{\mathrm{local}}$ vs p when the cells agree, log colour scale.

Figure 10: Geometry and projection validation (test sample).

7.3 Same-event vs mixed-event matching (test sample)

737 720 QA tracks are projected and tested against 3 372 677 partner events in total. Integrated over p_T:

	P_{SE}	P_{ME} (%)	$f_{\mathrm{rand}} = P_{\mathrm{ME}}/P_{\mathrm{SE}}$ (%)
cell	0.682	2.34	3.4
cell±1 (production-like)	0.759	5.24	6.9

The production match rate for the same projected tracks is 0.734. The statistical uncertainty on P_{ME} is $\leq 0.01\%$ (absolute).

Table 6: Matching probabilities vs p_T for projected QA tracks (generated table `tables/me_eff_test.tex`). P_{prod}: production match with $\beta > 0$. Statistical errors on P_{ME}; those on P_{SE} and P_{prod} are below 0.003.

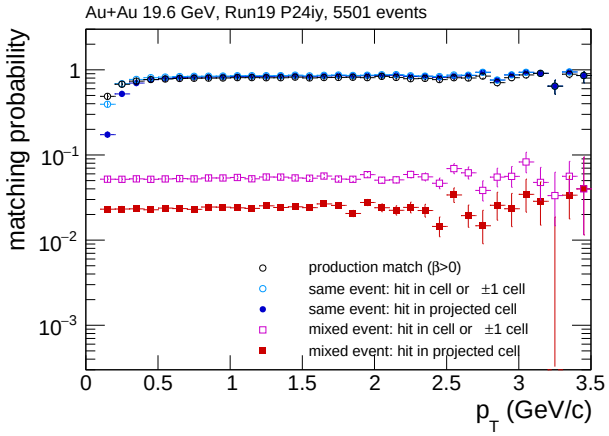
			projected cell			cell or ±1 (production-like)		
p_T (GeV/c)	N_{proj}	P_{prod}	P_{SE}	P_{ME} (%)	f_{rand} (%)	P_{SE}	P_{ME} (%)	f_{rand} (%)
0.15–0.30	203 994	0.625	0.427	2.316 ± 0.016	5.4	0.608	5.184 ± 0.023	8.5
0.30–0.50	229 351	0.751	0.729	2.304 ± 0.015	3.2	0.788	5.215 ± 0.022	6.6
0.50–0.70	135 791	0.785	0.800	2.365 ± 0.019	3.0	0.827	5.252 ± 0.028	6.3
0.70–1.00	101 000	0.798	0.823	2.380 ± 0.022	2.9	0.842	5.312 ± 0.033	6.3
1.00–1.50	53 358	0.810	0.838	2.420 ± 0.031	2.9	0.855	5.408 ± 0.046	6.3
1.50–2.00	11 319	0.810	0.844	2.489 ± 0.068	3.0	0.859	5.425 ± 0.100	6.3
2.00–3.00	2 789	0.808	0.850	2.292 ± 0.133	2.7	0.864	5.361 ± 0.200	6.2

Main observations:

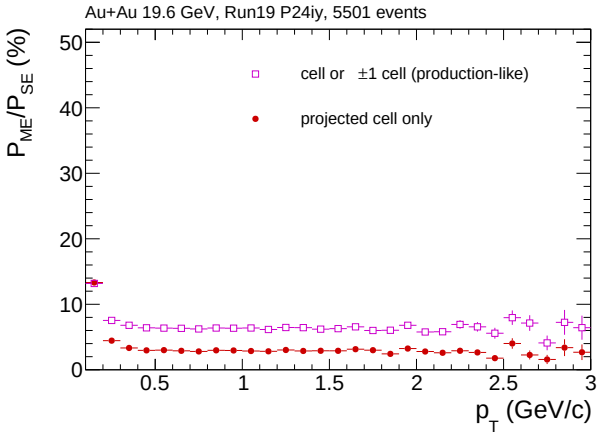
- P_{ME} **is the hit occupancy.** P_{ME} grows linearly with the number of hits N_{hits} in the partner event (Fig. 11c). A fit $P_{\mathrm{ME}} = a\, N_{\mathrm{hits}}$ gives $a = 4.31 \times 10^{-5}$, which is 0.99 times the uniform-occupancy expectation 1/23040. Averaged over trials, $P_{\mathrm{ME}}/(\langle N_{\mathrm{hits}} \rangle / 23040) = 0.99$.
- P_{ME} **is nearly flat** (2.3–2.5% for the cell definition) in p_T, η and in the local coordinates (Figs. 11a, 11e, 11f, 11g), as expected for hits unrelated to the track; it rises with refMult (Fig. 11d). Because random hits do not depend on the track, P_{ME} does not depend on the projection precision.
- **Random-match fraction.** For the production-like definition f_{rand} is 6.3% for $0.5 < p_T < 2\,\mathrm{GeV}/c$ and 8.5% for $0.15 < p_T < 0.3\,\mathrm{GeV}/c$ (Fig. 11b); the increase at low p_T comes from the lower P_{SE} there.
- **Same-event shape.** P_{SE} falls towards the cell edges in y_{local} (0.726 for $|y| < 1\,\mathrm{cm}$, 0.604 for $1.3 < |y| < 1.6\,\mathrm{cm}$) and at $|z_{\mathrm{local}}| \gtrsim 2.8\,\mathrm{cm}$; there the true hit is often in the neighbouring cell or module.
- First-order true matching efficiency: $\varepsilon_{\mathrm{true}} = 0.746$ (cell±1) and 0.674 (cell).

Caveat

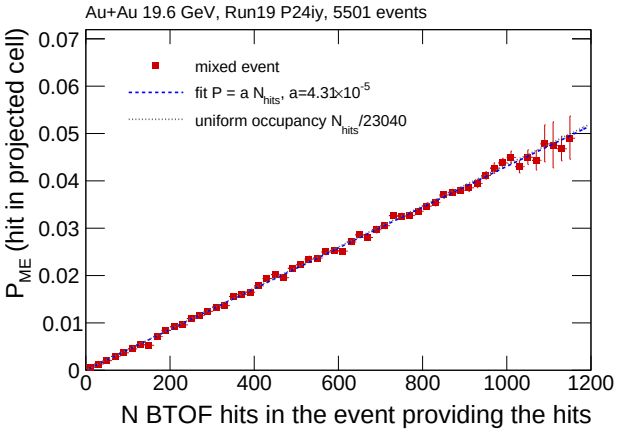
f_{rand} treats every projected track as equally exposed to random hits. Tracks whose own hit is present can still pick up a random hit in a neighbouring cell, which production would resolve differently. Production also resolves hits claimed by several tracks (flag 2) and requires a valid time, while P_{SE} here does not; this is why $P_{\mathrm{SE}}^{\pm 1}$ exceeds P_{prod} above $0.3\,\mathrm{GeV}/c$.



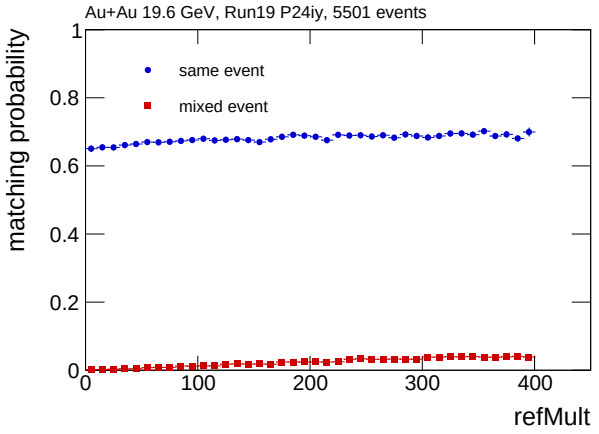
(a) Matching probabilities vs p_T (log scale): production (open black), SE cell±1 (open blue), SE cell (blue), ME cell±1 (open magenta), ME cell (red).



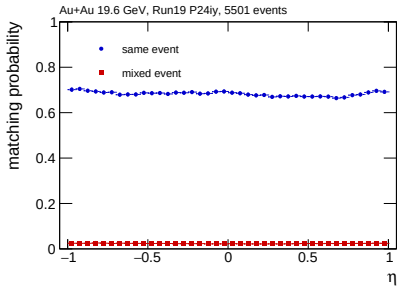
(b) $f_{\text{rand}} = P_{\text{ME}}/P_{\text{SE}}$ vs p_T (%) for cell±1 (open magenta) and cell (red).



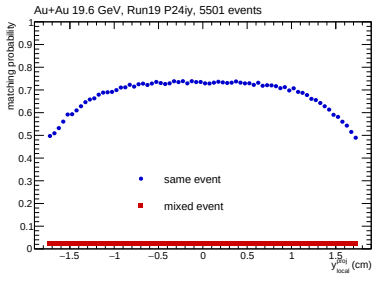
(c) P_{ME} (cell) vs number of BTOF hits in the partner event (red); dashed blue: linear fit through the origin; dotted grey: uniform occupancy $N_{\text{hits}}/23040$.



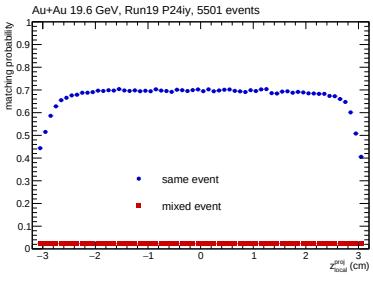
(d) P_{SE} (blue) and P_{ME} (red), cell definition, vs refMult of the track's event.



(e) P_{SE} and P_{ME} (cell) vs η.



(f) P_{SE} and P_{ME} (cell) vs projected y_{local}.



(g) P_{SE} and P_{ME} (cell) vs projected z_{local}.

Figure 11: Same-event (blue) and mixed-event (red) matching, test sample. Errors: 68% binomial intervals.

7.4 Mixed-event results with the 1M-event sample

Sample and processing. The large sample is 15 runs evenly spaced over the run range (20057003–20091008), one HTCondor job per run (`scripts/select_runs.py`, `scripts/me_v3.sub`, cluster 790125). Runs are taken only if they are not in the StRefMultCorr bad-run list for 2019 and if their largest files give at least 60 000 events. Files are deduplicated by name, because the catalog lists every replica as a separate entry (99 273 entries, 79 596 unique files; `logs/catalog_unique_P24iy.txt`). Files with a replica on the XRootD server qgp017 were excluded, because jobs reading from it stalled repeatedly. All 15 jobs finished normally (`scripts/merge_v3.sh`): 1 011 835 events read and 423 775 selected. With one run per job the 126 mixing buffers fill once per job, and 97.8% of the selected events have all 5 mixing partners. The jobs use the geometry fitted on the earlier large pass (`geom/btof_geom_1M.txt`).

Caveat

An earlier 1M-event pass (one file per job, draft v0.3 of this note) processed 5 files twice because of the duplicate catalog entries. It is superseded by this sample. Its integrated P_{ME} values (2.33% and 5.21%) agree with the ones below within 0.03 percentage points.

Caveat

The selected-event fraction is 42% here, against 55% in the test file, because the V_z distribution differs between runs. All rates below are per projected track or per mixed trial, so they are not affected, but event-level averages are not directly comparable with Sec. 4.

Geometry stability. The geometry refitted from the merged sums of the 15 jobs (`geom/btof_geom_v3.txt`) has 3697 fitted modules, fit residual 0.0029 cm and pitch 3.450 cm. For the 3696 modules fitted in both, the module centres differ from the test-run geometry by 0.0005 cm on average (at most 0.006 cm). The BTOF geometry is stable across the runs used, and the test-run geometry is adequate.

Projection and matching. With 71 105 376 QA tracks and 268 305 132 mixed trials, the projection-level numbers of the test sample are reproduced: 76.8% of the tracks are projected, and for production-matched tracks 92.8% (cell) and 98.2% (module) agree above 1.5 GeV/c. The integrated probabilities are

	P_{SE}	P_{ME} (%)	$f_{\mathrm{rand}} = P_{\mathrm{ME}}/P_{\mathrm{SE}}$ (%)
cell	0.679	2.34	3.4
cell±1 (production-like)	0.757	5.24	6.9

with statistical uncertainties below 10^{-4}, compared with 2.34% and 5.24% in the test sample. f_{rand} (cell±1) is 6.3% for $0.5 < p_T < 2$ GeV/c and 8.5% for $0.15 < p_T < 0.3$ GeV/c (Table 7, Fig. 12b). P_{ME} equals the trial-averaged hit occupancy ($P_{\mathrm{ME}}/(\langle N_{\mathrm{hits}}\rangle/23040) = 1.00$; slope 1.00 times 1/23040). These are probabilities that a random hit is *present* in the matching cells. How often such a hit is actually chosen instead of the track's own hit, and which m^2 it then gives, is the subject of Sec. 7.5.

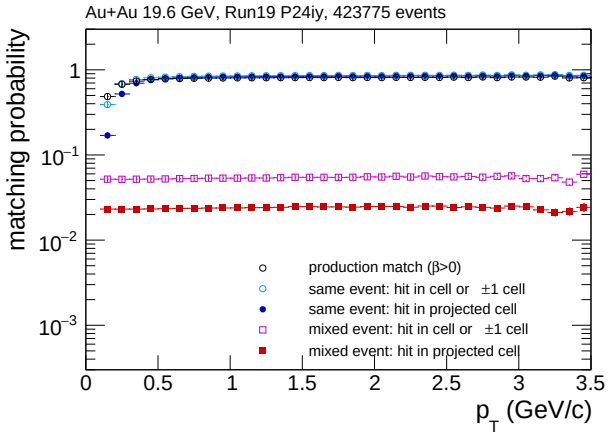
Caveat

P_{ME} is not exactly flat in p_T (Table 7). The origin is not understood. Candidates: harder tracks come more often from higher-refMult events within a mixing class, or point preferentially to higher-occupancy regions. This was not checked.

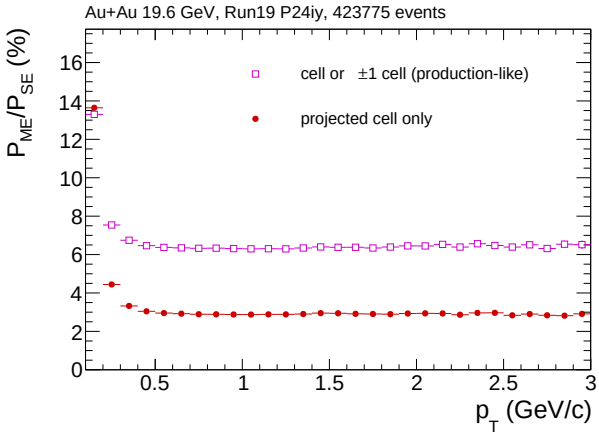
Table 7: Matching probabilities vs p_T, 1M-event sample (`tables/me_eff_v3.tex`); notation as in Table 6.

			projected cell			cell or ±1 (production-like)		
p_T (GeV/c)	N_{proj}	P_{prod}	P_{SE}	P_{ME} (%)	f_{rand} (%)	P_{SE}	P_{ME} (%)	f_{rand} (%)
0.15–0.30	15 119 485	0.625	0.426	2.314 ± 0.002	5.4	0.607	5.184 ± 0.003	8.5
0.30–0.50	17 034 547	0.751	0.726	2.319 ± 0.002	3.2	0.786	5.201 ± 0.002	6.6
0.50–0.70	10 030 992	0.785	0.798	2.346 ± 0.002	2.9	0.825	5.249 ± 0.003	6.4
0.70–1.00	7 454 599	0.798	0.822	2.373 ± 0.003	2.9	0.841	5.319 ± 0.004	6.3
1.00–1.50	3 943 131	0.806	0.834	2.411 ± 0.003	2.9	0.851	5.372 ± 0.005	6.3
1.50–2.00	826 470	0.812	0.842	2.459 ± 0.008	2.9	0.857	5.471 ± 0.011	6.4
2.00–3.00	195 367	0.814	0.847	2.472 ± 0.016	2.9	0.860	5.563 ± 0.023	6.5

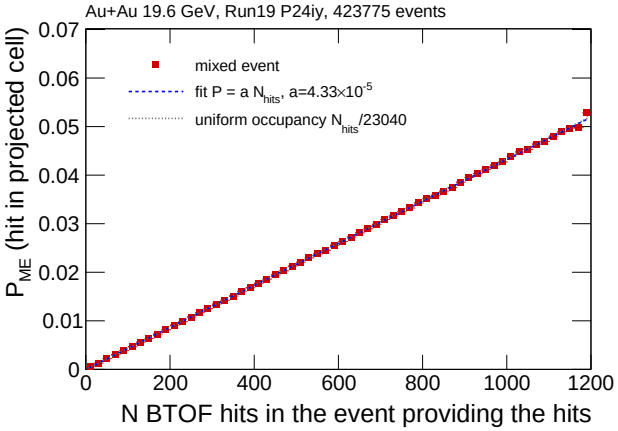
Run dependence. Per run, P_{ME} (cell±1) ranges from 5.10% to 5.38% (Fig. 12e), following the occupancy: the ratio $P_{\mathrm{ME}}/(\langle N_{\mathrm{hits}}\rangle/23040)$ stays between 0.99 and 1.01 (Fig. 12f). Per-run values: `logs/me_runs_v3.txt` (`scripts/make_me_runs.C`).



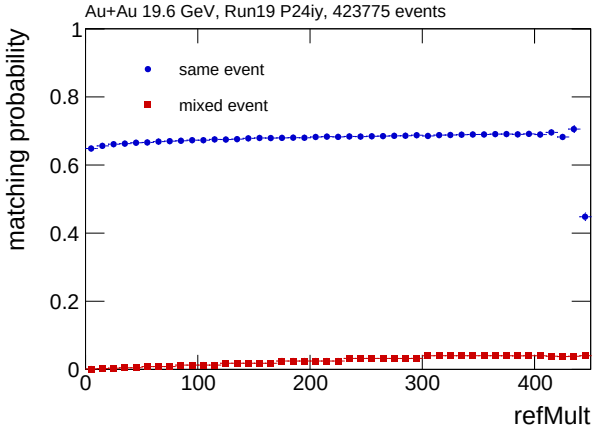
(a) Matching probabilities vs p_T, 1M sample; markers as in Fig. 11a.



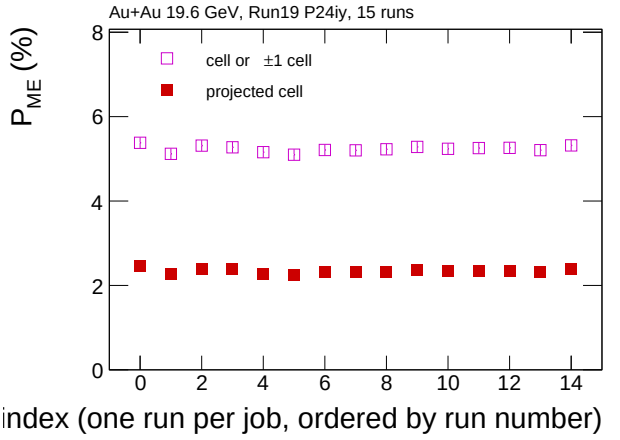
(b) $f_{\rm rand}$ vs p_T, 1M sample: cell±1 (open magenta), cell (red).



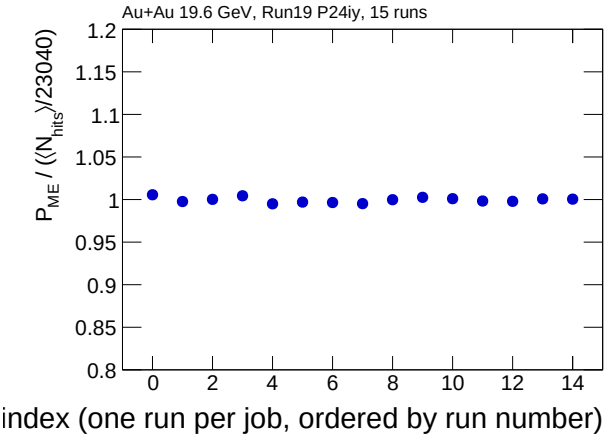
(c) $P_{\rm ME}$ (cell) vs number of BTOF hits in the partner event, with linear fit (dashed) and uniform occupancy (dotted).



(d) $P_{\rm SE}$ (blue) and $P_{\rm ME}$ (red) vs refMult, cell definition.

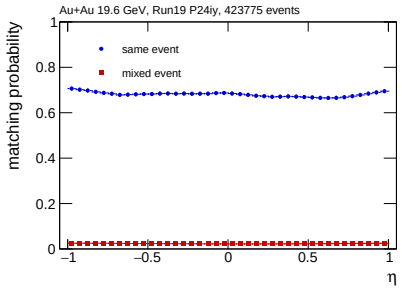


(e) $P_{\rm ME}$ per run (one job per run, ordered by run number): cell±1 (open magenta) and cell (red); statistical errors are smaller than the markers.

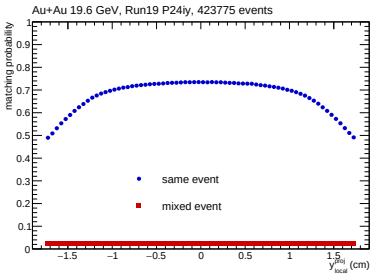


(f) $P_{\rm ME}$ (cell) per run divided by the trial-averaged occupancy $\langle N_{\rm hits}\rangle/23040$.

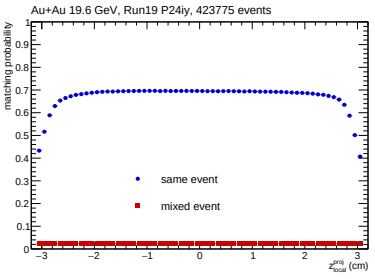
Figure 12: Mixed-event results, 1M sample (423 775 selected events from 15 runs).



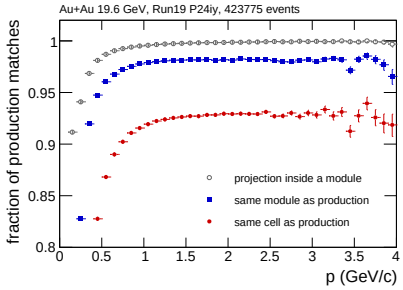
(a) P_{SE}, P_{ME} (cell) vs η.



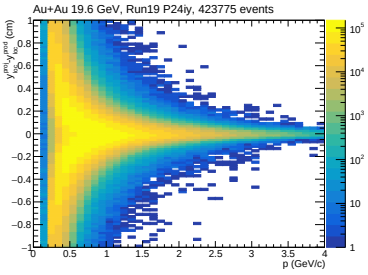
(b) P_{SE}, P_{ME} (cell) vs projected y_{local}.



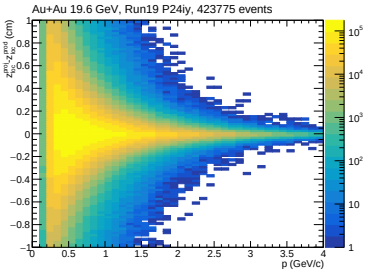
(c) P_{SE}, P_{ME} (cell) vs projected z_{local}.



(d) Projection vs production (as Fig. 10d).



(e) $\Delta y_{\mathrm{local}}$ vs p, same cell.



(f) $\Delta z_{\mathrm{local}}$ vs p, same cell.

Figure 13: Differential mixed-event and projection plots, 1M sample (same blue/red convention as Sec. 7.3).

7.5 m^2 of same-event and mixed-event matches per centrality

This section uses the 1M-event sample of Sec. 7.4 (`StRoot/StTofMEMaker` version 3, `output/me_v3.root`). It answers the question: what does the m^2 distribution of random TPC–BTOF matches look like, and how much of the measured m^2 distribution does it account for, in each momentum bin and centrality class? Numbers: `logs/m2_numbers_v3.txt` from `scripts/make_m2.C`.

7.5.1 Method

Centrality. Centrality is taken from StRefMultCorr (Au+Au 19.6 GeV 2019 parameters: V_z- and luminosity-corrected refMult, nine classes from 0–5% to 70–80%). Of the 423775 selected events, 873 are flagged as pile-up by its refMult–nBTOFMatch cut, 40647 are beyond 80%, and 497 have no BTOF hit with a time (no valid start time). The remaining 381758 events enter the m^2 histograms. Bad runs are excluded at the run selection. The mixing classes are unchanged (raw refMult).

Hit times. A picoDst BTOF hit carries only its cell. A time of flight exists for a hit only if a picoDst track was matched to it in production (`StPicoBTofPidTraits::btof()`, with $\beta > 0$). On average 35.3% of the hits per event have a time. Each cell therefore has one of three states: no hit, hit with time t, hit without time.

β and m^2. For each projected QA track the hit is chosen as in production: the projected cell, otherwise the nearer, then the farther neighbour, preferring hits that have a time (Sec. 7). The path length is $L =$ `tofPathLength`(primary vertex, projected point, helix curvature), the same function as

`StBTofMatchMaker`, and

$$\beta = \frac{L}{c\,t}, \qquad m^2 = p^2\left(\frac{1}{\beta^2} - 1\right), \tag{3}$$

with p the primary momentum. For tracks whose own production hit is chosen, L agrees with production ($L - \beta_{\mathrm{prod}} c\, t_{\mathrm{prod}}$: mean 0.001 cm, RMS 0.26 cm) and m^2 differs from the production value by $0.0007\,\mathrm{GeV}^2/c^4$ RMS. The same-event (SE) m^2 is therefore the production m^2 of the projected tracks.

Mixed event with overlay. A random hit only produces a wrong m^2 when it is chosen *instead of* the track's own hit. The mixed event (ME) is therefore built as an overlay: the hits of a partner event (same run, V_z and refMult class, up to 5 partners) are added on top of the hits of the track's own event, and the hit choice above is redone. When a partner hit is chosen (projected cell free in this event, or same cell with an earlier partner time), its time is combined with the track's L and p to give an m^2. Without the overlay, every partner hit present in the matching cells would count. That version (kept for reference, `hM2ME`) exceeds the SE yield in the π–K and K–p valleys by a factor 3.8–5.5 in 0–5% ($0.4 < p < 1.5\,\mathrm{GeV}/c$), which is impossible. The overlay keeps 34% of those entries.

Normalization. No fit is used. In each centrality class c and fine momentum bin j ($0.05\,\mathrm{GeV}/c$)

$$\frac{dN_{\mathrm{rand}}}{dm^2}(c,j) = \frac{dN_{\mathrm{ME}}}{dm^2}(c,j)\,\frac{N_{\mathrm{proj}}(c,j)}{N_{\mathrm{trials}}(c,j)}\,\frac{n_{\mathrm{t}} + n_{\mathrm{u}}}{n_{\mathrm{t}}}, \tag{4}$$

where N_{proj} counts projected tracks, N_{trials} the (track, partner) pairs, and n_{t} (n_{u}) the trials whose chosen partner hit has (has no) time. The last factor, 2.32 on average (2.41 in 70–80%, 2.32 in 0–5%), restores the random hits that have no stored time: only 43.2% of the partner hits chosen in the mixed events have a time. Coarse momentum bins are sums of the normalized fine bins.

Choice

All BTOF hits are included in the random-match estimate (user's decision). Random hits without a stored time are assumed to have the same time distribution as random hits with a time.

Caveat

Hits without a time are those no picoDst track was matched to: secondaries, particles not reconstructed in the TPC, noise. Their times are probably later on average than those of matched hits, so the shape of the random m^2 distribution may be biased towards small m^2; its normalization would be lower by a factor 2.32 if only timed hits were counted.

7.5.2 Results

Caveat

The overlay used here always keeps the track's own hit in the projected cell. Production instead picks the hit with the largest ToT among the projected and neighbouring cells, so the random matches below are underestimated. Sec. 7.6 studies this and the late background the picoDst cannot describe.

Figures 15a–18a overlay the normalized ME distribution on the SE m^2 distribution in nine momentum bins for 0–5%, 30–40%, 70–80% and 0–80%; the other centralities are in Appendix A. Table 8 and Fig. 14a give the random fraction under the π, K and p peaks.

- **Size.** Integrated over m^2 and $0.2 < p < 3\,\mathrm{GeV}/c$, random matches are 2.5% of the SE matches in 0–80%, 4.2% in 0–5%, 1.3% in 30–40% and 0.2% in 70–80%. This is smaller than f_{rand} of Sec. 7.4, because the track's own hit is usually chosen first.

- **Under the peaks (0–5%).** At $0.2 < p < 0.4\,\mathrm{GeV}/c$ the random fraction is 6.9% (π), 3.9% (K) and 3.0% (p); at $0.8 < p < 1.0\,\mathrm{GeV}/c$ it is 2.1%, 3.0% and 1.0%. In 30–40% the values are about three times smaller (0.6%, 0.9%, 0.3% at $0.8 < p < 1.0\,\mathrm{GeV}/c$), and in 70–80% they are below 0.4%. The kaon fraction stays roughly flat with p, because random m^2 values pile up between the pion and proton peaks.

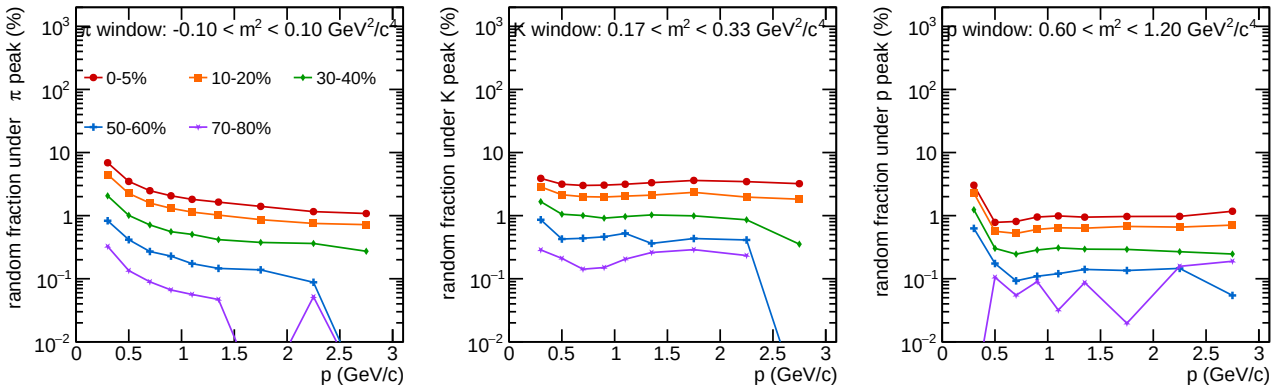
- **Shape (read by eye).** The random distribution peaks just above $m^2 = 0$ and falls steadily towards large m^2. At low momentum it follows the SE distribution in the π–K valley; at higher momentum it lies well below the SE distribution everywhere except in the valleys.

- **Consistency.** In 0–5%, $0.4 < p < 1.5\,\mathrm{GeV}/c$, the largest ME/SE ratio in any m^2 bin ($0.04\,\mathrm{GeV}^2/c^4$, at least 50 SE counts) is 1.07. In the lowest momentum bin (0.2–0.4 GeV/c) it is 1.21 (Table 8), so the estimate is somewhat too large there. The excess probably comes from the all-hits scaling, or from the less precise projection at low momentum (Sec. 7).

- **Tail above the proton.** For $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$ (between p and d) the SE yield is 7.41 times the random estimate in 0–80% (2–9 times in 0–5% above 0.6 GeV/c, Table 8). Random matches account for only a small part of this tail.

Caveat

Most of the SE yield between the proton and deuteron peaks is not from random hits. Candidates: decays in flight (the track's p does not belong to the particle that fired the cell), wrong primary vertex or out-of-time pile-up, and wrongly matched neighbour hits of the same event. Not studied.

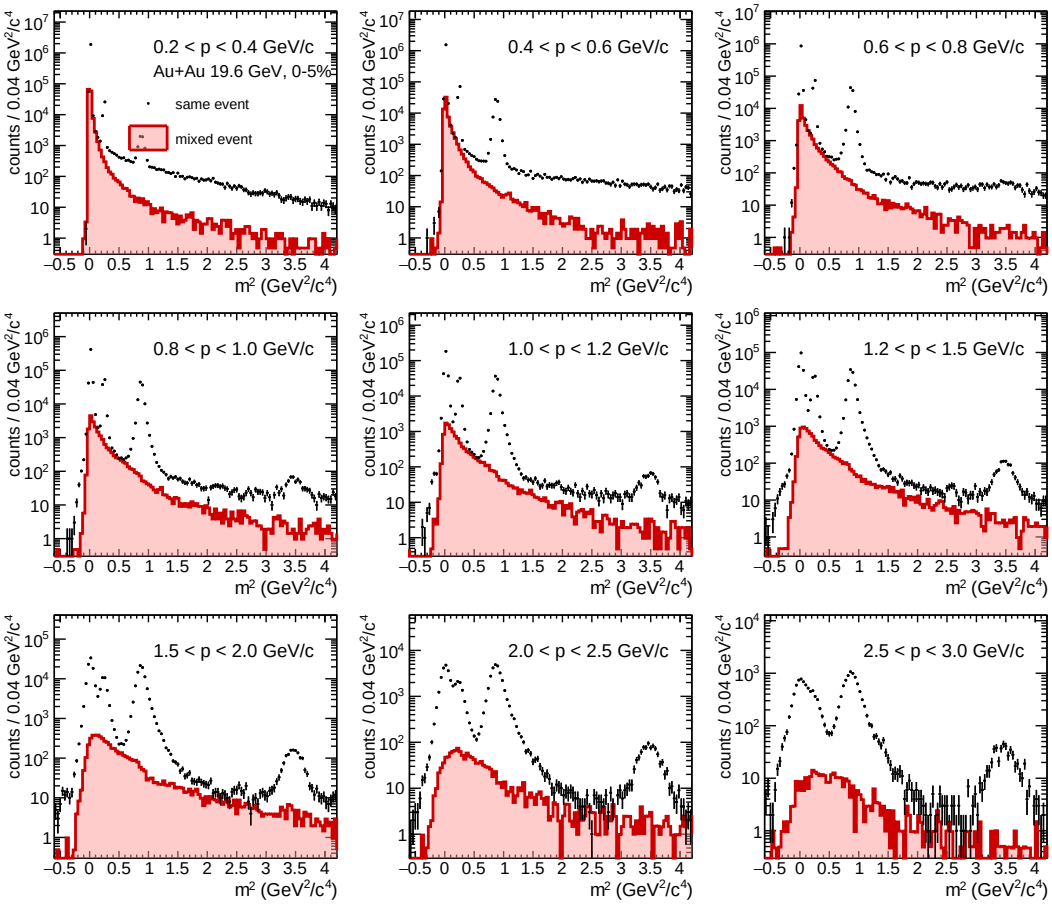


(a) Random-match fraction under the π (left), K (middle) and p (right) peaks vs momentum for 0–5% (red), 10–20% (orange), 30–40% (green), 50–60% (blue) and 70–80% (violet). Points with fewer than 100 SE counts in the window are omitted. ME: overlay, all hits.

Figure 14: Random fraction under the PID peaks.

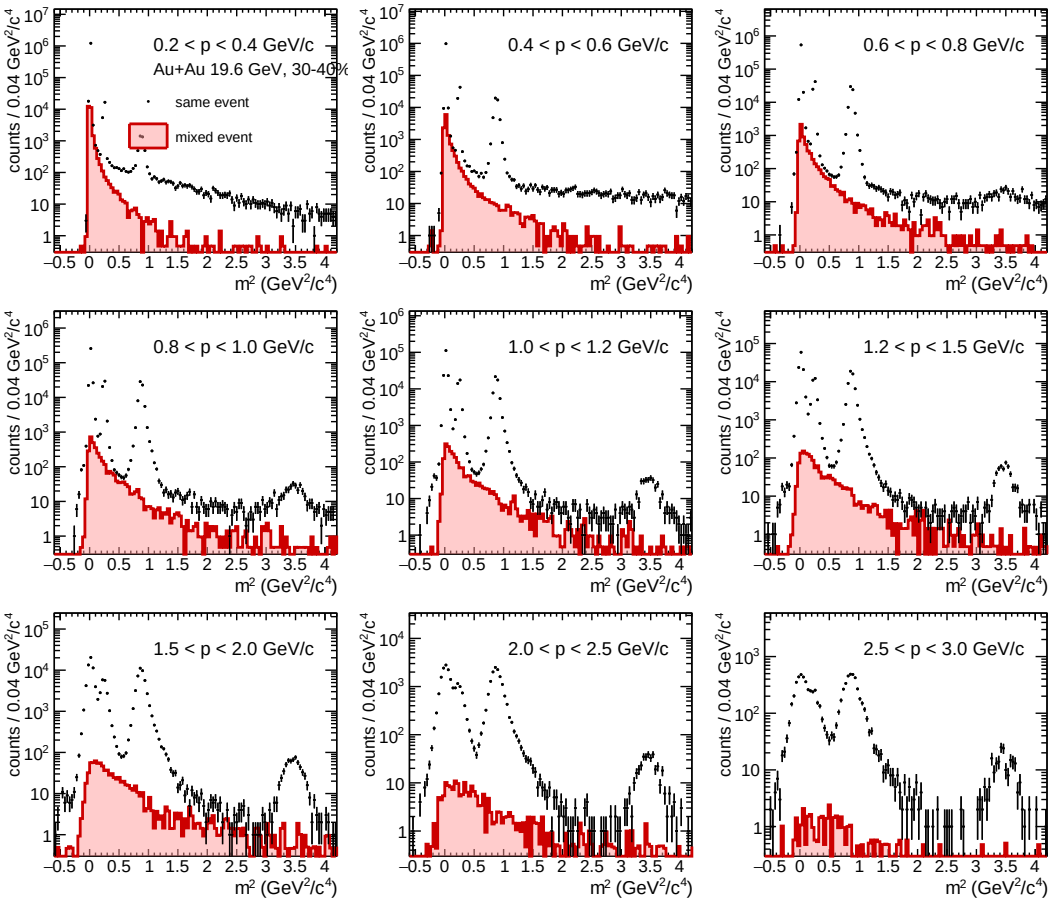
Table 8: Random-match fraction (%) of the SE yield in the π ($-0.1 < m^2 < 0.1$), K ($0.17 < m^2 < 0.33$) and p ($0.6 < m^2 < 1.2\,\mathrm{GeV}^2/c^4$) windows, SE/ME ratio between the p and d peaks, and the largest ME/SE ratio over m^2 bins with ≥ 50 SE counts in $-0.3 < m^2 < 3$. ME: overlay, all hits. Statistical errors not shown; they are large in 70–80% above 1 GeV/c. Generated: `tables/m2_contam_v3.tex`.

		random fraction under peak (%)			SE/ME	max
centrality	p (GeV/c)	π	K	p	$1.6 < m^2 < 2.8$	ME/SE
0–5%	0.2–0.4	6.88	3.89	3.03	22.4	1.21
	0.4–0.6	3.49	3.15	0.78	25.6	1.07
	0.6–0.8	2.48	3.01	0.81	8.6	0.98
	0.8–1.0	2.06	3.04	0.95	5.0	1.04
	1.0–1.2	1.81	3.14	0.99	3.7	0.91
	1.2–1.5	1.64	3.33	0.95	2.6	0.82
	1.5–2.0	1.40	3.61	0.97	2.1	0.59
	2.0–2.5	1.16	3.46	0.97	2.9	0.31
	2.5–3.0	1.08	3.21	1.17	4.9	0.16
30–40%	0.2–0.4	2.06	1.66	1.24	65.8	0.81
	0.4–0.6	1.01	1.05	0.30	28.8	0.82
	0.6–0.8	0.71	1.00	0.25	7.9	0.70
	0.8–1.0	0.56	0.91	0.29	6.6	0.70
	1.0–1.2	0.50	0.97	0.31	3.9	0.53
	1.2–1.5	0.42	1.03	0.29	3.5	0.56
	1.5–2.0	0.38	0.99	0.29	3.0	0.27
	2.0–2.5	0.36	0.86	0.27	5.4	0.13
	2.5–3.0	0.27	0.35	0.25	23.5	0.03
70–80%	0.2–0.4	0.33	0.29	0.00	111.3	0.17
	0.4–0.6	0.13	0.21	0.11	71.4	0.09
	0.6–0.8	0.09	0.14	0.05	24.7	0.06
	0.8–1.0	0.07	0.15	0.09	0.0	0.01
	1.0–1.2	0.06	0.20	0.03	6.7	0.04
	1.2–1.5	0.05	0.26	0.09	0.0	0.01
	1.5–2.0	0.00	0.29	0.02	0.0	0.01
	2.0–2.5	0.05	0.23	0.16	6.3	0.01
	2.5–3.0	0.00	0.00	0.19	0.0	0.00



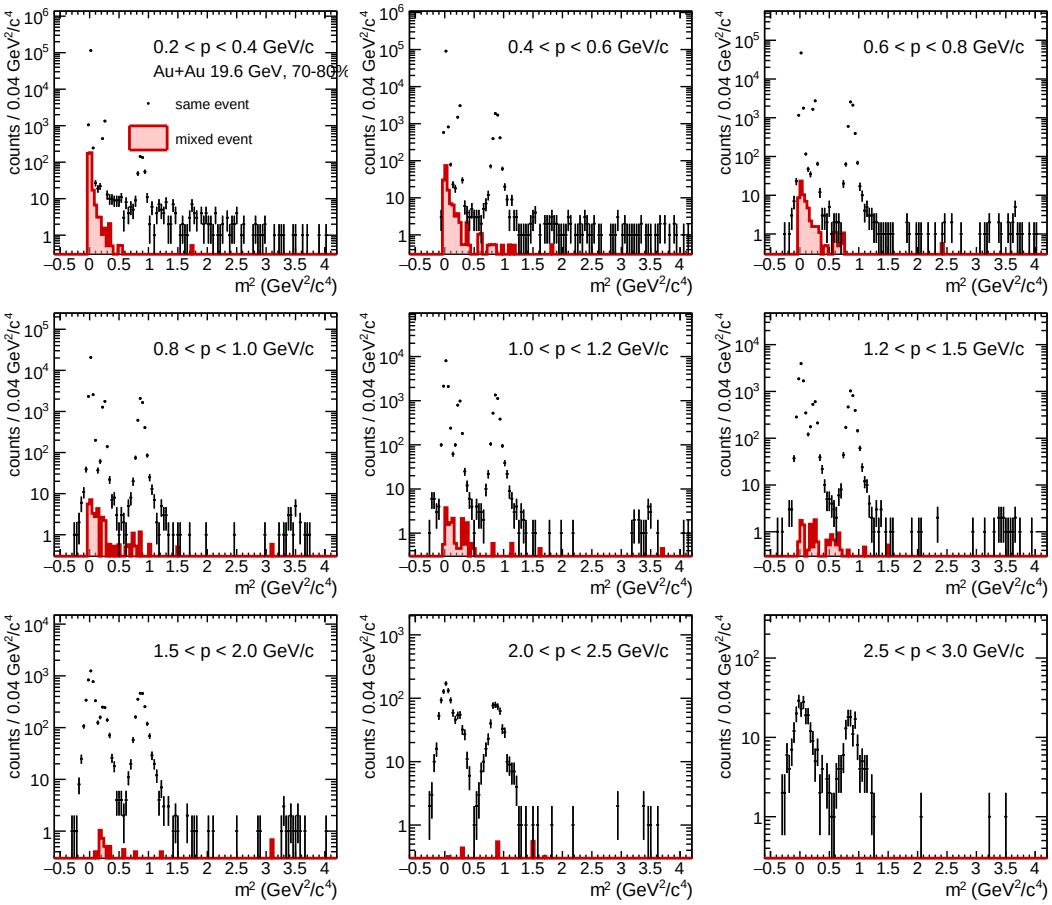
(a) m^2 of projected QA tracks, 0–5%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 15: SE vs ME m^2, 0–5%.



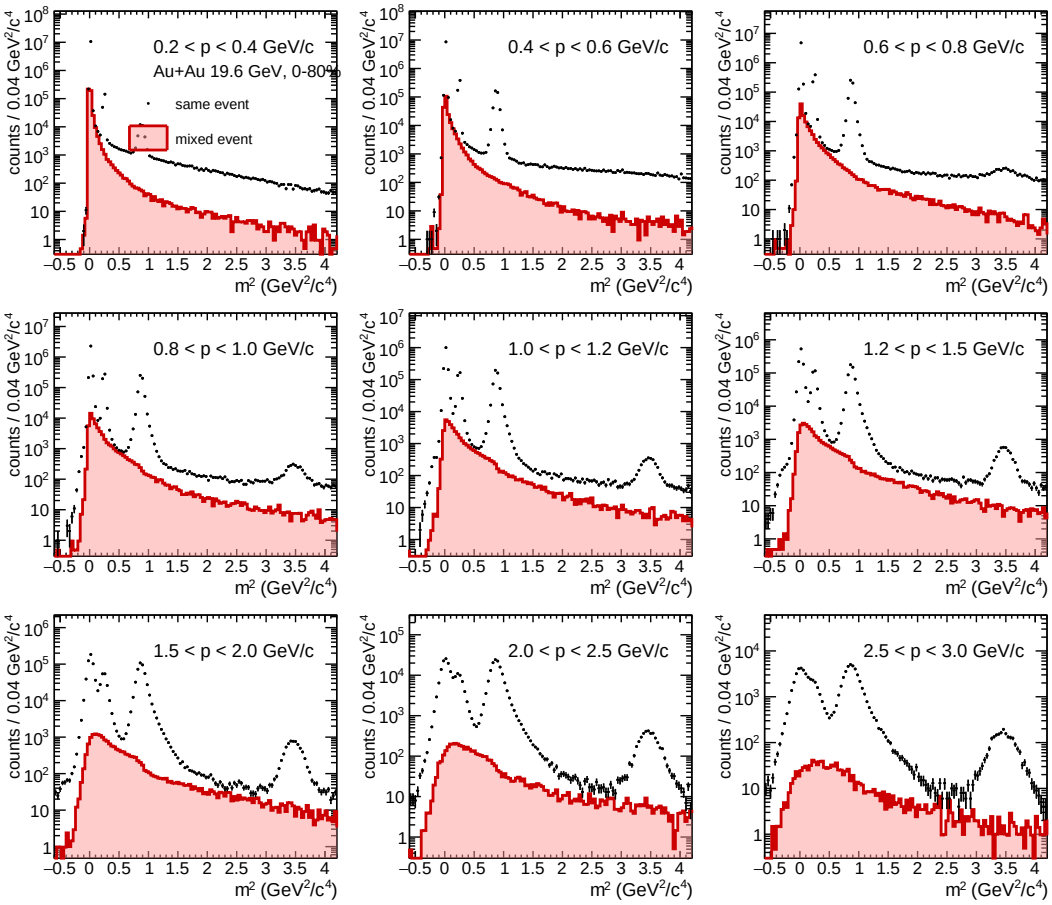
(a) m^2 of projected QA tracks, 30–40%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 16: SE vs ME m^2, 30–40%.



(a) m^2 of projected QA tracks, 70–80%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 17: SE vs ME m^2, 70–80%.



(a) m^2 of projected QA tracks, 0–80% (sum of the nine classes), nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 18: SE vs ME m^2, 0–80%.

7.6 Origin of the same-event background in m^2

Sec. 7.5 found that the overlay mixed event does not describe the same-event (SE) m^2 away from the peaks. In particular, the SE yield between the proton and deuteron peaks ($1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$, called HI below) is several times larger than the mixed-event (ME) estimate. This section reports the study of that mismatch. It uses four extra passes of `StRoot/StTofMEMaker` over the same 15 runs and events as Sec. 7.4:

- v4: SE and ME split by 12 track, event and hit variables;
- v5: the same quantities per projected BTOF cell;
- v6: details of neighbour-cell matches, on the first 25 000 events of each run;
- v7, v8: two emulations of the production hit choice.

Scripts: `scripts/make_bkg.C`, `make_cell.C`, `make_nb.C`, `make_m2.C`, `make_late.C`; logs `logs/bkg_v4.txt`, `cell_v5.txt`, `nb_v6.txt`, `m2_numbers_v7.txt`, `m2_numbers_v8.txt`, `late_v8.txt`. All comparisons are for 0–80% centrality unless stated otherwise.

7.6.1 Where the HI excess is

Table 9 lists the HI yield per 10^4 projected tracks for SE and for the overlay ME of Sec. 7.5, split by each variable.

- **No dependence:** the SE/ME ratio does not depend on the difference between the TPC and VPD vertex (≈ 6 in every class, including events without a VPD vertex), on $|z_{\mathrm{local}}|$, on η or on the track charge. This rules out start-time errors and event pile-up as the main source (Fig. 19b, 19c).
- **Mild dependence:** the ratio rises with the DCA ($5.9 \rightarrow 9.0$) and is larger for short tracks. Tracks with $n\sigma_{\mathrm{p}} > 8$ (light-nucleus candidates) show a strong excess, but they are a small share.
- **Strong dependence, the chosen cell:** the HI excess comes from tracks matched to a hit in a *neighbouring* cell (Fig. 19a). Neighbour-cell matches are 6% of the SE matches but 82% of the SE HI entries (v6). SE/ME in the HI window is 10.5 for them and 2.0 for projected-cell matches.

The neighbour-cell excess is not caused by dead or inefficient cells. Cells with an efficiency above 0.8 (79% of the projected tracks) carry 70%+13% of the HI entries, and cells below 0.5 carry about 1% (Table 10, Fig. 19d). It is not a projection effect either. The rate of HI neighbour matches is about 7×10^{-4} per SE match whether the track crosses the centre or the edge of the projected cell (Fig. 20a). At the centre the neighbour cell is $\geq 1.7\,\mathrm{cm}$ away, against a projection precision of about $0.05\,\mathrm{cm}$ at these momenta. Finally, in 219063 of the 350881 neighbour matches (v6, $0.6 < p < 1.5\,\mathrm{GeV}/c$) the projected cell *did* contain a hit, but that hit has no time in the picoDst: production did not match it to any track and gave our track the neighbour hit instead. The HI fraction of these matches is 15.3 per mille of the peak yield, against 6.1 per mille when the projected cell was empty.

Table 9: HI yield ($1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$) per 10^4 projected tracks for SE and overlay ME (projected cell first, all hits), by category. "share" is the category's share of all SE matches. *Hit variables: absolute counts instead of rates, because projected tracks cannot be assigned to them. Generated: `tables/bkg_cat_v4.tex`.

		$0.6 < p < 1.5\,\mathrm{GeV}/c$			$1.5 < p < 3\,\mathrm{GeV}/c$			
variable	category	SE	ME	SE/ME	SE	ME	SE/ME	share
gDCA	<0.5 cm	6.94	1.18	5.9	15.91	5.15	3.1	75.9%
	0.5-1 cm	7.29	1.13	6.4	23.89	5.46	4.4	15.7%
	1-2 cm	8.65	1.23	7.1	59.15	5.32	11.1	6.1%
	2-3 cm	10.92	1.22	9.0	175.75	7.43	23.7	2.3%
$\lvert V_z - V_z^{\mathrm{VPD}}\rvert$	<3 cm	6.70	1.11	6.0	19.36	4.89	4.0	51.7%
	3-6 cm	6.93	1.15	6.0	18.80	5.32	3.5	22.1%
	>6 cm	7.83	1.30	6.0	21.13	5.41	3.9	23.0%
	no VPD	9.61	1.54	6.2	20.02	7.72	2.6	3.2%
$\lvert z_{\mathrm{local}}\rvert$	<1 cm	7.11	1.04	6.8	19.65	4.75	4.1	36.1%
	1-2 cm	7.15	1.08	6.6	19.55	4.86	4.0	35.4%
	2-2.6 cm	7.01	1.14	6.2	19.43	4.82	4.0	19.5%
	>2.6 cm	7.12	1.95	3.6	20.49	8.22	2.5	8.9%
$\lvert y_{\mathrm{local}}\rvert$	<0.6 cm	6.51	0.78	8.4	17.51	3.32	5.3	36.0%
	0.6-1.2 cm	6.86	0.82	8.3	18.73	3.67	5.1	35.5%
	>1.2 cm	8.06	2.10	3.8	23.22	9.43	2.5	28.5%
nHitsFit	15-24	14.03	5.14	2.7	24.75	9.11	2.7	0.2%
	25-34	8.57	1.84	4.7	32.52	5.58	5.8	2.0%
	≥35	7.01	1.13	6.2	19.05	5.11	3.7	97.8%
$\lvert\eta\rvert$	<0.3	6.49	1.15	5.6	24.03	5.41	4.4	33.1%
	0.3-0.6	7.24	1.19	6.1	20.15	5.24	3.8	35.1%
	>0.6	7.43	1.16	6.4	17.89	5.04	3.6	31.8%
centrality	0-10%	9.83	1.70	5.8	24.52	7.55	3.2	34.2%
	10-40%	6.23	1.03	6.1	17.65	4.34	4.1	52.2%
	40-80%	3.13	0.35	8.8	12.49	1.11	11.3	13.6%
hit cell	projected	2378.00	1168.69	2.0	1437.00	537.28	2.7	84.9%*
	neighbour	10912.00	1034.81	10.5	2417.00	481.07	5.0	15.1%*
hit owner	own	13180.00	0.00	0.0	3830.00	0.00	0.0	99.7%*
	own, shared	0.00	0.00	0.0	0.00	0.00	0.0	0.0%*
	other track	110.00	2203.49	0.0	24.00	1018.34	0.0	0.3%*
	other, shared	0.00	0.00	0.0	0.00	0.00	0.0	0.0%*
hits in cell	1	13290.00	2203.49	6.0	3854.00	1018.34	3.8	100.0%*
	≥2	0.00	0.00	0.0	0.00	0.00	0.0	0.0%*
$n\sigma_{\mathrm{p}}$	<-3	6.61	1.08	6.1	34.77	6.57	5.3	85.4%
	-3..3	8.21	1.63	5.0	18.93	5.03	3.8	14.2%
	3..8	12.45	1.51	8.2	26.05	9.81	2.7	0.3%
	>8	117.38	5.91	19.8	109.97	15.90	6.9	0.0%
charge	-	6.96	1.22	5.7	17.19	5.53	3.1	45.9%
	+	7.21	1.16	6.2	20.82	5.13	4.1	54.1%

Table 10: HI yield by cell efficiency ε_c (fraction of tracks projected to the cell for which the cell has a hit), $0.6 < p < 3\,\mathrm{GeV}/c$. "nb. match": share of the projected tracks matched to a neighbour cell. Generated: `tables/cell_eff_v5.tex`.

ε_c	cells	tracks (%)	nb. match (%)	nb. cell HI (10^{-4})		proj. cell HI (10^{-4})		share of HI (%)
				SE	ME	SE	ME	
0.0–0.2	935	4.1	0.5	1.08	0.35	0.24	0.00	0.6
0.2–0.5	77	0.3	4.2	10.97	2.35	5.09	2.16	0.6
0.5–0.8	3636	16.2	4.6	5.23	0.93	2.84	1.04	15.8
0.8–0.9	15284	66.8	5.6	6.87	0.74	1.79	0.86	69.8
0.9–1.0	3108	12.6	5.3	7.39	0.59	1.31	0.70	13.2

7.6.2 Production chooses among several hits by ToT

In `StBTofMatchMaker` (SL24y, lines 782–833) a track with hits in more than one candidate cell (projected cell and neighbours) is matched to the hit with the *largest ToT* (match flag 2). Only for equal ToT is the hit closest in y used (flag 3). So a random hit in a neighbour cell can take the place of the track's own hit. The overlay ME of Sec. 7.5 always kept the own hit in the projected cell and therefore misses these cases. The picoDst has no ToT, so two emulations were run:

- **Equal chance** (v7): ToT independent of the hit's origin. Each of the hits in the candidate cells, from this event and from the partner, wins with probability $1/(n_A + n_B)$. Untimed partner hits are included through the scaling of Sec. 7.5.
- **Earliest time wins** (v8; ToT replaced by the time, smaller time wins): the hit chosen for the track in this event competes with the timed partner hits in the candidate cells. Untimed partner hits cannot take part.

	proj. cell first	equal chance	earliest time
random matches / SE matches, 0–80% (%)	2.5	4.3	1.9
0–5% (%)	4.2	7.3	3.3
70–80% (%)	0.2	0.3	0.1
max ME/SE per m^2 bin, 0–5%, $0.4 < p < 1.5$	1.07	3.08	0.83
SE/ME, HI window, 0–80%, $0.2 < p < 3$	7.41	1.71	11.63

Under the peaks in 0–5% at $0.8 < p < 1\,\mathrm{GeV}/c$ the random fraction is 4.0% (π), 8.3% (K) and 3.3% (p) with equal chance, and 1.9%, 2.5% and 0.7% with the earliest-time rule (Figs. 21a, 22a).

- Equal chance roughly doubles the random matches. It describes the HI region above $p \approx 0.8\,\mathrm{GeV}/c$, but it exceeds the SE yield in the valleys next to the peaks by up to a factor 3.08, which is impossible. The track's own hit must therefore win more often than an equal-chance draw gives.
- Earliest time wins never exceeds the SE yield (largest ratio 0.83). By construction, though, a random hit wins only when it is earlier than the own hit, so it fills the low-m^2 side and leaves the HI excess (SE/ME = 11.63).
- Production selects on ToT, not on time, so the two emulations bracket the in-time random matches that come from hits with a time: 3.3–7.3% of the TOF matches in 0–5%.

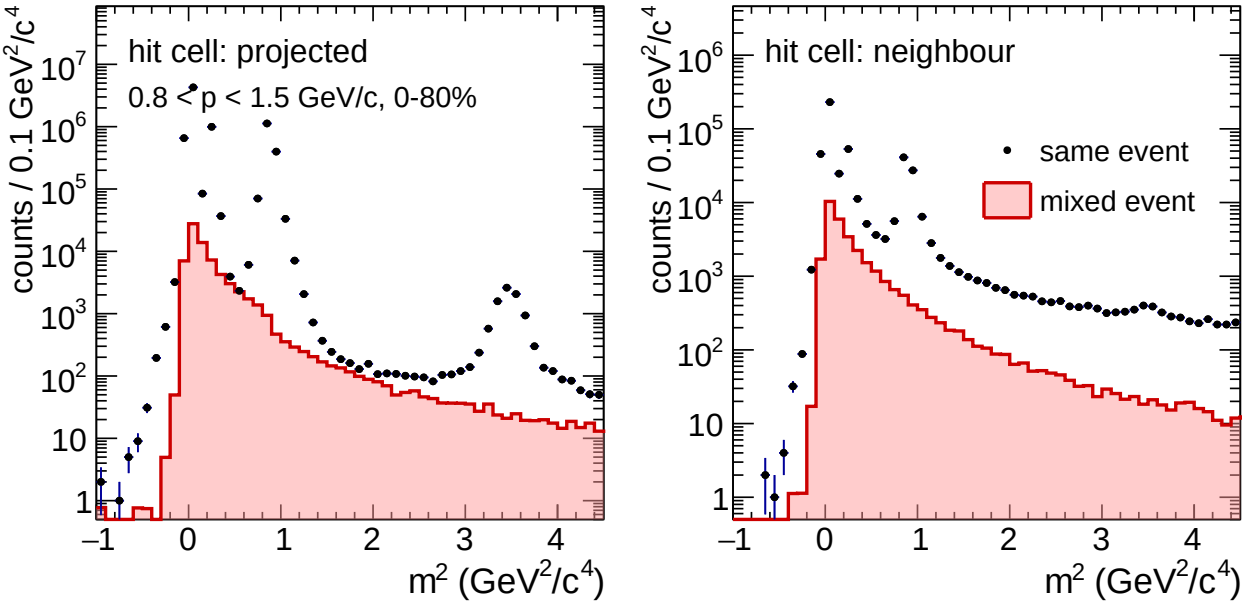
7.6.3 A late component flat in time

In time units the HI excess is a nearly flat plateau. Between 16 and 26 ns after the pion arrival time the SE rate falls only slowly, while the ME falls by about a factor 4. Only 0.45% of the hits that have a time in the picoDst arrive later than 25 ns. The plateau hits must therefore be among the hits without a time, i.e. hits that production did not match to any track in their own event. Such a hit enters the SE when production matches it to the track under study, because the track's BTofPidTraits then carries its time. In a partner event it has no time and cannot give an m^2, so no picoDst-based mixed event can contain it.

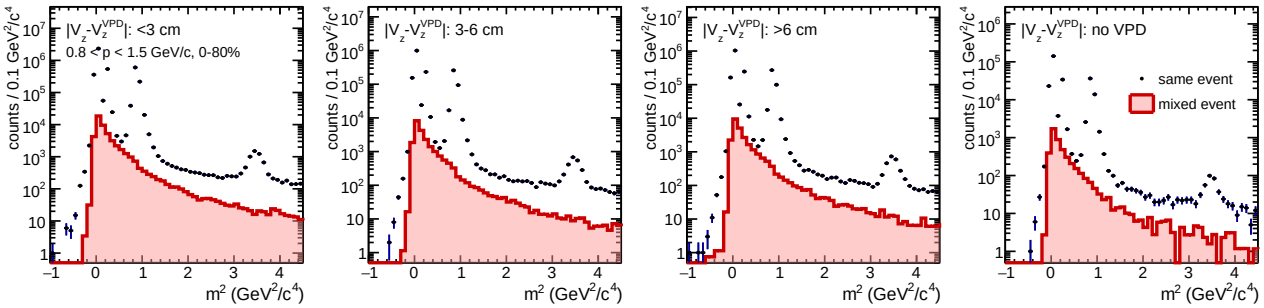
The plateau rate per projected track (Fig. 20b) rises only from 25 to 89 hits per ns per 10^6 tracks (factor 3.5) from 70–80% to 0–5% for $0.6 < p < 1.5\,\mathrm{GeV}/c$, and from 166 to 349 (factor 2.1) for $0.2 < p < 0.6\,\mathrm{GeV}/c$. The BTOF hit multiplicity, which follows refMult (Fig. 2a), grows by more than an order of magnitude over the same range (read from the refMult range of the classes, not computed per class). Most of the plateau is therefore unrelated to the triggered collision: electronic noise or hits from other bunch crossings in the TOF readout window.

Caveat

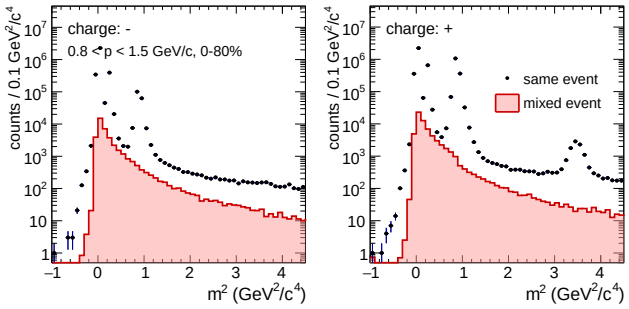
The picoDst cannot provide a complete mixed-event description of random matches: (1) it has no ToT, so the production choice among several hits can only be bracketed; (2) 65% of the BTOF hits have no time, among them the flat late component. Both are available in the MuDst (`StMuBTofHit`), where every hit has a time and a ToT; a MuDst test is being set up (Sec. 10).



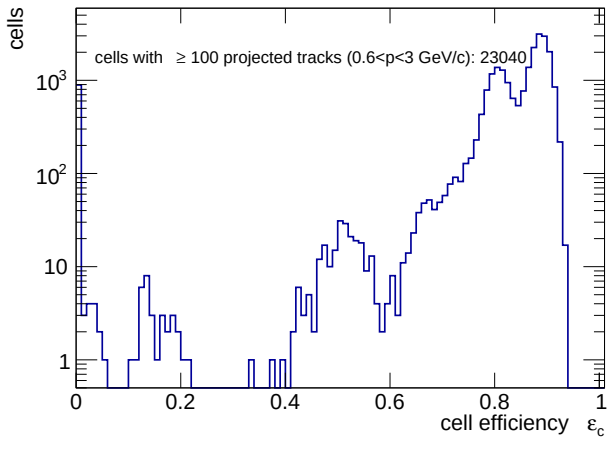
(a) SE (points) and overlay ME (red) m^2, $0.8 < p < 1.5\,\mathrm{GeV}/c$, 0–80%, for hits in the projected cell (left) and in a neighbour cell (right). Per 10^4 projected tracks is not defined for hit categories; ME is normalized per projected track in total.



(b) Same, split by $|V_z - V_z^{\mathrm{VPD}}|$: < 3, 3–6, $> 6\,\mathrm{cm}$ and no VPD vertex. ME normalized per projected track of each class.

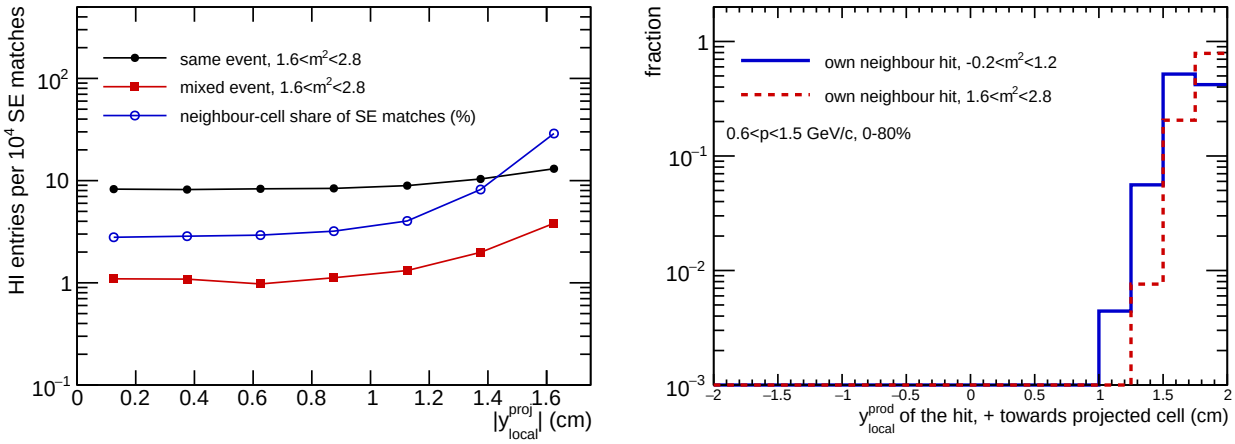


(c) Same, split by charge.

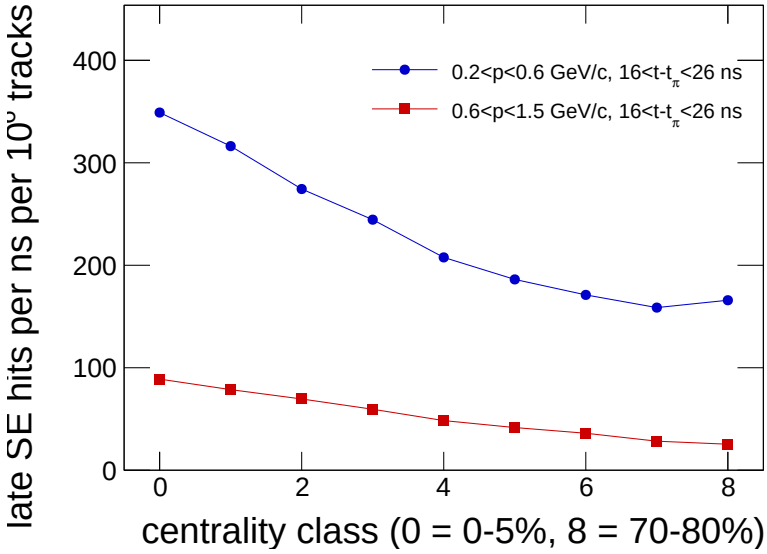


(d) Distribution of the cell efficiency ε_c ($0.6 < p < 3\,\mathrm{GeV}/c$), log scale.

Figure 19: Background diagnostics (v4, v5).

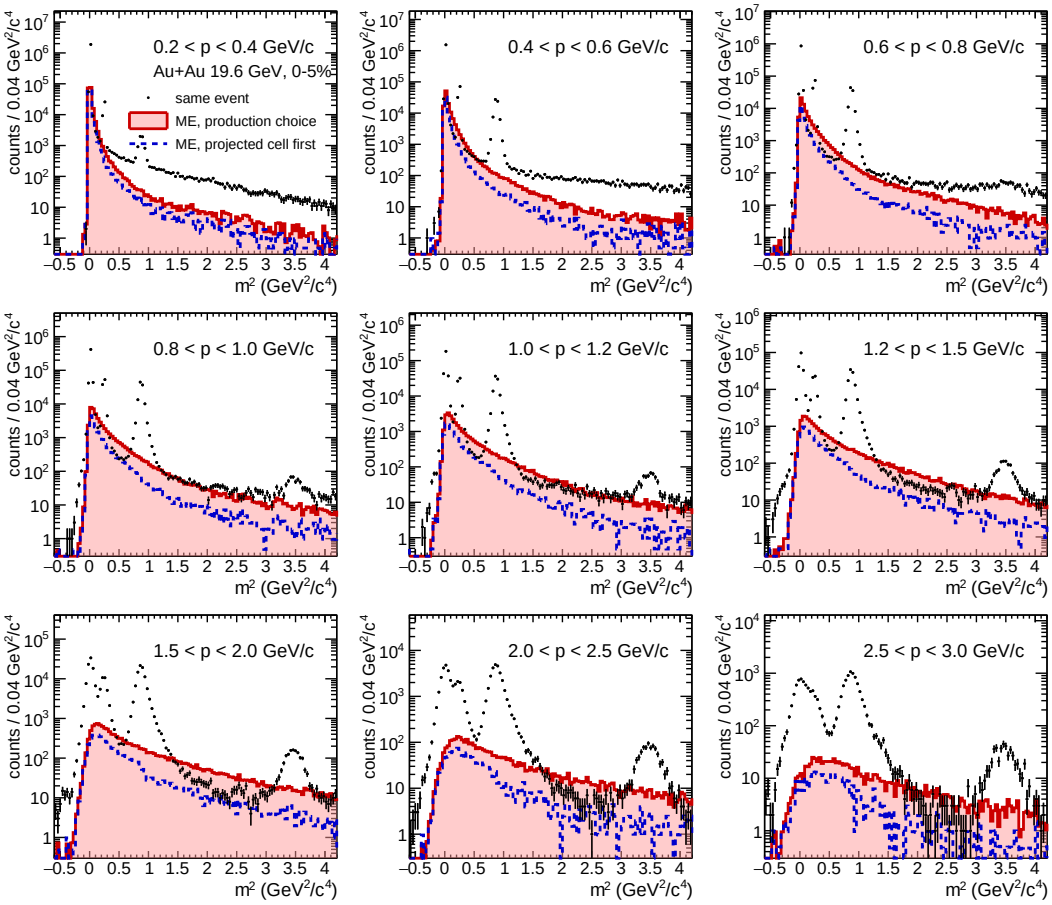


(a) Left: HI entries per 10^4 SE matches vs $|y_{\mathrm{local}}|$ of the projection for SE (black) and overlay ME (red), and the share of neighbour-cell matches (open blue, %). Right: production y_{local} of own neighbour hits (read by eye; the quantity is not used in the conclusions).



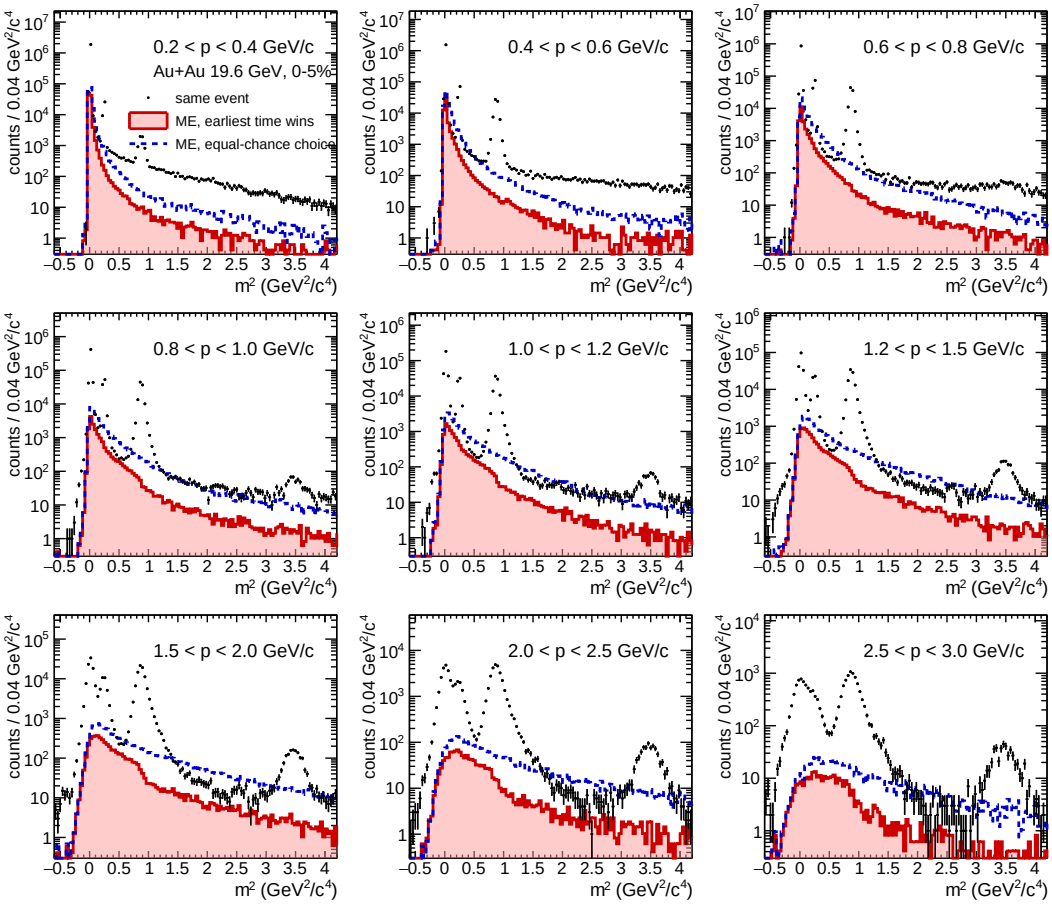
(b) SE hits arriving 16–26 ns after the pion time, per ns and per 10^6 projected tracks, vs centrality class, for $0.2 < p < 0.6$ (blue) and $0.6 < p < 1.5\,\mathrm{GeV}/c$ (red).

Figure 20: Neighbour-cell matches (v6) and the late plateau (v8).



(a) m^2 of projected QA tracks, 0–5%, nine momentum bins (log scale). Black: same event. Red filled: overlay ME with the equal-chance production choice, all hits, normalized per projected track. Blue dashed: overlay ME with the projected cell first (Sec. 7.5).

Figure 21: SE vs ME m^2, 0–5%, equal chance.



(a) m^2 of projected QA tracks, 0–5%, nine momentum bins (log scale). Black: same event. Red filled: overlay ME with the earliest time winning, timed partner hits only, normalized per projected track. Blue dashed: equal-chance choice.

Figure 22: SE vs ME m^2, 0–5%, earliest time wins.

7.7 Mixed event on MuDst

The picoDst stores a time only for BTOF hits that production matched to a track (35.3% of the hits, Sec. 7.6). The unmatched hits therefore cannot compete in a picoDst mixed event, and the late same-event background could not be reproduced (best picoDst result: SE/ME = 9.25 between the p and d peaks). The MuDst keeps every `StMuBTofHit` with its leading and trailing edge, so this section repeats the mixed event on MuDst. Two methods are used: an emulation of the production matching in our own maker (Sec. 7.7.3), and, as the main result, a re-run of the production makers themselves on the overlaid events (Sec. 7.7.4).

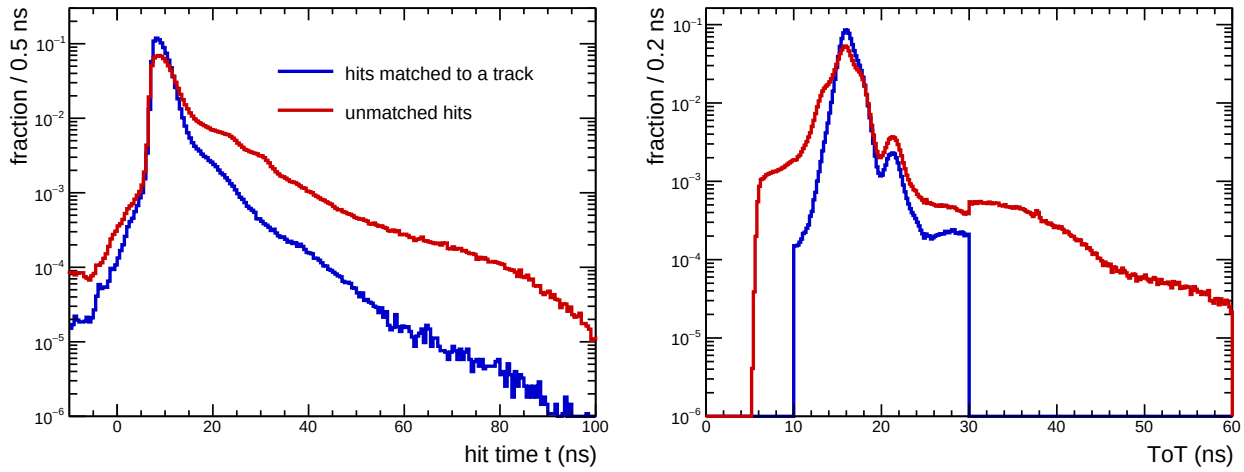
Sample. The P24iy MuDst are on HPSS only. Six MuDst files of three runs were restored through the Data Carousel (`scripts/mudst_restore.sh`, list `filelists/mudst/restore_test.txt`); they are the raw files of the picoDst sample. The two files of run 20088015 are not used: reading them floods the log with trigger-data decoding errors and the jobs stall. The four files used are `st_physics_20060065_raw_1000011`, `_5500010`, `st_physics_20073019_raw_4500013` and `_7000013`: 68459 events, of which 21160 pass the event selection of Sec. 7.5 and have a VPD start time. The m^2 numbers in this section use these events only; they are a test sample of two runs. Statistical uncertainties only.

7.7.1 Times of matched and unmatched hits

`macros/probeMuBTof.C` reads all hits of the four files (23653 selected events) in two passes. Pass 1 measures a per-cell offset as the median of $t_{\mathrm{LE}} - t_{\mathrm{tof}} - T_0$ over hits matched by production, where t_{tof} is the production time of flight and T_0 the event median of $t_{\mathrm{LE}} - t_{\mathrm{tof}}$. Pass 2 gives every hit the time $t = t_{\mathrm{LE}} - \mathrm{off}_{\mathrm{cell}} - T_0$. On matched hits t reproduces the production time of flight with an RMS of 0.54 ns (97% within 1 ns); 0.2% of the unmatched hits are in cells without an offset and are dropped. Of the hits, 2761120 are matched and 3929867 (59%) are unmatched.

Unmatched hits are systematically later (Fig. 23a). The fraction at 8–10 ns is 42% for matched and 27% for unmatched hits; at 15–26 ns it is 5.5% and 15.9%, and above 26 ns 1.3% and 9.5%. Per ns, unmatched hits are 7 times more frequent than matched hits at 30–60 ns. The late fraction of unmatched hits falls only mildly with the BTOF hit multiplicity (12% for the lowest, 8% for the highest classes). Their ToT distribution has the same mean as for matched hits (16.6 vs 16.4 ns). This is the behaviour of slow particles that are not reconstructed in the TPC, not of electronic noise; the hits are in the same bunch crossing by construction of the production.

All matched hits have $10 < \mathrm{ToT} < 30$ ns (0.00% outside, Fig. 23a right), while 3% of the unmatched hits are below 10 ns and 3.1% above 30 ns. Hits outside this window cannot end up as a calibrated match, so they never contribute to random matches.



(a) BTOF hits of the four MuDst files, normalized to unit area. Left: calibrated hit time t (0.5 ns bins, log scale). Right: ToT (0.2 ns bins). Blue: hits matched to a track by production; red: unmatched hits. The legend of the left panel applies to both. Unmatched hits have a long late tail: 9.5% above 26 ns against 1.3% for matched hits.

Figure 23: Time and ToT of BTOF hits in MuDst (`scripts/make_probe.C`).

7.7.2 Closure test: production matching and calibration re-run on MuDst

The SL24y makers `StBTofMatchMaker`, `StVpdCalibMaker` and `StBTofCalibMaker` can run on MuDst (`setMuDstIn()`). The chain (`macros/runTofClosure.C`) is `StMuDstMaker`, `St_db_Maker` (current database), the three makers with the y2019 geometry, and `StRoot/StTofClosureMaker`. That maker stores the production matching of every global track before the re-run and compares it afterwards. On 300 events of `st_physics_20060065_raw_1000011`:

- all 16451 production matches are found again in the same cell; there are 0 matches in another cell, 0 only in production and 0 only in the re-run;

- the re-run time of flight differs from production by a median of -14 ps (68% between -28 and 1 ps, 98% within ±133 ps); 88% of the tracks agree in $1/\beta$ within 0.005;
- the event start time differs by a median of 4 ps, and 51% of the events agree within 5 ps.

The residual differences are small compared with the BTOF resolution; they are attributed to calibration tables updated after the production (not verified: the database time stamp was not pinned to the production).

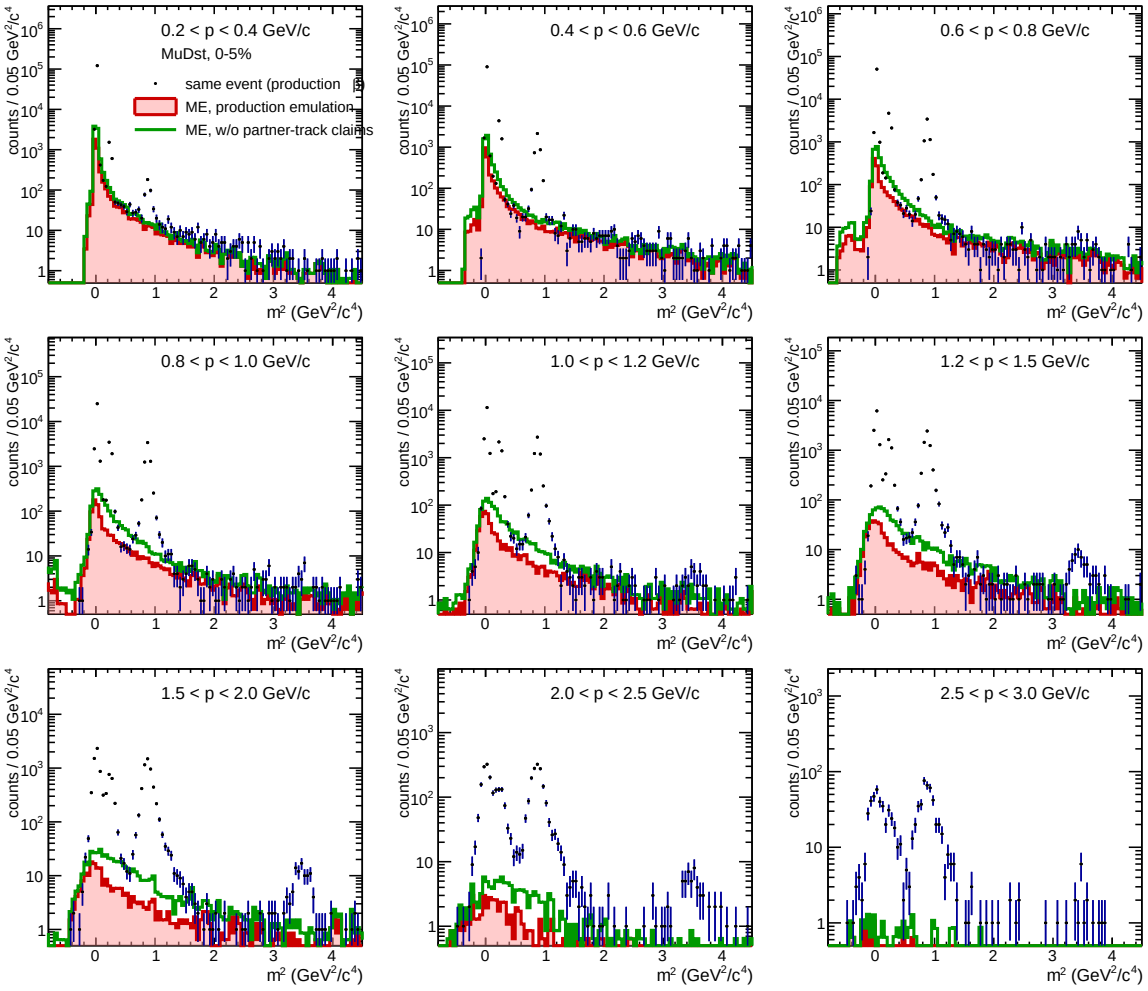
Choice

The re-run of the production BTOF makers on MuDst reproduces the production matching exactly and the time of flight to within tens of ps. It is therefore used for the mixed event in Sec. 7.7.4, so that same event and mixed event share the same matching and calibration.

7.7.3 Method 1: emulation of the production matching

`StRoot/StTofMEMuMaker` (derived from `StTofMEMaker`, same projection and mixing classes) emulates the SL24y matching rule on the overlaid hit map. Every production-valid global track (flag, $n_{\mathrm{HitsFit}} \geq 15$, ratio ≥ 0.52, outer helix) is projected; candidates are the projected cell and its two neighbours in the module. A cell claimed by more than one track is discarded, and among several candidates the largest ToT (below 40 ns) wins. Hit times are our own calibration (t_{LE} minus per-cell offset minus event T_0, RMS 0.43 ns against production). On the same event, the emulation picks the production hit for 98.3% of the production matches.

In the mixed event, the hits of a partner event are overlaid on the hit map of the event, and hits of the two events in the same cell are merged (earlier leading edge, larger ToT). The partner's own production-valid tracks also claim the cells of the partner hits. A partner hit made by a reconstructed particle is thus discarded as shared, exactly as in its own event. Without these claims the random fraction is 6.0% in 0–5% (green line in Fig. 24a). With them it is 2.7% (0–5%) and 1.6% (0–80%). The same-event histogram uses the production β.



(a) m^2 of QA tracks, 0–5%, nine momentum bins, four MuDst files (log scale). Black points: same event (production β). Red filled: mixed event with the production emulation including the partner-track claims, normalized per QA track. Green line: the same without partner-track claims. Between the p and d peaks SE/ME = 1.29 (0.2–3 GeV/c).

Figure 24: MuDst mixed event, emulation of the production matching, 0–5% (`scripts/make_mu.C`).

Caveat

The emulation uses our own hit times (RMS 0.43 ns, no slewing or z_{local} correction). Random matches to pion-like partner hits are smeared about three times wider in m^2 than in the same event. In the thin bins just below and beside the pion and kaon peaks the mixed event exceeds the same event bin by bin, up to ME/SE = 3.70 in 0–5% ($0.4 < p < 1.5$ GeV/c). This is the motivation for method 2.

7.7.4 Method 2 (main result): production makers re-run on overlaid events

`StRoot/StTofMEOverlayMaker` (macro `macros/runTofMEOverlay.C`, one HTCondor job per file and setup) runs after the chain of Sec. 7.7.2, which first re-runs the three production makers on the event itself; this gives the same event. Mixing classes are 14 V_z bins × 9 refMult classes per run, with the last 5 events of a class as partners. For each partner:

1. the partner hits are appended to the `StMuBTofHit` array of the event, with leading and trailing

edges shifted by $t_{\mathrm{start}}^{\mathrm{event}} - t_{\mathrm{start}}^{\mathrm{partner}}$ (VPD start times from `StBTofHeader`), so that each partner hit keeps its time of flight. A partner hit in a cell that already has an overlapping hit ($|\Delta t_{\mathrm{LE}}|$ below the ToT of the earlier hit) is merged into it: earlier leading edge, later trailing edge;

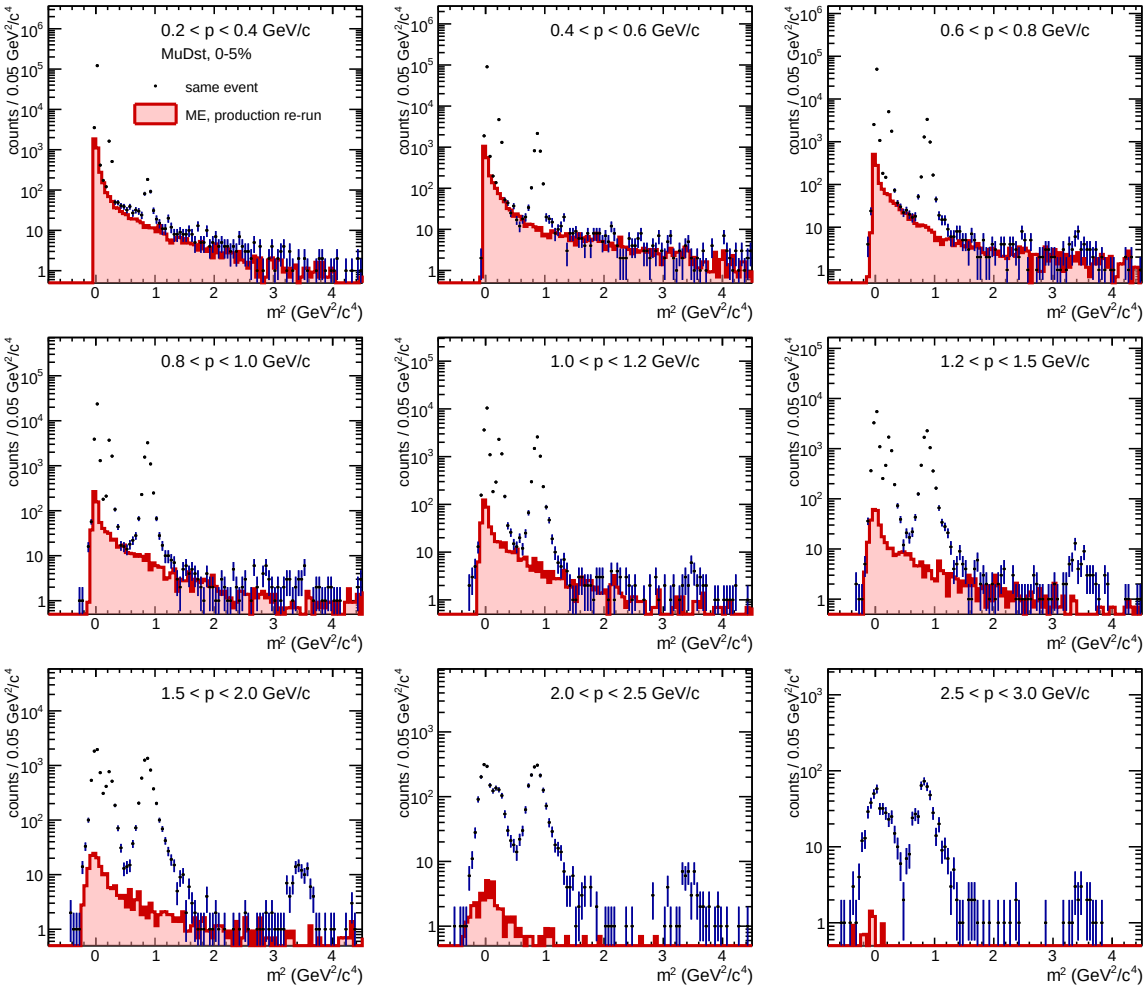
2. the partner's global tracks with $n_{\mathrm{HitsFit}} \geq 15$ and $0 < \mathrm{flag} < 1000$ (on average 578 per partner) are appended to the global-track array. They have no primary-track link and are not counted; their role is to claim the cells of the partner hits, so that a partner hit made by a reconstructed particle is discarded as shared, as in its own event;

3. `StBTofMatchMaker`, `StVpdCalibMaker` and `StBTofCalibMaker` are run again on the overlaid event. The match maker resets the BTOF information of every track and hit at the start of each pass;

4. a QA track of the event matched to an appended hit, or to a merged hit whose leading edge came from the partner, is a random match. Its m^2 is computed from the re-run time of flight and path length, exactly as for the same event;

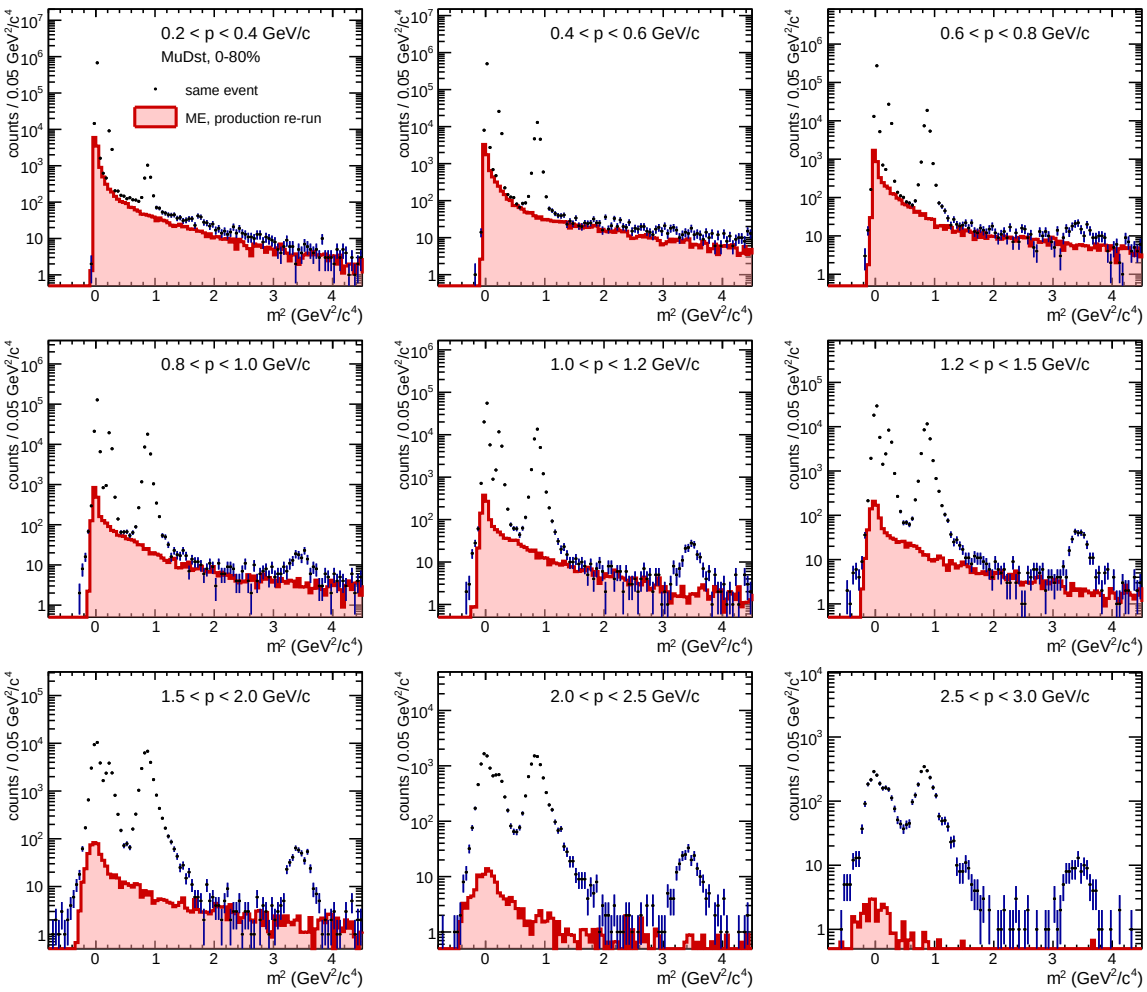
5. the appended hits and tracks are removed and the hit array of the event is restored.

The mixed event is normalized per QA track in each (centrality, p) bin. The same event and the mixed event therefore share the production matching rule (shared cells, largest ToT), the production calibration and the same path-length formula.



(a) m^2 of QA tracks, 0–5%, nine momentum bins, four MuDst files (log scale). Black points: same event. Red filled: mixed event with the production makers re-run on the overlay of partner hits and partner tracks, normalized per QA track. Both use the re-run calibration. Random matches are 2.62% of the TOF matches; SE/ME between the p and d peaks is 1.39 (0.2–3 GeV/c) and 1.06–1.39 per bin at 0.4–2 GeV/c.

Figure 25: MuDst mixed event (main result), production makers re-run on overlaid hits and tracks, 0–5% (`scripts/make_ov.C`).



(a) As Fig. 25a for 0–80% centrality. Random matches 1.57%; SE/ME between the p and d peaks 1.55 (0.2–3 GeV/c) and 1.19–1.50 per bin at 0.4–2 GeV/c.

Figure 26: MuDst mixed event (main result), 0–80%.

Random matches are 2.62% of the TOF matches in 0–5% and 1.57% in 0–80% ($0.2 < p < 3$ GeV/c). This agrees with the emulation of Sec. 7.7.3 (2.7% and 1.6%). With the common calibration the spike of the emulation beside the pion peak is gone: the largest bin-by-bin ME/SE at $0.4 < p < 1.5$ GeV/c is 1.18 in 0–5%, and the mixed event follows the shape of the same-event background at all m^2, including the valleys and the region between the p and d peaks (read by eye, Fig. 25a). The fractions under the PID peaks are listed per momentum bin in Table 11.

Effect of the partner tracks on the event's own matches. The added tracks also compete for the hits of the event, which the overlay makes denser than a real event. Of the QA tracks matched in the same event, 95.2% keep their own hit in the overlays, 1.2% are matched to a partner hit instead and 3.7% lose their match, mostly because a partner track shares the cell. Without partner tracks the overlay matches 56.9% of the QA tracks (same event 56.4%); with them 54.6%.

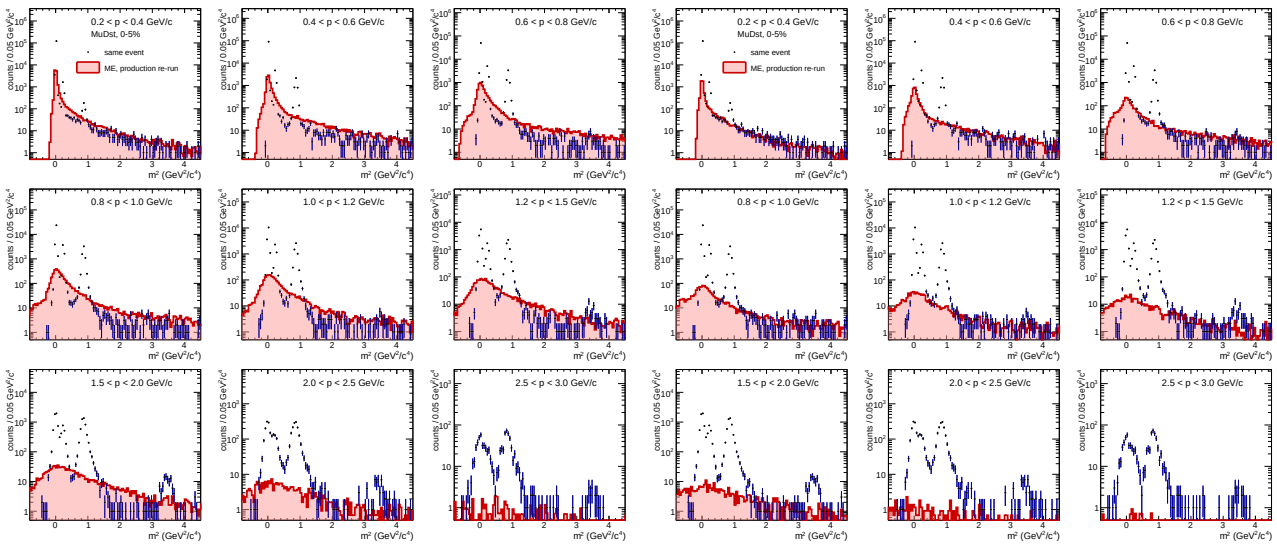
Caveat

Between the p and d peaks the mixed event is still 1.06–1.39 times below the same event per momentum bin at 0.4–2 GeV/c (0–5%), with 25–129 same-event entries per bin. Below 0.4 GeV/c and above 2 GeV/c the ratio is larger (Table 11), and these bins are sparse. The overlay removes 3.7% of the event's own matches, which slightly reduces the competition and is a possible bias of the same order. Light-nucleus candidates and other real tracks in the window are not subtracted. The larger sample is needed to tell these apart.

7.7.5 Cross-checks: overlay without partner tracks, and the pool method

The same maker was run in three other setups on the same events:

- **Overlay, partner hits only:** random matches 5.98% (0–5%) and 3.56% (0–80%), SE/ME between the p and d peaks 0.92. Without their tracks, partner hits of reconstructed particles are never discarded as shared, so the random matches are overestimated by a factor 2.3 (0–5%).
- **Pool** (the conventional mixed event: the event's own hits removed, its tracks matched to the partner hits only): 10.44% and 5.90%, SE/ME 0.43. Every track is treated as if its own hit were missing, so the result is the random-hit probability for tracks without their own hit, not the random-match fraction. The overshoot reaches ME/SE = 20.37 in single valley bins (0–5%).
- **Pool with partner tracks:** 3.93% and 2.14%, SE/ME 0.79. The partner tracks remove 62% of the pool matches (0–5%): the hits left free are mostly those of unreconstructed particles.



(a) Pool: event hits removed, partner hits only.

(b) Pool with partner tracks.

Figure 27: Pool setups, 0–5%; encodings as in Fig. 25a (`scripts/make_ov.C`).

Choice

The pool method cannot give the random-match fraction without the fraction of tracks that have no own hit, which is not measurable (the observed matches contain random ones), and it misses random hits that beat the track's own hit by ToT. The overlay keeps the own hit and lets the production algorithm decide both cases; it is used for the result.

7.7.6 Comparison and current result

Table 11: MuDst random matches per momentum bin. Main result: production makers re-run on the overlay of partner hits and partner tracks (Sec. 7.7.4). N_{HI} = SE entries for $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$ (HI); f = ME/SE of all TOF matches (%); HI = SE/ME in HI; π/K/p = ME/SE under the peaks (%). Cross-checks: emulation with partner-track claims (Sec. 7.7.3), overlay of partner hits only, pool (partner hits only, own hits removed) and pool with partner tracks (Sec. 7.7.5).

			overlay + partner tracks					emulation		f of other setups		
cent.	p (GeV/c)	N_{HI}	f	HI	π	K	p	f	HI	hits only	pool	pool+trk
0-5%	0.2–0.4	118	3.2	1.76	2.6	10.5	21.7	3.2	2.01	7.3	11.7	4.6
0-5%	0.4–0.6	129	2.6	1.24	1.9	4.5	1.9	2.7	1.21	5.9	10.5	4.1
0-5%	0.6–0.8	76	2.3	1.23	1.7	2.8	1.2	2.5	0.98	5.3	9.9	3.7
0-5%	0.8–1	54	2.3	1.26	1.8	2.0	1.0	2.4	1.07	5.2	9.7	3.5
0-5%	1–1.2	43	2.2	1.20	1.9	1.6	0.6	2.3	1.03	5.0	9.3	3.2
0-5%	1.2–1.5	46	2.1	1.39	1.8	1.3	0.5	2.2	1.19	4.9	8.8	2.9
0-5%	1.5–2	25	1.9	1.06	1.6	1.5	0.4	2.0	1.10	4.6	8.1	2.4
0-5%	2–2.5	19	1.8	3.18	1.4	1.5	0.3	1.9	2.61	4.3	7.4	1.9
0-5%	2.5–3	13	1.8	5.57	2.0	1.3	0.3	1.9	5.56	4.3	6.9	2.0
0-80%	0.2–0.4	474	1.9	1.94	1.5	6.9	13.6	1.9	2.16	4.3	6.7	2.6
0-80%	0.4–0.6	533	1.5	1.50	1.1	2.7	1.2	1.6	1.42	3.4	6.0	2.3
0-80%	0.6–0.8	292	1.4	1.34	1.0	1.7	0.7	1.5	1.10	3.2	5.5	1.9
0-80%	0.8–1	186	1.4	1.33	1.0	1.2	0.5	1.4	1.13	3.1	5.4	1.8
0-80%	1–1.2	133	1.3	1.19	1.1	1.0	0.4	1.4	1.05	3.0	5.2	1.7
0-80%	1.2–1.5	146	1.3	1.41	1.1	0.8	0.3	1.4	1.23	3.0	5.0	1.5
0-80%	1.5–2	106	1.2	1.45	1.1	0.8	0.2	1.3	1.41	2.9	4.6	1.3
0-80%	2–2.5	68	1.1	3.70	0.9	1.0	0.2	1.1	3.17	2.6	4.2	1.1
0-80%	2.5–3	34	1.1	5.86	1.0	1.1	0.2	1.2	5.52	2.7	3.8	0.9

Decision needed

Current result: the overlay of partner hits and tracks with the production makers re-run (method 2). Random matches are 2.62% of the TOF matches in 0–5% and 1.57% in 0–80% ($0.2 < p < 3\,\mathrm{GeV}/c$), with the emulation (2.7% and 1.6%) as the cross-check. To be confirmed on the 98 restored MuDst files of 14 runs.

The MuDst mixed event describes the same-event background between the p and d peaks far better than any picoDst mixed event: SE/ME there is 1.39 in 0–5%, against 9.25 on picoDst. The late background is therefore made of random matches to real, mostly unreconstructed slow particles, which have no time in the picoDst. Above $2\,\mathrm{GeV}/c$ and below $0.4\,\mathrm{GeV}/c$ the region has few same-event entries per bin (N_{HI} in Table 11); these bins need the larger sample (98 MuDst files restored).

7.8 Large MuDst sample: method 1 (emulation)

Sample and production. Restored from HPSS (`scripts/mudst_restore100.sh`, list `filelists/mudst/restore_100.txt`): 98 MuDst files, 7 per run for the 14 runs of the picoDst sample without 20088015. Together with the four files of Sec. 7.7 the sample has 102 files (`filelists/mudst/tags_all.txt`) and 1858212 events, of which 643650 pass the selection of Sec. 7.5 and are used for m^2. One HTCondor job per file:

1. calibration pass: `macros/probeMuBTof.C` pass 1 on the first 4000 events of each file (`scripts/probe1.sub`); the outputs of a run are merged into the per-cell offsets `output/mudst/off_<run>.root` (`scripts/off_big.sh`);

2. method 1, `StTofMEMuMaker` v4 (`scripts/mubig.sub`, `scripts/mu_file.sh`): production emulation with partner-track claims, as in Sec. 7.7.3. Mixing buffers are per run and file;

3. method 2, `StTofMEOverlayMaker` v3 with partner tracks (`scripts/ovbig.sub`); results in Sec. 7.9.

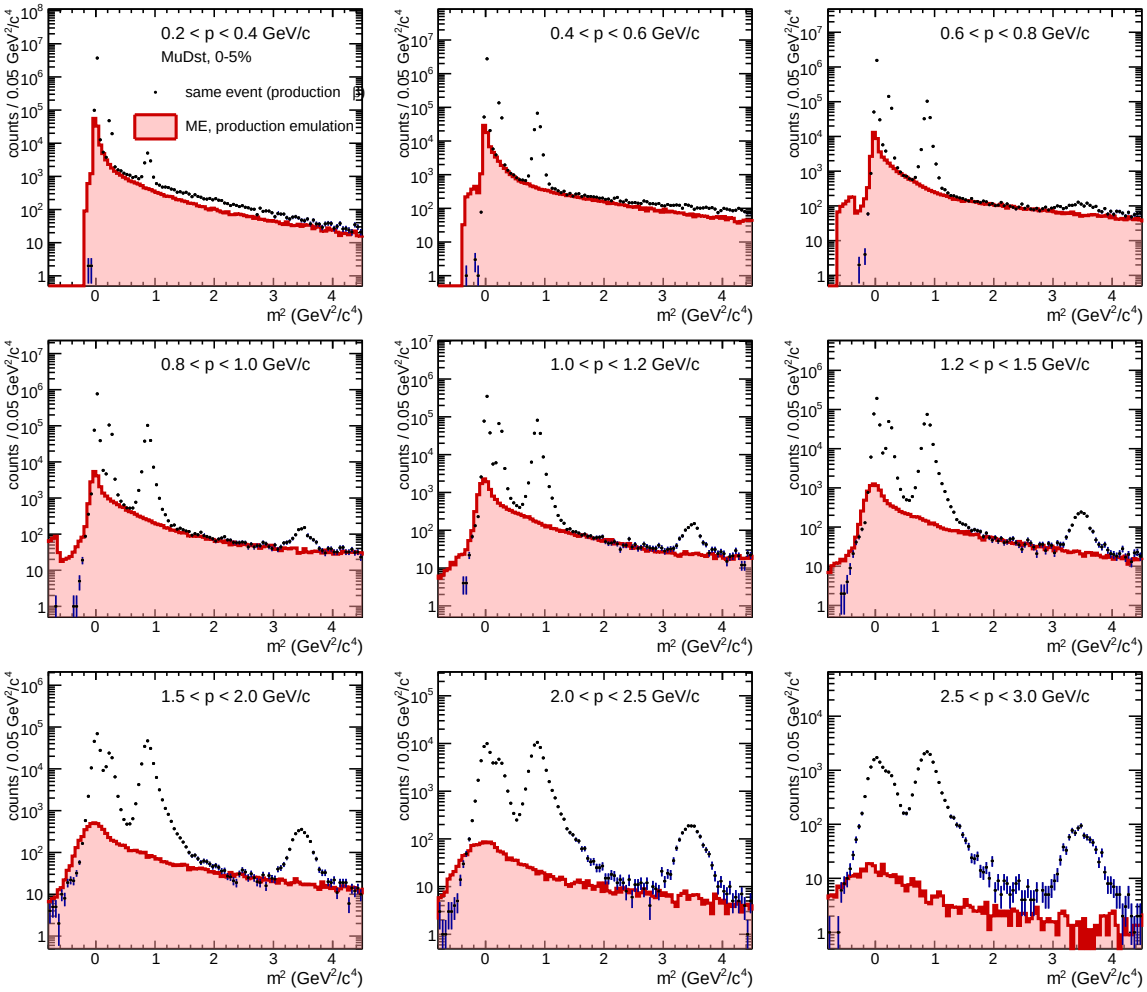
Both makers (v4 and v3) additionally store the charge-signed $q \times m^2$ and the m^2 of negative tracks for SE and ME; the positive-track histograms are the difference. The 102 method-1 outputs are merged into `data/mu_big.root`. Scripts: `scripts/make_mu.C` (tag `big`), `scripts/make_qm2.C`, `scripts/make_fit.C`, `scripts/make_big_tables.py`; logs `logs/mubig_numbers.txt`, `logs/qm2_mubig.txt`, `logs/fit_mubig*.txt`. Statistical uncertainties only.

Caveat

All numbers of this section are from method 1, which uses our own hit times (RMS 0.43 ns against production; Sec. 7.7.3). Method 2 and the comparison follow in Sec. 7.9.

7.8.1 Random matches

The emulation picks the production hit for 98.2% of the production matches, as in the four-file test. Random matches are 2.8% of the TOF matches in 0–5% and 1.7% in 0–80% ($0.2 < p < 3\,\mathrm{GeV}/c$), consistent with the four-file values (2.7% and 1.6%). Between the p and d peaks SE/ME is 1.32 (0–5%) and 1.50 (0–80%). Per momentum bin (Table 12) it is 1.05–1.32 at 0.6–2 GeV/c (0–5%). The excess that remains is at $0.2 < p < 0.4\,\mathrm{GeV}/c$ (1.88) and above 2 GeV/c (2.3–4.4). The largest bin-by-bin ME/SE values are in nearly empty bins just below the pion peak; they are the timing artefact of method 1 (red box in Sec. 7.7.3).



(a) m^2 of QA tracks, 0–5%, nine momentum bins, 102 MuDst files (log scale). Black points: same event (production β). Red filled: mixed event, method 1 with partner-track claims, normalized per QA track. Random matches 2.8%; SE/ME between p and d 1.32.

Figure 28: Large sample, method 1, SE vs ME, 0–5% (`scripts/make_mu.C`).

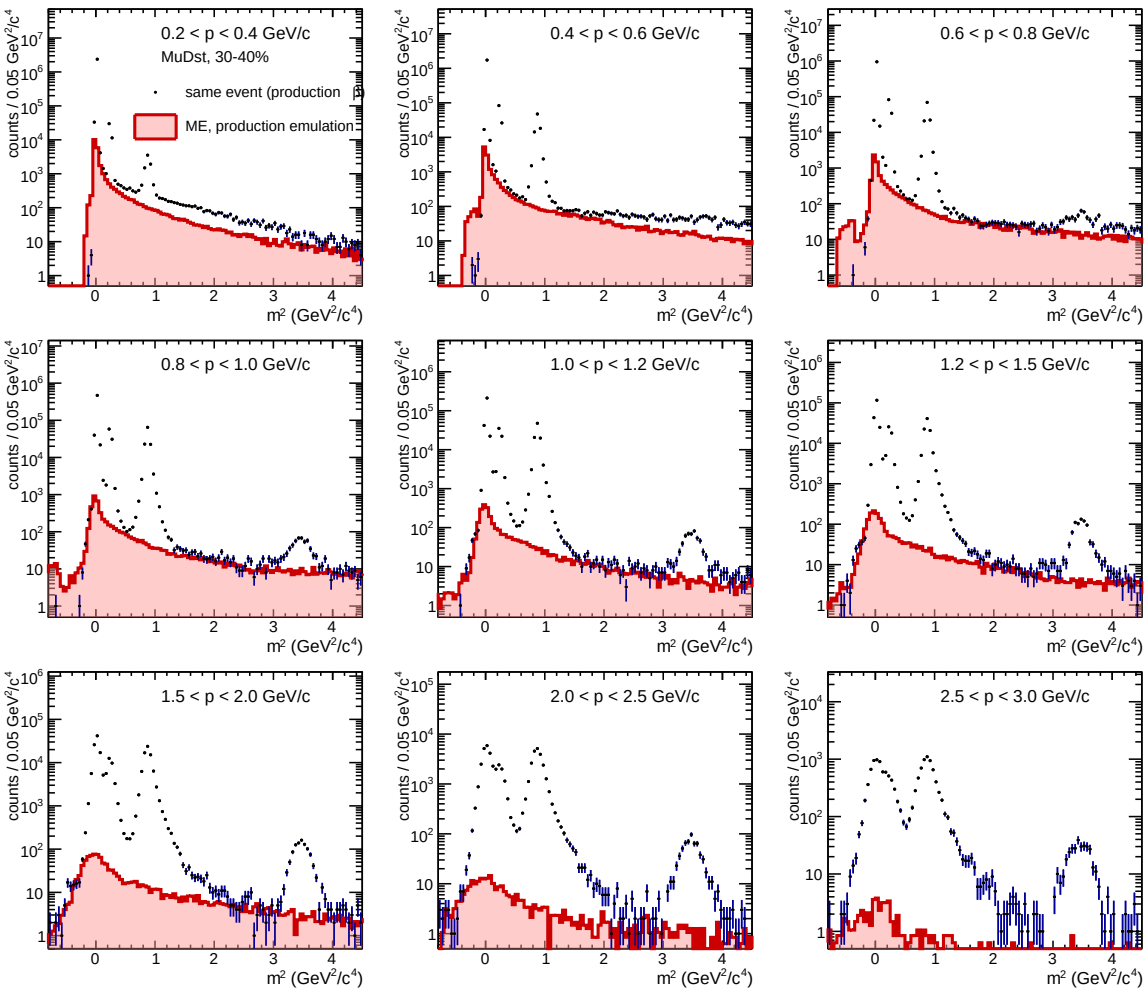
Table 12: Large sample, method 1 (emulation with partner-track claims): random matches per momentum bin. f = ME/SE of all TOF matches (%); HI = SE/ME for $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$; max = largest bin-by-bin ME/SE (bins with ≥ 20 SE entries); π/K/p = ME/SE under the peaks (%); f^-, f^+ = f for negative and positive tracks (from $q \times m^2$).

cent.	p (GeV/c)	f	HI	max	π	K	p	f^-	f^+
0-5%	0.2–0.4	3.26	1.88	1.0	2.59	11.85	23.45	3.22	3.31
0-5%	0.4–0.6	2.76	1.30	13.8	1.93	4.86	2.60	2.83	2.70
0-5%	0.6–0.8	2.55	1.07	5.7	1.72	2.99	1.33	2.70	2.42
0-5%	0.8–1	2.42	1.12	2.2	1.63	2.12	0.99	2.66	2.24
0-5%	1–1.2	2.29	1.05	3.8	1.53	1.72	0.74	2.69	2.09
0-5%	1.2–1.5	2.21	1.10	3.8	1.40	1.53	0.58	2.80	1.95
0-5%	1.5–2	2.12	1.32	4.1	1.25	1.49	0.45	2.95	1.81
0-5%	2–2.5	2.00	2.31	1.5	1.14	1.54	0.36	3.16	1.62
0-5%	2.5–3	1.83	4.41	0.4	1.03	1.42	0.35	3.14	1.44
0-80%	0.2–0.4	1.94	2.27	0.9	1.51	7.72	15.48	1.92	1.96
0-80%	0.4–0.6	1.64	1.48	7.7	1.13	3.00	1.53	1.68	1.60
0-80%	0.6–0.8	1.52	1.12	16.2	1.01	1.83	0.80	1.61	1.45
0-80%	0.8–1	1.45	1.14	3.9	0.96	1.30	0.59	1.59	1.35
0-80%	1–1.2	1.39	1.12	2.7	0.91	1.05	0.46	1.61	1.27
0-80%	1.2–1.5	1.35	1.21	2.6	0.83	0.94	0.35	1.66	1.20
0-80%	1.5–2	1.31	1.53	2.3	0.73	0.94	0.27	1.76	1.13
0-80%	2–2.5	1.23	3.09	2.9	0.66	0.93	0.22	1.89	1.01
0-80%	2.5–3	1.15	6.54	1.0	0.61	0.88	0.22	1.94	0.91

Other centralities. Figures 29a and 30a show the same comparison for mid-central (30–40%) and peripheral (60–80%) events, and Table 13 gives the numbers per momentum bin. Random matches fall with centrality: 2.8% (0–5%), 0.84% (30–40%) and 0.16% (60–80%) of the TOF matches at $0.2 < p < 3\,\mathrm{GeV}/c$. The mixed event describes the region between the p and d peaks best in central events; SE/ME there is 1.32 (0–5%), 1.8 (30–40%) and 4.9 (60–80%). In the lowest momentum bin it is 1.88, 3.0 and 9.6: the SE continuum above the ME at $0.2 < p < 0.4\,\mathrm{GeV}/c$ is seen in all three centralities and is relatively largest in peripheral events, where the random background is smallest (read from the figures: in 60–80% the SE continuum at 0.2–0.6 GeV/c lies several times above the ME, cf. the HI column of Table 13). In 60–80% above 1.2 GeV/c the ME between p and d is close to zero and the HI ratio is set by a few SE counts.

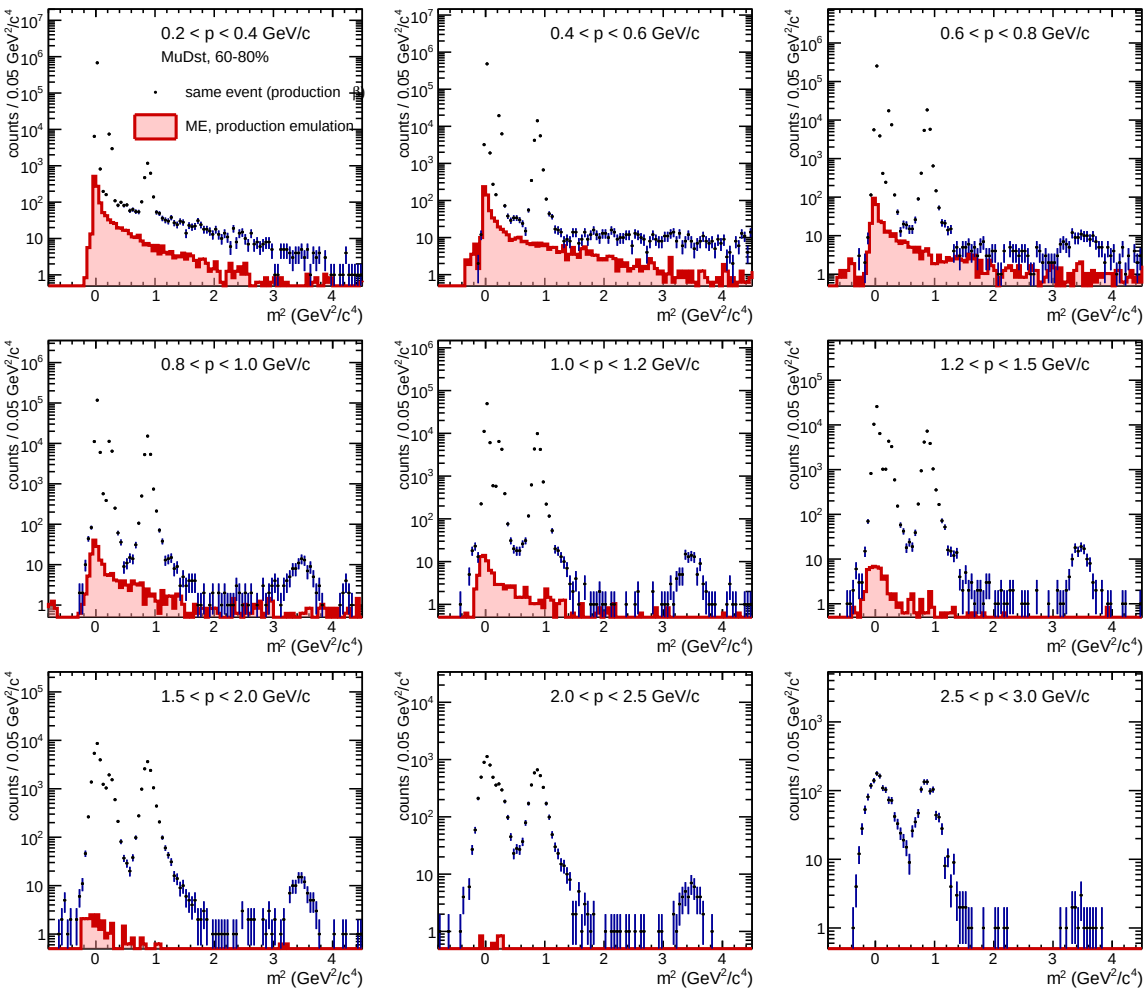
Caveat

A source of SE entries above the ME that scales with the number of tracks rather than with the hit density (e.g. late hits correlated with the track itself) would give exactly this centrality pattern. It is studied in Sec. 7.9.3.



(a) m^2 of QA tracks, 30–40%, nine momentum bins, 102 MuDst files (log scale). Black points: same event; red filled: mixed event, method 1 with partner-track claims. Random matches 0.84%; SE/ME between p and d 1.8.

Figure 29: Large sample, method 1, SE vs ME, 30–40% (`scripts/make_mu.C`).



(a) m^2 of QA tracks, 60–80%, nine momentum bins, 102 MuDst files (log scale). Encodings as in Fig. 29a. Random matches 0.16%; SE/ME between p and d 4.9.

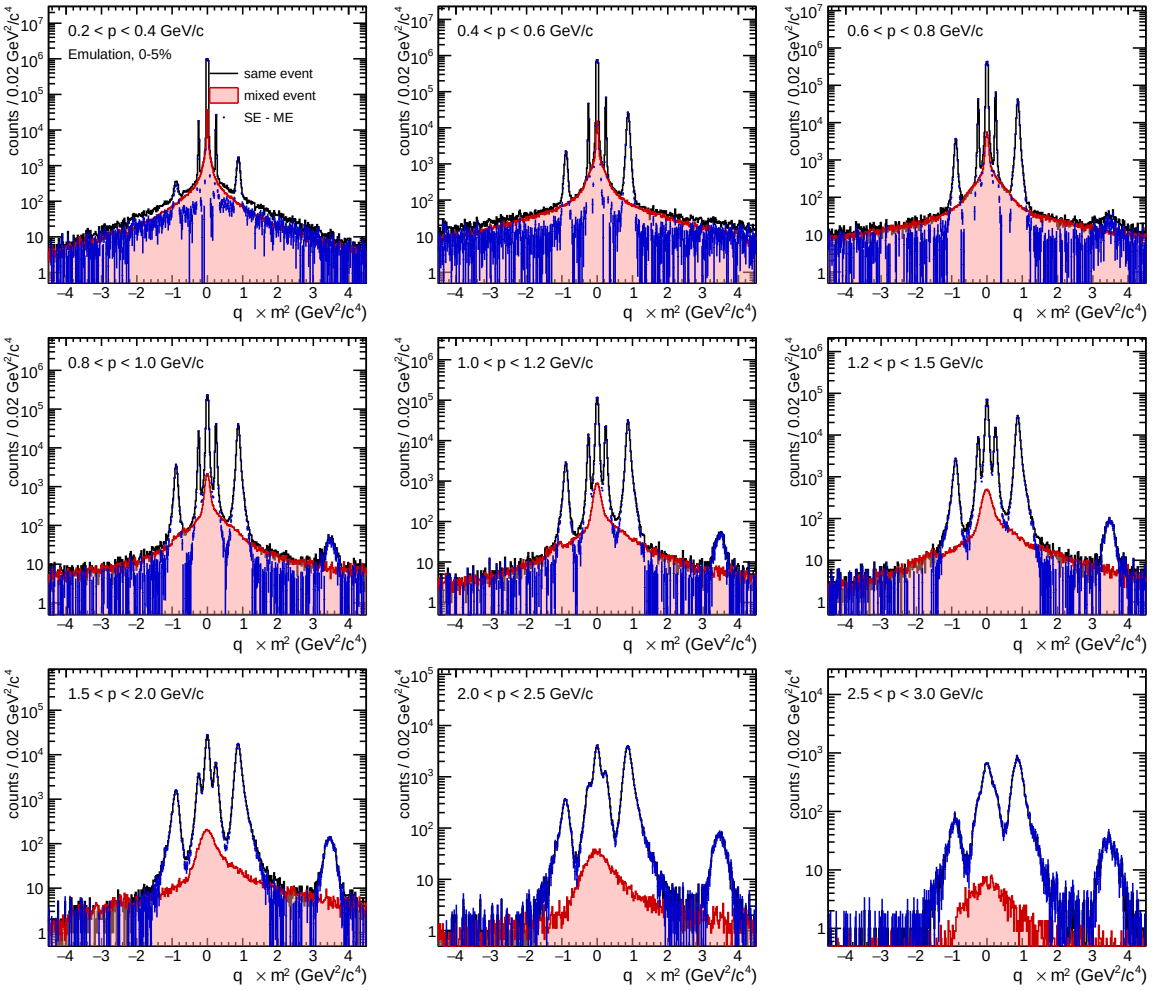
Figure 30: Large sample, method 1, SE vs ME, 60–80% (`scripts/make_mu.C`).

Table 13: Large sample, method 1, mid-central (30–40%) and peripheral (60–80%) events: random matches f (ME/SE of all TOF matches, %), ME/SE under the proton peak (%) and SE/ME between the p and d peaks (HI, $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$; – = no ME entries) per momentum bin.

	30–40%			60–80%		
p (GeV/c)	f	p	HI	f	p	HI
0.2–0.4	1.00	9.18	3.00	0.19	2.11	9.62
0.4–0.6	0.84	0.83	1.80	0.15	0.20	4.15
0.6–0.8	0.77	0.41	1.17	0.14	0.08	2.57
0.8–1	0.72	0.30	1.26	0.13	0.06	2.02
1–1.2	0.67	0.23	1.27	0.12	0.05	1.57
1.2–1.5	0.64	0.17	1.45	0.11	0.03	4.96
1.5–2	0.61	0.14	2.09	0.10	0.03	11.19
2–2.5	0.58	0.10	4.79	0.11	0.01	15.19
2.5–3	0.53	0.10	17.04	0.07	0.03	–

7.8.2 Charge-separated $q \times m^2$

Fig. 31a shows SE, ME and SE−ME versus $q \times m^2$, which separates $\pi^{\pm}$, $K^{\pm}$ and $p/\bar{p}$; the deuteron appears on the positive side only (read by eye). The random fraction is similar for both charges at low momentum and larger for negative tracks at high momentum (f^-, f^+ in Table 12). The random hits do not depend on the track charge; the fraction is larger for negatives because there are fewer real heavy negative particles to normalize to.



(a) $q \times m^2$ of QA tracks, 0–5%, 102 MuDst files, $0.02\,\mathrm{GeV}^2/c^4$ bins (log scale). Black line: same event; red filled: mixed event (method 1); blue points: SE−ME.

Figure 31: Large sample, method 1, charge-signed m^2, 0–5% (`scripts/make_qm2.C`).

7.8.3 Signal extraction: SE−ME and line-shape fits

In each centrality group and momentum bin, the SE histogram ($0.01\,\mathrm{GeV}^2/c^4$ bins) is fitted in $-0.6 < m^2 < 4.8\,\mathrm{GeV}^2/c^4$ with a Poisson likelihood. The model is signal + constant + the normalized ME, with the ME held fixed; this is the SE−ME subtraction with correct statistics. The function is integrated over each bin. The signal has four peaks (π, K, p, d), and four line shapes are compared, each with free yield, mean and width per species:

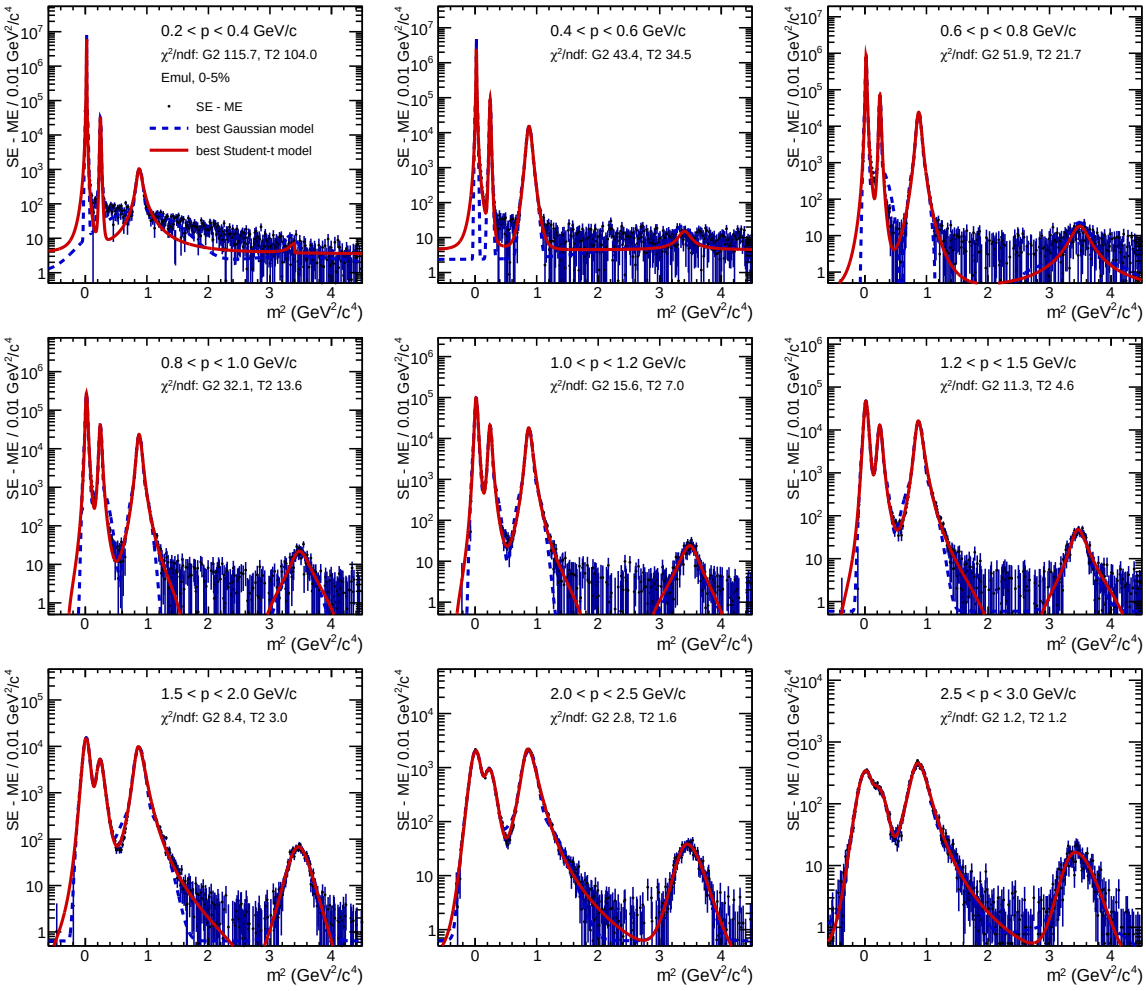
- G: Gaussian;
- G2: double Gaussian (common mean, core and wide width, core fraction);

- T: Student-t, $A/\sigma\, t_\nu((m^2-\mu)/\sigma)$, free ν per species;
- T2: split Student-t, separate widths below and above the mean, free ν per species.

The deuteron width is bounded so that it cannot absorb a continuum. The model with the smallest Pearson χ^2/ndf of SE against the model is called best.

Choice

The Student-t family describes the peaks better than the Gaussian family. The split Student-t (T2) is the best model in 9 of 9 momentum bins in 0–5% and 9 of 9 in 0–80% (all tracks). Mean χ^2/ndf in 0–5%: G 208, G2 31, T 34, T2 21. The asymmetric shape is needed for the high-m^2 tail of the proton peak. T2 is used for the yields.



(a) SE−ME m^2 (points), 0–5%, all tracks, 102 MuDst files, method 1 (log scale), with the signal part of the best Gaussian-family model (blue dashed) and the best Student-t-family model (red). The χ^2/ndf of both are given in each panel.

Figure 32: Large sample, method 1: SE−ME and line-shape fits, 0–5% (`scripts/make_fit.C`).

Table 14: Large sample, method 1, all tracks: Pearson χ^2/ndf of the four line-shape models (G Gaussian, G2 double Gaussian, T Student-t, T2 split Student-t), the best model, the yields of the best model and the sum of yields over SE−ME.

cent.	p (GeV/c)	G	G2	T	T2	best	Y_π	Y_K	Y_p	Y_d	$\sum Y/S$
0-5%	0.2–0.4	247	116	156	104	T2	3.71e+06	6.3e+04	1.32e+04	47.4	0.995
0-5%	0.4–0.6	508	43.4	40.5	34.5	T2	2.78e+06	1.8e+05	1.21e+05	366	1.000
0-5%	0.6–0.8	447	51.9	41.4	21.7	T2	1.61e+06	2.06e+05	1.82e+05	778	1.002
0-5%	0.8–1	288	32.1	30.5	13.6	T2	8.74e+05	1.68e+05	1.96e+05	1.01e+03	1.002
0-5%	1–1.2	177	15.6	17.0	7.0	T2	4.63e+05	1.18e+05	1.77e+05	1.07e+03	1.002
0-5%	1.2–1.5	118	11.3	12.0	4.6	T2	3.17e+05	1.03e+05	1.94e+05	1.76e+03	1.003
0-5%	1.5–2	68.7	8.4	7.7	3.0	T2	1.59e+05	6.54e+04	1.6e+05	2.25e+03	1.002
0-5%	2–2.5	17.7	2.8	2.8	1.6	T2	3.4e+04	1.71e+04	5.14e+04	1.5e+03	1.002
0-5%	2.5–3	3.5	1.2	1.3	1.2	T2	7.86e+03	3.81e+03	1.5e+04	803	1.001
0-80%	0.2–0.4	1695	613	883	529	T2	2.08e+07	3.52e+05	7.71e+04	78.2	0.996
0-80%	0.4–0.6	3633	211	166	130	T2	1.54e+07	9.74e+05	7.11e+05	1.42e+03	0.999
0-80%	0.6–0.8	2917	266	202	92.5	T2	8.81e+06	1.08e+06	1.02e+06	3.97e+03	1.001
0-80%	0.8–1	1883	166	152	57.4	T2	4.75e+06	8.66e+05	1.07e+06	5.6e+03	1.001
0-80%	1–1.2	1062	76.1	80.2	26.4	T2	2.5e+06	5.98e+05	9.21e+05	5.72e+03	1.001
0-80%	1.2–1.5	618	51.0	52.0	15.3	T2	1.72e+06	5.16e+05	9.76e+05	7.25e+03	1.001
0-80%	1.5–2	346	35.0	31.4	8.9	T2	8.59e+05	3.27e+05	7.74e+05	1.03e+04	1.001
0-80%	2–2.5	83.3	8.9	8.6	2.5	T2	1.87e+05	8.35e+04	2.47e+05	6.73e+03	1.001
0-80%	2.5–3	12.8	2.1	1.9	1.2	T2	3.97e+04	2.28e+04	7.22e+04	3.46e+03	1.001

The fitted yields add up to SE−ME within 0.5% in every bin. With T2, χ^2/ndf is 1.2–1.6 above 2 GeV/c, 3.0–7.0 at 1–2 GeV/c, 14–35 at 0.4–1 GeV/c and 104 at 0.2–0.4 GeV/c (0–5%).

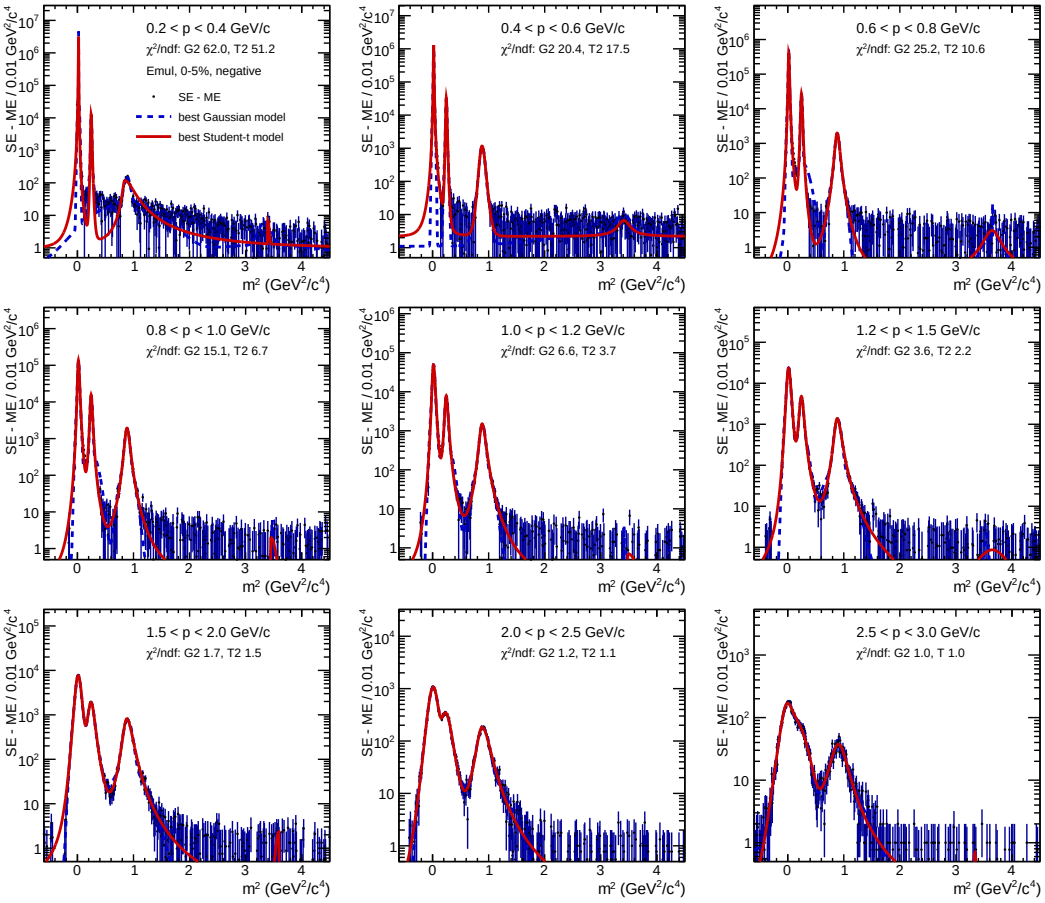
Caveat

No model is statistically acceptable below 2 GeV/c with 643650 events. Two limitations are seen in the figures (read by eye): (i) at $0.2 < p < 0.4$ GeV/c a continuum of about 100 counts per bin remains between the peaks after the subtraction, the same SE excess as in Table 12; it is under-subtracted background, not a line-shape problem; (ii) each momentum bin mixes tracks with different m^2 resolution (momentum, path length, centrality), so a single line shape per species cannot be exact at this precision. Finer momentum bins or a resolution-scaled variable ($1/\beta$ residuals) would address (ii).

Charge-separated fits. The same fits were done separately for negative and positive tracks (Figs. 33a and 34a). The Student-t family is better in 8 (negative) and 9 (positive) of 9 bins in 0–5%. The sum of the negative and positive yields reproduces the all-track yield within 0.1% (π), 0.6% (K) and 0.1% (p) at 0.4–2.5 GeV/c. In that range the ratios are $\pi^-/\pi^+ = 1.00$–1.08, $\mathrm{K}^-/\mathrm{K}^+ = 0.59$–$0.62$ and $\bar{\mathrm{p}}/\mathrm{p} = 0.089$–$0.104$ (0–5%, Table 15); these are raw yield ratios without efficiency corrections. At 0.2–0.4 GeV/c the $\bar{\mathrm{p}}/\mathrm{p}$ ratio is distorted by the continuum, and at 2.5–3 GeV/c the π and K peaks are no longer separated.

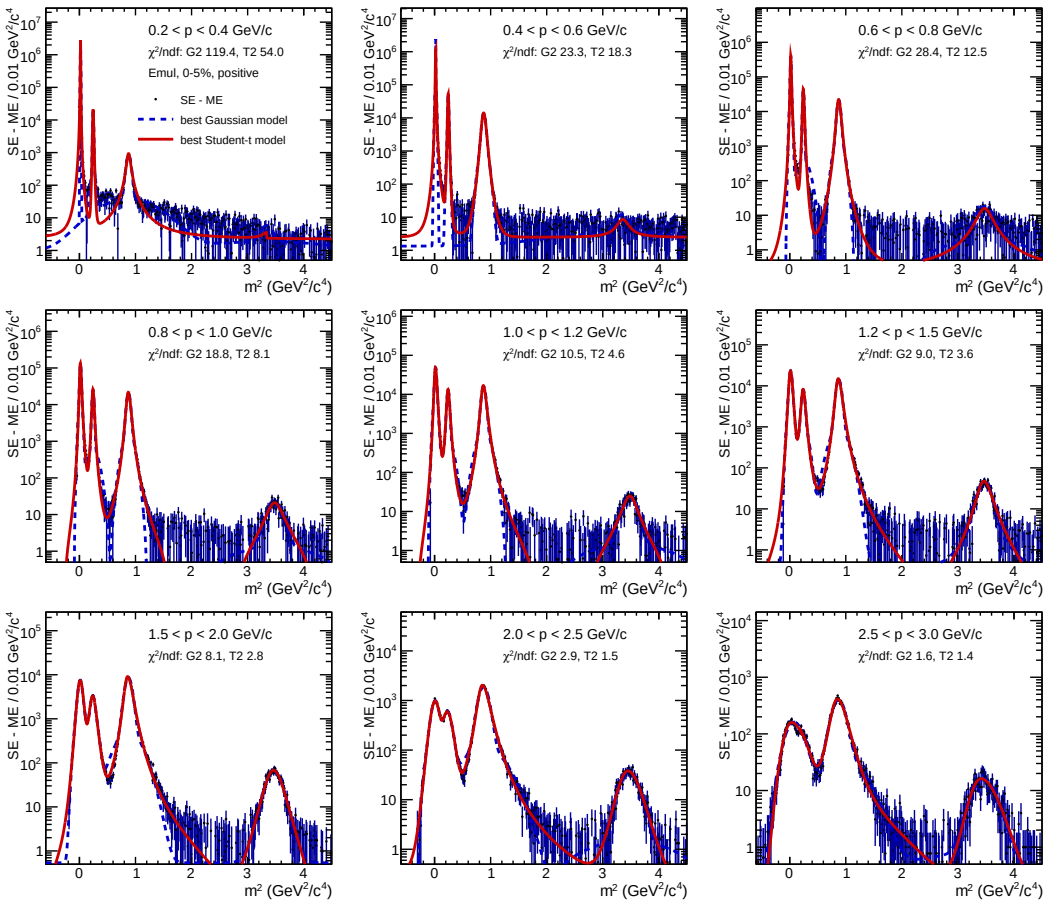
Table 15: Large sample, method 1, 0–5%: antiparticle/particle yield ratios from the separate fits of negative and positive tracks (best model each), and the closure of the charge-separated fits against the all-track fit, $(Y^- + Y^+)/Y$.

p (GeV/c)	π^-/π^+	$\mathrm{K}^-/\mathrm{K}^+$	$\bar{\mathrm{p}}/\mathrm{p}$	closure π	closure K	closure p
0.2–0.4	1.04	0.64	0.336	1.000	1.000	1.041
0.4–0.6	1.02	0.62	0.089	1.000	1.000	1.000
0.6–0.8	1.01	0.60	0.097	1.000	1.000	1.000
0.8–1	1.00	0.60	0.101	1.000	1.000	1.000
1–1.2	1.00	0.60	0.103	1.000	1.000	1.000
1.2–1.5	1.01	0.59	0.104	1.000	1.000	1.000
1.5–2	1.03	0.60	0.102	1.000	1.002	1.000
2–2.5	1.08	0.60	0.101	0.999	1.006	0.999
2.5–3	0.66	2.03	0.109	1.284	0.692	0.929



(a) Negative tracks: SE−ME m^2 and fits, 0–5%; encodings as in Fig. 32a.

Figure 33: Large sample, method 1: SE−ME and line-shape fits, negative tracks, 0–5%.



(a) Positive tracks: SE−ME m^2 and fits, 0–5%; encodings as in Fig. 32a.

Figure 34: Large sample, method 1: SE−ME and line-shape fits, positive tracks, 0–5%.

7.9 Large MuDst sample: method 2 (overlay) and comparison

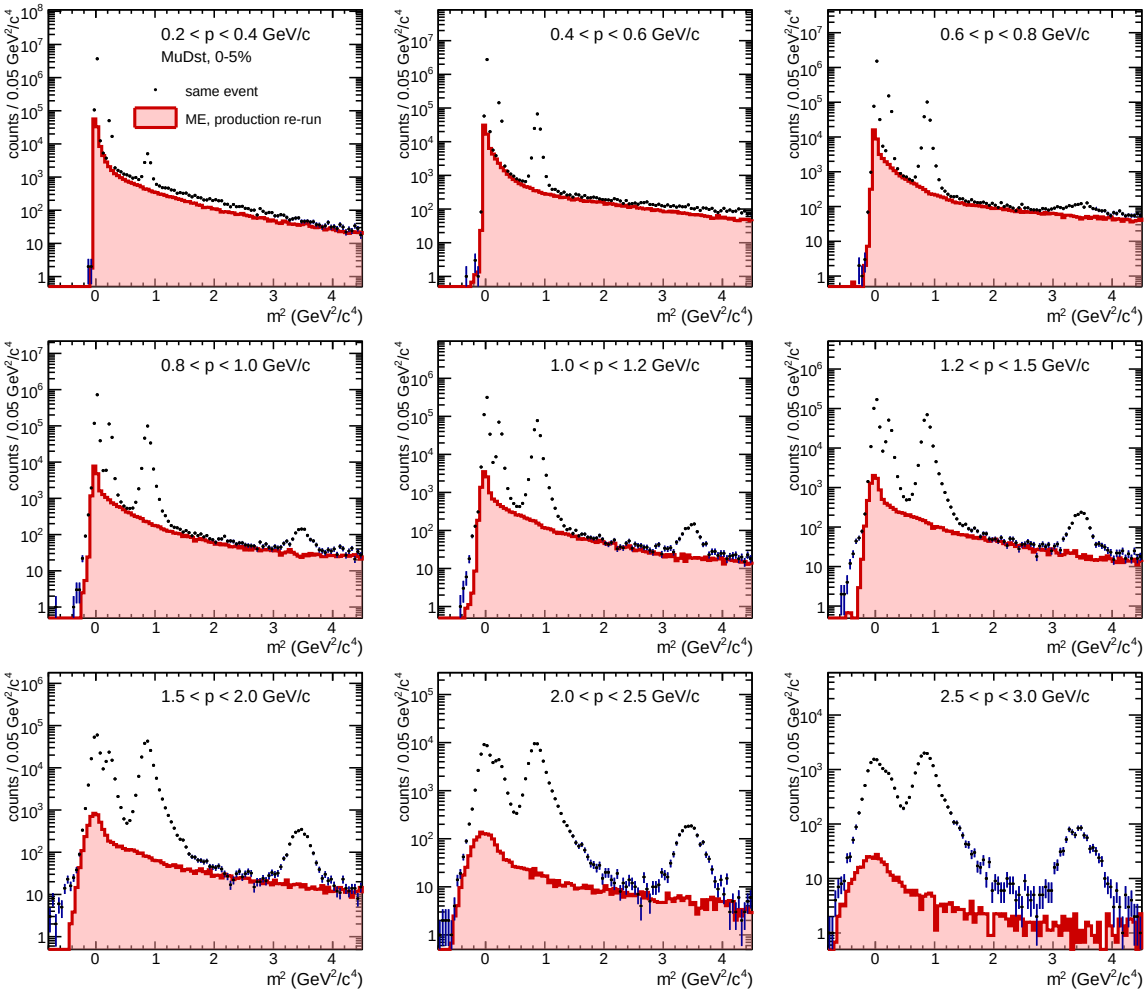
Method 2 (`StTofMEOverlayMaker` v3, mode 1: partner hits and partner global tracks overlaid on this event, production `StBTofMatchMaker`/`StBTofCalibMaker` re-run; Sec. 7.7) was run on the same 102 files, one job per file (`scripts/ovbig.sub`). The outputs are merged into `data/ov_big.root` (`scripts/merge_ovbig.sh`): 1858212 events, 643544 used for m^2, on average 564 partner global tracks added per overlay. The analysis chain is the one of Sec. 7.8: `scripts/make_ov.C` (tag `big`), `scripts/make_qm2.C`, `scripts/make_fit.C` (method 0) and `scripts/make_big_m2_tables.py`. In the overlay, 95.2% of the SE-matched tracks keep their own hit, 1.2% switch to a partner hit and 3.6% lose the match (four-file values in Sec. 7.7).

7.9.1 Random matches: method 2 vs method 1

Method 2 gives 2.64% random matches in 0–5% and 1.58% in 0–80% ($0.2 < p < 3\,\mathrm{GeV}/c$). The two methods agree within 9% (relative) in every centrality group (Table 16), although method 2 uses production calibration and production matching while method 1 emulates both. Between the p and d peaks SE/ME is 1.41 (0–5%), with 1.20–1.45 per momentum bin at 0.6–2 GeV/c and 1.71 at 0.2–0.4 GeV/c (Table 17). The largest bin-by-bin ME/SE at 0.4–1.5 GeV/c is 1.33 (0–5%), against 13.81 for method 1: method 2 has no timing artefact below the pion peak.

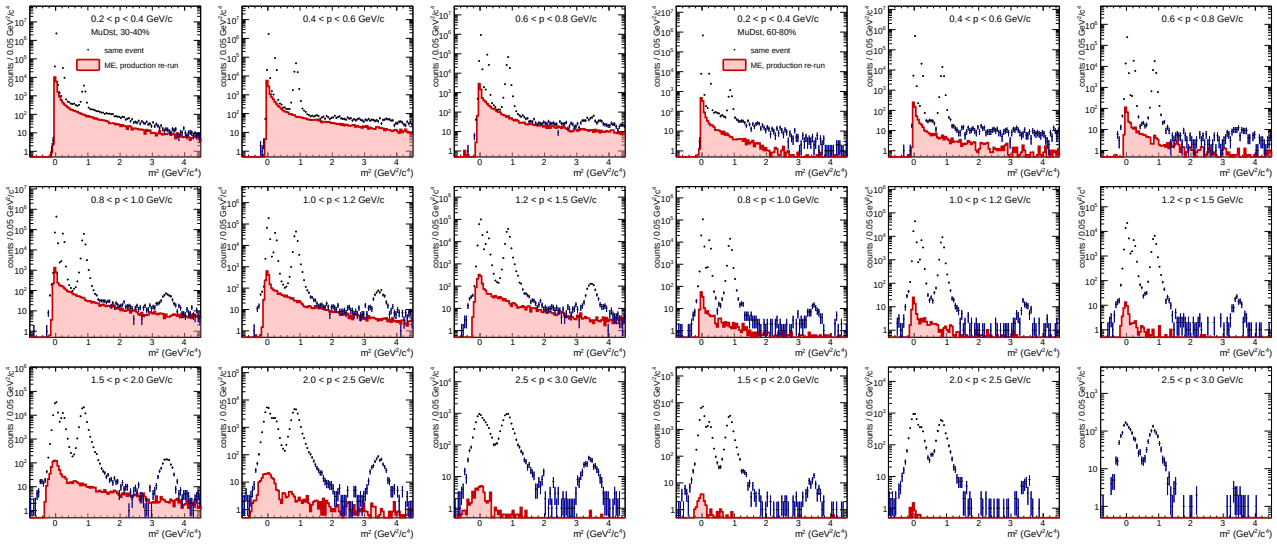
Table 16: Large sample: random matches f (% of TOF matches, $0.2 < p < 3\,\mathrm{GeV}/c$) and SE/ME between the p and d peaks (HI) per centrality group, methods 1 and 2.

	f (%)		HI	
centrality	m1	m2	m1	m2
0-5%	2.76	2.64	1.32	1.41
30-40%	0.84	0.80	1.85	2.05
60-80%	0.16	0.14	4.90	6.23
0-80%	1.65	1.58	1.50	1.62



(a) m^2 of QA tracks, 0–5%, nine momentum bins, 102 MuDst files (log scale). Black points: same event; red filled: mixed event, method 2 (overlay with partner tracks, production makers re-run), normalized per QA track. Random matches 2.64%; SE/ME between p and d 1.41.

Figure 35: Large sample, method 2, SE vs ME, 0–5% (`scripts/make_ov.C`).



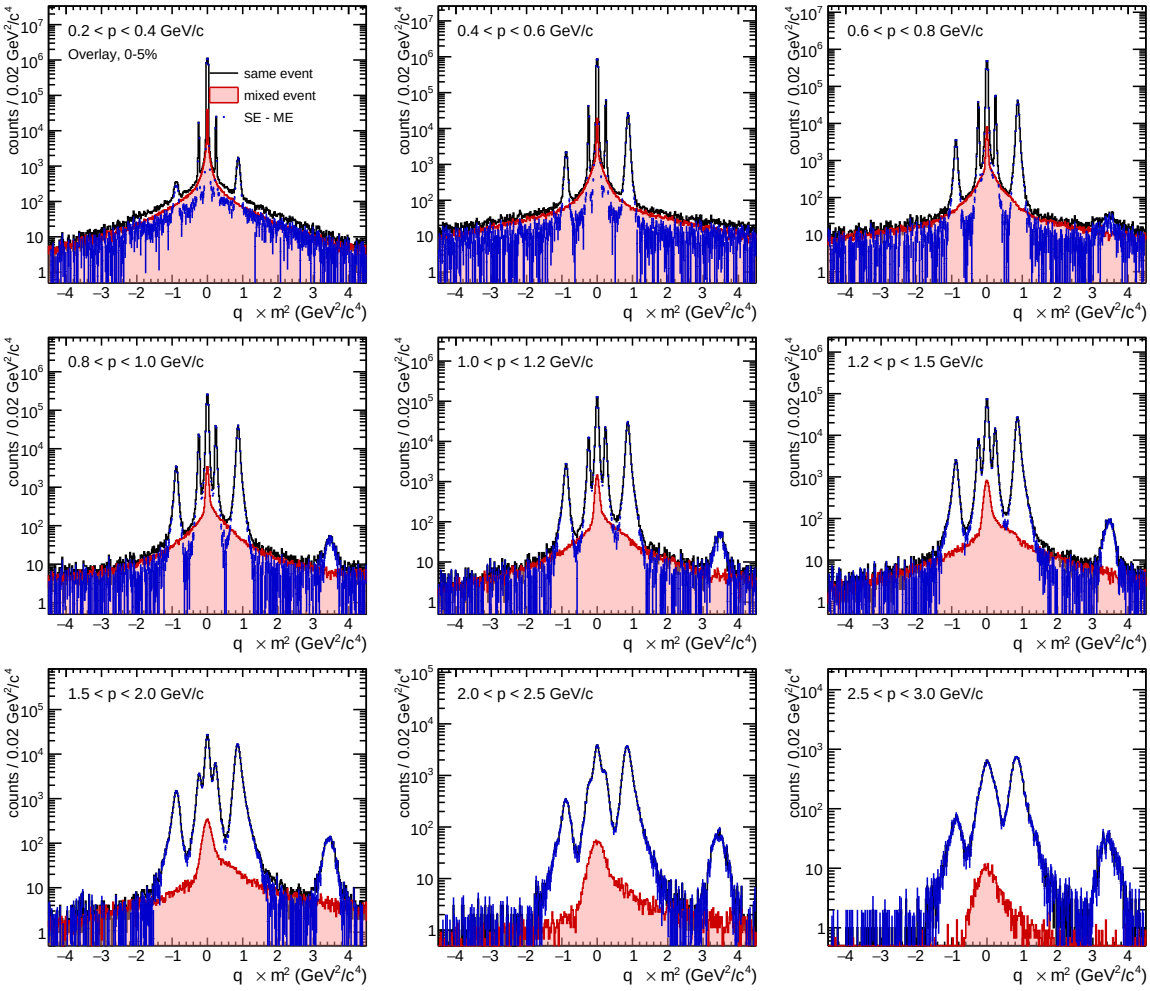
(a) 30–40%: random matches 0.80%, SE/ME between p and d 2.05.

(b) 60–80%: random matches 0.14%, SE/ME between p and d 6.23.

Figure 36: Large sample, method 2, SE vs ME in mid-central and peripheral events; encodings as in Fig. 35a.

Table 17: Large sample: method 2 (overlay with partner tracks, production makers re-run) per momentum bin, with method 1 for comparison. f = ME/SE of all TOF matches (%); HI = SE/ME for $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$; π/K/p = ME/SE under the peaks (%, method 2); f^-, f^+ = f of negative and positive tracks (method 2).

		f		HI						
cent.	p (GeV/c)	m1	m2	m1	m2	π	K	p	f^-	f^+
0-5%	0.2–0.4	3.26	3.19	1.88	1.71	2.57	10.64	23.10	3.14	3.24
0-5%	0.4–0.6	2.76	2.61	1.30	1.33	1.91	4.43	2.18	2.66	2.55
0-5%	0.6–0.8	2.55	2.38	1.07	1.26	1.74	2.79	1.19	2.49	2.30
0-5%	0.8–1	2.42	2.26	1.12	1.31	1.74	2.00	0.89	2.43	2.13
0-5%	1–1.2	2.29	2.17	1.05	1.20	1.81	1.59	0.68	2.44	2.00
0-5%	1.2–1.5	2.21	2.11	1.10	1.22	1.87	1.37	0.53	2.52	1.89
0-5%	1.5–2	2.12	2.03	1.32	1.45	1.86	1.30	0.41	2.65	1.75
0-5%	2–2.5	2.00	1.92	2.31	2.59	1.76	1.61	0.33	2.81	1.59
0-5%	2.5–3	1.83	1.80	4.41	5.67	1.75	1.79	0.29	2.90	1.42
0-80%	0.2–0.4	1.94	1.89	2.27	2.06	1.50	6.84	15.01	1.87	1.91
0-80%	0.4–0.6	1.64	1.54	1.48	1.55	1.11	2.72	1.28	1.58	1.51
0-80%	0.6–0.8	1.52	1.42	1.12	1.36	1.02	1.70	0.71	1.48	1.37
0-80%	0.8–1	1.45	1.35	1.14	1.34	1.03	1.22	0.53	1.45	1.28
0-80%	1–1.2	1.39	1.31	1.12	1.29	1.07	0.97	0.42	1.46	1.21
0-80%	1.2–1.5	1.35	1.28	1.21	1.36	1.10	0.82	0.32	1.51	1.16
0-80%	1.5–2	1.31	1.25	1.53	1.70	1.09	0.78	0.26	1.59	1.09
0-80%	2–2.5	1.23	1.19	3.09	3.34	1.05	0.96	0.20	1.71	0.98
0-80%	2.5–3	1.15	1.14	6.54	7.30	1.05	1.18	0.17	1.80	0.90



(a) $q \times m^2$ of QA tracks, 0–5%, 102 MuDst files, 0.02 GeV$^2/c^4$ bins (log scale). Black line: same event; red filled: mixed event (method 2); blue points: SE−ME.

Figure 37: Large sample, method 2, charge-signed m^2, 0–5% (`scripts/make_qm2.C`).

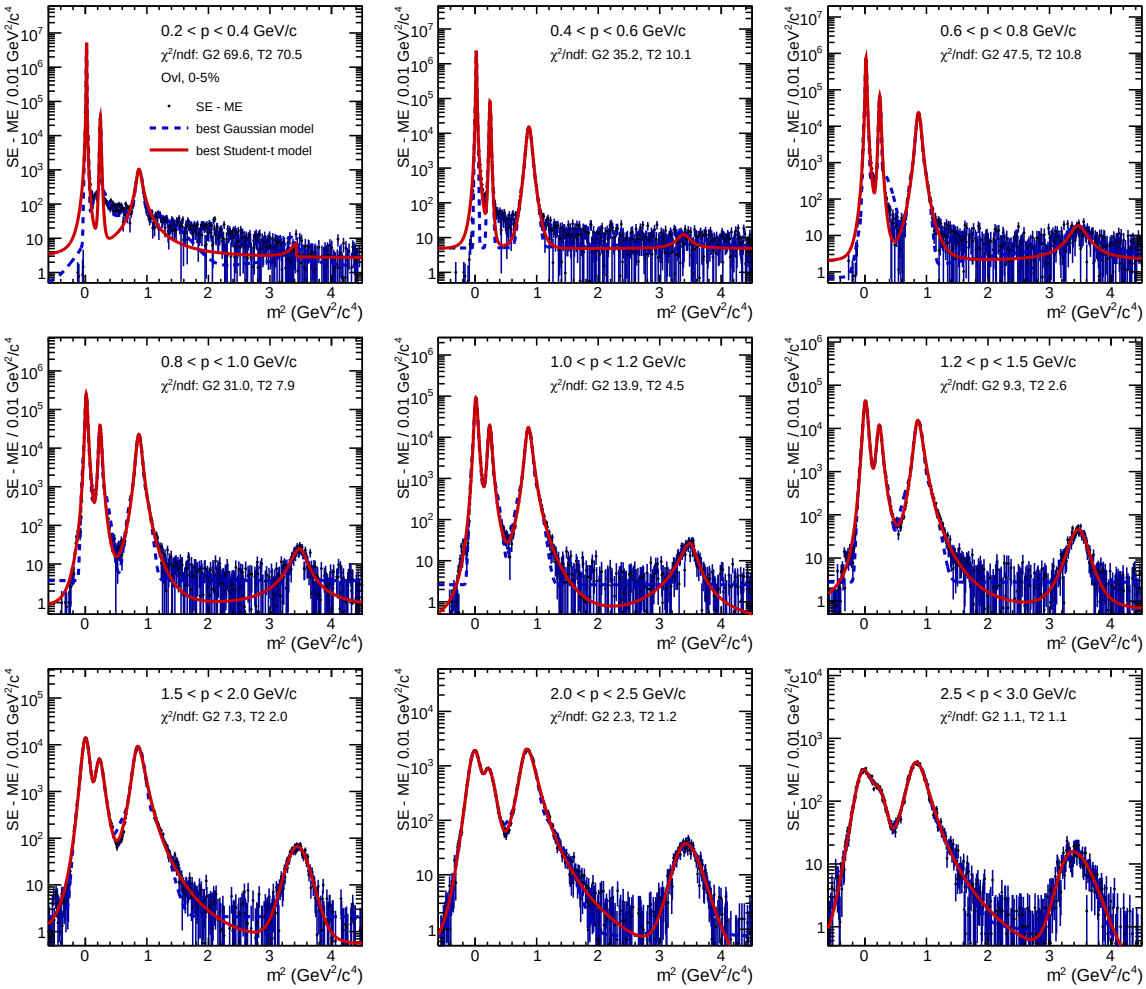
7.9.2 Signal extraction with method 2

The fits of Sec. 7.8.3 were repeated with the method-2 ME (Fig. 38a, Table 18). The split Student-t (T2) is again best: in 8 of 9 bins in 0–5% and 8 of 9 in 0–80%. The exception in both is 0.2–0.4 GeV/c, where the double Gaussian is marginally better and no model is acceptable. Mean χ^2/ndf in 0–5% is G2 24, T 23, T2 12, lower than with method 1 (T2 21). The fitted yields add up to SE−ME within 0.7%. The yields of the two methods agree within 1.8% for π, K and p at 0.4–2 GeV/c; at 0.2–0.4 GeV/c the method-2 proton yield is 15% higher (0–5%), because method 2 subtracts less background there.

Choice

For the signal yields the choice between methods 1 and 2 matters only below 0.4 GeV/c and for K above 2 GeV/c. Method 2 is the baseline (production calibration and matching); method 1 is a cross-check.

Per charge (method 2, 0–5%), the Student-t family is better in 9 (negative) and 8 (positive) of 9 bins. The raw ratios at 0.4–2.5 GeV/c are $\pi^-/\pi^+ = 1.00$–1.07, $\mathrm{K}^-/\mathrm{K}^+ = 0.60$–$0.62$ and $\bar{\mathrm{p}}/\mathrm{p} = 0.092$–$0.104$, the same as with method 1.



(a) SE−ME m^2 (points), 0–5%, all tracks, method 2 ME (log scale), with the signal part of the best Gaussian-family model (blue dashed) and the best Student-t-family model (red).

Figure 38: Large sample, method 2: SE−ME and line-shape fits, 0–5% (`scripts/make_fit.C`).

Table 18: Large sample, method 2, all tracks: Pearson χ^2/ndf of the four line-shape models, the best model, and the ratio of the method-2 to method-1 yields of the best models (Y_2/Y_1).

cent.	p (GeV/c)	G	G2	T	T2	best	Y_2/Y_1 π	K	p
0-5%	0.2–0.4	153	69.6	101	70.5	G2	0.995	1.050	1.152
0-5%	0.4–0.6	558	35.2	19.3	10.1	T2	0.997	1.002	1.003
0-5%	0.6–0.8	498	47.5	33.5	10.8	T2	0.996	1.001	1.000
0-5%	0.8–1	319	31.0	25.2	7.9	T2	0.996	0.997	0.999
0-5%	1–1.2	184	13.9	13.9	4.5	T2	0.996	0.998	0.998
0-5%	1.2–1.5	103	9.3	8.7	2.6	T2	0.992	1.009	0.997
0-5%	1.5–2	65.4	7.3	6.2	2.0	T2	0.990	1.009	0.998
0-5%	2–2.5	16.6	2.3	2.2	1.2	T2	1.010	0.958	1.001
0-5%	2.5–3	2.9	1.1	1.1	1.1	T2	0.682	1.602	1.009
0-80%	0.2–0.4	1303	471	727	480	G2	0.997	1.043	1.139
0-80%	0.4–0.6	124923	219	92.2	54.2	T2	0.998	1.002	1.002
0-80%	0.6–0.8	3215	258	166	53.2	T2	0.998	1.001	1.000
0-80%	0.8–1	2032	164	122	36.5	T2	0.998	0.998	1.000
0-80%	1–1.2	1080	69.7	62.4	17.7	T2	0.998	0.999	0.999
0-80%	1.2–1.5	548	44.2	36.7	9.1	T2	0.995	1.009	0.999
0-80%	1.5–2	332	30.2	23.5	5.7	T2	0.990	1.018	0.999
0-80%	2–2.5	78.6	6.3	5.9	1.7	T2	1.009	0.961	1.003
0-80%	2.5–3	9.5	1.4	1.5	1.1	T2	0.963	1.032	1.008

7.9.3 The SE excess at low momentum

Both methods leave SE/ME ≈ 2 between the p and d peaks at 0.2–0.4 GeV/c in central events, and more in peripheral events. `scripts/make_lowp.C` (log `logs/lowp_numbers.txt`) tests whether this excess behaves like random matches.

Centrality dependence. Fig. 39a shows SE, ME and SE−ME in $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$ per 10^4 tracks (method 1: projected tracks; method 2: QA tracks) in the nine centrality classes. From 70–80% to 0–5% the ME per track rises by a factor 28 (method 2, 0.2–0.4 GeV/c), as random matches should. SE−ME stays flat: 2.1–2.3 per 10^4 tracks at 0.2–0.4 GeV/c, 1.7–2.7 at 0.4–0.6 and 0.9–1.7 at 0.6–1 GeV/c (method 2). Method 1 gives 2.7–3.4, 2.2–3.2 and 0.6–1.2 (per projected track). The excess is a fixed rate per track, independent of the hit density. It is therefore not under-subtracted random background, which would scale like the ME. It stands out at low momentum and in peripheral events only because the random background is small there.

Charge. Table 19 gives the ratio of SE−ME of negative to positive tracks in $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$. Both methods agree. Two components are visible:

- at 0.3–0.6 GeV/c outside 60–80% the ratio is 0.75–0.92 (method 2): most of the excess is charge-symmetric;
- at 0.2–0.25 GeV/c the ratio drops to 0.04–0.2: the excess is almost only positive. A natural candidate is slow protons that lose energy in the material before the TOF, arrive late and are reconstructed at larger m^2 (the high-m^2 tail of the proton peak that also requires the split Student-t). The ratio below 1 at 0.6–1 GeV/c is the same proton tail.

The charge-symmetric part excludes kaon decays in flight, Λ decays and deuterons as its source.

Time structure. Fig. 40a shows $t - t_\pi$ of SE and ME (method 2, 0–80%). Beyond the particle peaks the SE−ME forms a flat plateau up to the end of the window (27 ns), at a level comparable to the ME. In 15–27 ns the SE/ME is 2.3 at 0.2–0.3 GeV/c and 1.8 at 0.6–1 GeV/c, with a similar SE−ME per ns

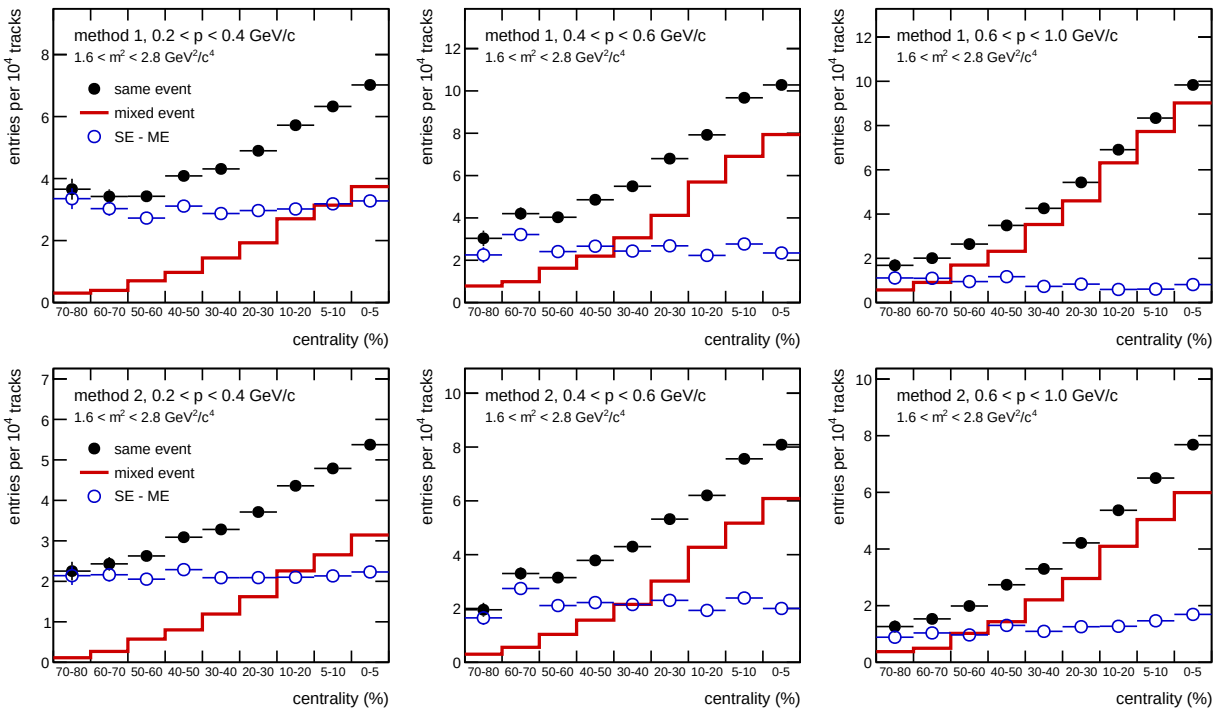
in all momentum bins (`logs/lowp_numbers.txt`). There is no peak at a fixed delay. The m^2 window $1.6 < m^2 < 2.8$ maps to delays of 15–35 ns at 0.3 GeV/c but to a few ns at 1 GeV/c, where the proton peak dominates; so a delay-flat component appears in the m^2 window mainly at low momentum.

Table 19: SE−ME in $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$ of negative over positive tracks, per momentum bin and centrality group, for both methods. A charge-symmetric source gives about 1 ($\pi^-/\pi^+ \approx 1$); a source related to protons gives values well below 1.

	method 1			method 2		
p (GeV/c)	60–80%	20–60%	0–20%	60–80%	20–60%	0–20%
0.20–0.25	0.21	0.12	0.24	0.21	0.04	0.13
0.25–0.30	0.31	0.55	0.62	0.38	0.54	0.56
0.30–0.35	0.57	0.83	0.86	0.60	0.82	0.86
0.35–0.40	0.93	0.81	0.95	0.88	0.78	0.92
0.40–0.60	0.90	0.75	0.85	0.85	0.75	0.83
0.60–1.00	0.64	0.46	0.66	0.66	0.60	0.74

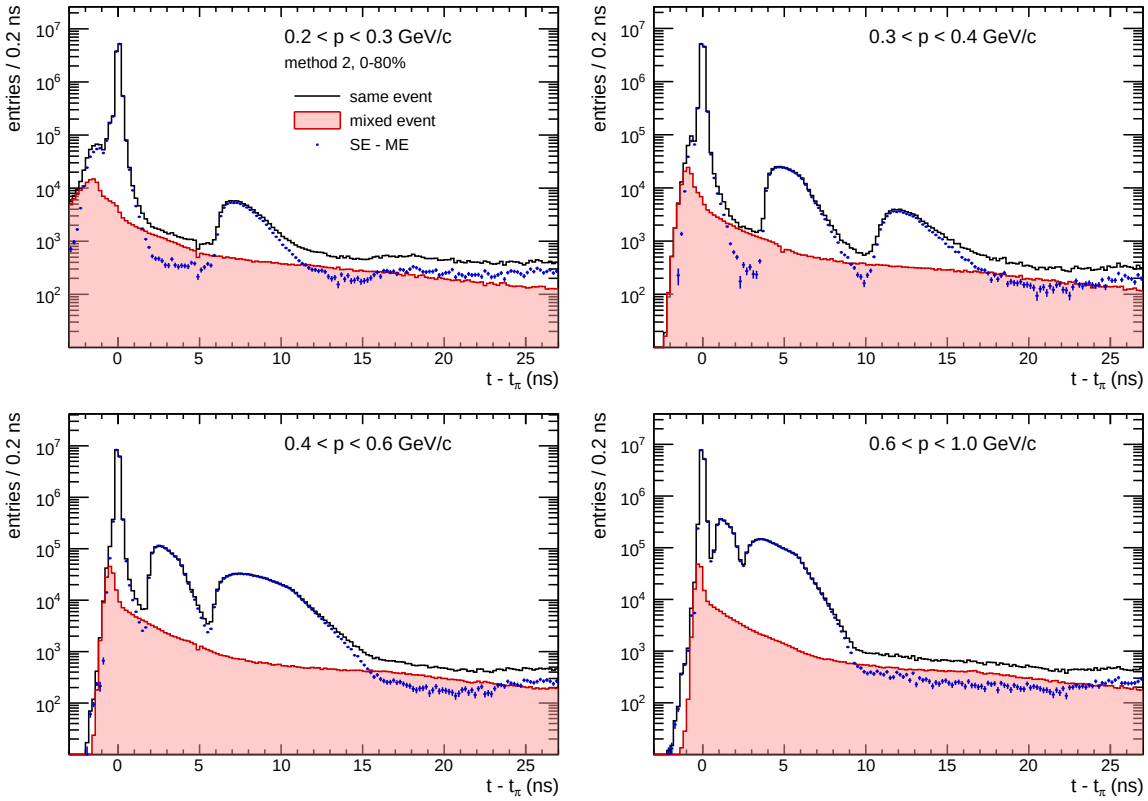
Caveat

Interpretation (not proven). The positive-only part at the lowest momenta is most likely the proton energy-loss tail. The charge-symmetric part is a late hit correlated with the track itself, in or next to its cell, broad in time. Candidates are secondaries from hadronic interactions of the particle in the TOF trays and the material behind them (slow nucleons, fragments, back-splash), or delayed signals in neighbouring channels. A fixed-delay crosstalk is disfavoured by the flat time distribution. No ME can model such a correlated component. If needed, it can be estimated as a per-track template, for instance from peripheral events where the ME is negligible. Not done; see Sec. 10.



(a) SE (black), ME (red) and SE−ME (blue open) in $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$ per 10^4 tracks versus centrality, for 0.2–0.4, 0.4–0.6 and 0.6–1 GeV/c. Top: method 1 (per projected track); bottom: method 2 (per QA track).

Figure 39: Large sample: centrality dependence of the SE excess between the p and d peaks (`scripts/make_lowp.C`).



(a) $t - t_\pi$ of TOF-matched QA tracks, 0–80%, four momentum bins (log scale). Black line: same event; red filled: mixed event (method 2, normalized per momentum bin); blue points: SE−ME. Peaks: electrons just below 0, π at 0, K and p at positive delays.

Figure 40: Large sample, method 2: time structure of the SE excess (`scripts/make_lowp.C`).

7.10 Full simulation: origin of the low-momentum excess

The SE excess over the ME between the p and d peaks at low momentum is a fixed rate per track, flat in centrality and flat in time up to 27 ns (Sec. 7.9.3). Two track-correlated sources were suspected: decays in flight and particles scattered or produced in the detector material. Both are present in a full simulation, where every BTOF hit can be traced to the particle that made it.

Simulation set-up.

- Generator: HIJING 1.383, Au+Au at $\sqrt{s_{NN}} = 19.6$ GeV, impact parameter 0–14.5 fm, no jet quenching, weak decays (π^0, K^0_S, Λ, ...) left to GEANT (`sim/starsim_hijing.C`). All particles go to GEANT (no η cut). Vertex Gaussian with $\sigma_{x,y} = 0.1$ cm, $\sigma_z = 50$ cm.
- Detector: starsim (GEANT3) with the y2019a AgML geometry (iTPC, BTOF, VPD, magnet).
- Reconstruction: `sim/reco_sim.C` with the bfc options `y2019a`, `fzin`, `AgML`, `useXgeom`, `iTpcIT`, `MakeEvent`, `ITTF`, `VFMinuit`, `tpcDB`, `btofsim`, `vpdSim`, `btofMatch`, `btofCalib`, `vpdCalib`, `IdTruth`, `McEvent`, `CMuDst` (full list in `sim/sim_job.sh`), plus either `tfs` (fast TPC simulator) or `TpcRS`, `TpxClu`, `TpcHitMover` (full TPC response). The BTOF matching and calibration are the production makers.
- BTOF simulator: the SL24y `StBTofSimMaker` in fast mode, with a local patch (`StRoot/StBTofSimMaker`). The SL24y maker sets the truth id of a hit only for crosstalk hits; the patch

sets it for every hit to the GEANT track of the earliest contribution in the cell, i.e. the one that defines the hit time.

- Truth analysis: `StRoot/StTofSimAnaMaker` (`macros/runTofSimAna.C`) on the simulation MuDst. Same event and track selection as the data (Table 4). The matched hit of every TOF-matched QA track is classified with the MC record (`StMuMcTrack`, `StMuMcVertex`): *own* (the track's particle made the hit); *decay* (a descendant through a decay: weakly decaying parent and a muon daughter, or a decay vertex at $R < 190\,\mathrm{cm}$); *secondary* (a descendant through any other process, mostly interactions); *relative* (an ancestor or sibling); *unrelated* (no common ancestry, the part a mixed event describes). m^2 is computed with the true event time (hit time minus the simulation clock offset of 19.01 ns).

- Samples: 49857 events with the fast TPC simulator (999 jobs of 50 events; one job hung in HIJING and was removed), 40496 after the vertex cuts, 5995969 QA tracks; and the same 1000 GEANT events reconstructed both with the fast simulator and with TpcRS. Production scripts `sim/sim_job.sh`, `sim/sim_job2.sh`, `sim/sim.sub`, `sim/sim2.sub`; numbers from `scripts/make_sim.C` (`logs/sim_numbers.txt`).

Choice

The fast TPC simulator is used for the large sample. The full TPC response costs about 3 minutes per peripheral and 10 minutes per mid-central event (98% of the CPU), against a few seconds. On the same 1k events the two give the same fractions of all match categories within $1.6\,\sigma$ (Table 20).

Table 20: The same 1000 GEANT events reconstructed with the fast TPC simulator (tfs) and with the full TPC response (TpcRS): QA tracks, TOF-matched QA tracks per category, and correlated entries in $1.6 < |q\,m^2| < 2.8\,\mathrm{GeV}^2/c^4$, for $0.2 < p < 1\,\mathrm{GeV}/c$. Δ/σ compares the fractions (matches/QA tracks) with binomial-free Poisson errors.

	tfs	TpcRS	Δ/σ
QA tracks	103984	101532	
own particle	56666	55521	-0.6
decay	1148	1078	+0.9
secondary	1706	1682	-0.3
relative	0	0	+0.0
unrelated	416	452	-1.6
corr. in p–d window	19	19	

Caveat

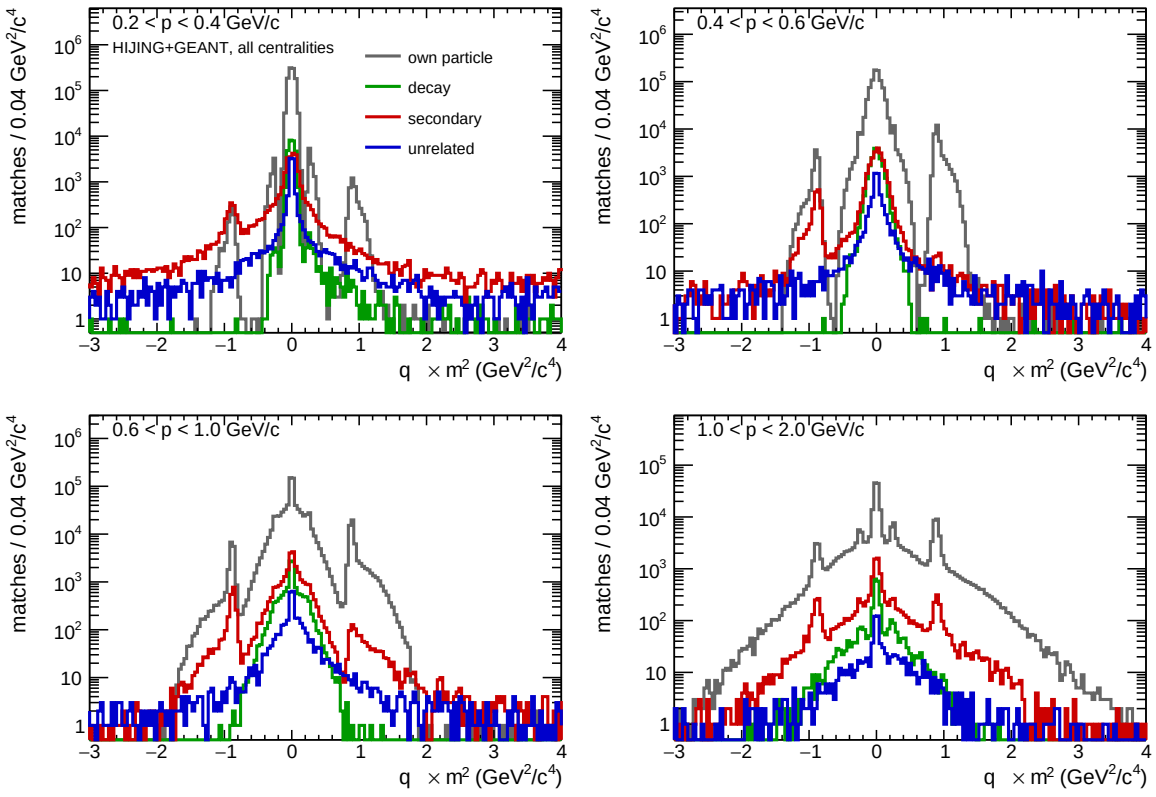
Limitations of the simulation: (i) the fast BTOF simulator gives every hit ToT = 0, so the production rule "largest ToT wins" between several candidate hits cannot act and ties go to the next criterion; in data a slow, heavily ionising secondary has a large ToT and is preferred. (ii) Production of slow protons and nuclear fragments in the material depends on the GEANT3 hadronic package and its cuts (not tuned here). (iii) Centrality: HIJING plus reconstruction was not tuned to the data refMult; simulated events are classified by their refMult percentile within the simulated sample ($b < 14.5\,\mathrm{fm}$), an approximate match to the data classes. (iv) HIJING has no light nuclei (no deuteron peak). (v) No noise hits and no pile-up.

7.10.1 Where the matched hits come from

Table 21 gives the categories per momentum bin. At 0.2–0.4 GeV/c, 93.9% of the matches are own hits, 2.4% decay muons, 2.7% secondaries and 1.0% unrelated hits. Fig. 41a shows how they distribute in $q \times m^2$ (read by eye): decay muons stay close to the pion peak; secondaries that arrive together with the track's own particle sit at its m^2 (pion and proton peaks), and the late ones form a broad continuum; at 0.2–0.4 GeV/c this continuum is above the unrelated matches for $|q\,m^2| > 1$.

Table 21: Simulation (40496 HIJING events, fast TPC simulator): origin of the matched BTOF hit of TOF-matched QA tracks, in % of all matches, per momentum bin (all centralities). Last two columns: correlated (decay + secondary + relative) and unrelated matches in $1.6 < |q\,m^2| < 2.8\,\mathrm{GeV}^2/c^4$ per 10^4 QA tracks.

p (GeV/c)	own	decay	secondary	relative	unrelated	corr. (p–d)	unrel. (p–d)
0.2–0.4	93.85	2.43	2.72	0.000	1.00	2.71	0.95
0.4–0.6	94.76	1.96	2.67	0.000	0.61	1.16	1.16
0.6–1.0	94.45	1.52	3.53	0.000	0.50	1.52	0.86
1.0–1.5	94.05	1.12	4.40	0.000	0.42	3.41	0.94
1.5–2.5	94.06	0.84	4.71	0.000	0.38	16.35	1.23

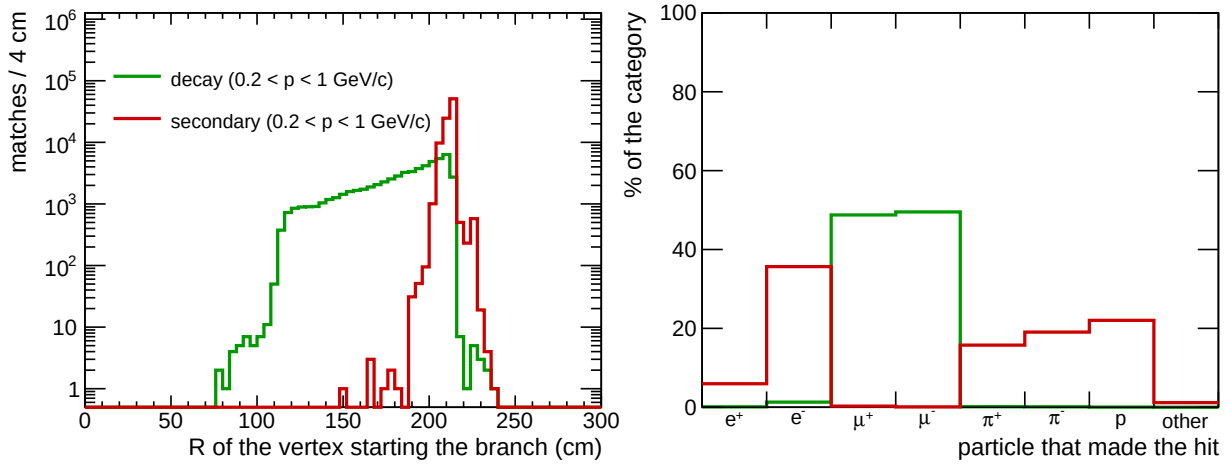


(a) $q \times m^2$ of TOF-matched QA tracks in HIJING+GEANT (40496 events, all centralities, fast TPC simulator), by origin of the matched hit: own particle (gray), decay (green), secondary (red), unrelated (blue). The relative category is empty. 0.04 GeV$^2/c^4$ bins, log scale.

Figure 41: Simulation: origin of the matched hit versus $q \times m^2$ (`scripts/make_sim.C`).

Fig. 42a shows where the branch leading to the hit starts. 100% of the secondaries (0.2–1 GeV/c) start at $200 < R < 240$ cm, i.e. in the BTOF trays themselves; the decays are spread along the path

from $R \approx 110\,\text{cm}$ to the trays (read by eye). The secondaries that make the hit are $e^{\pm}$ (42%), $\pi^{\pm}$ (35%) and protons (22%); the decay hits are muons.



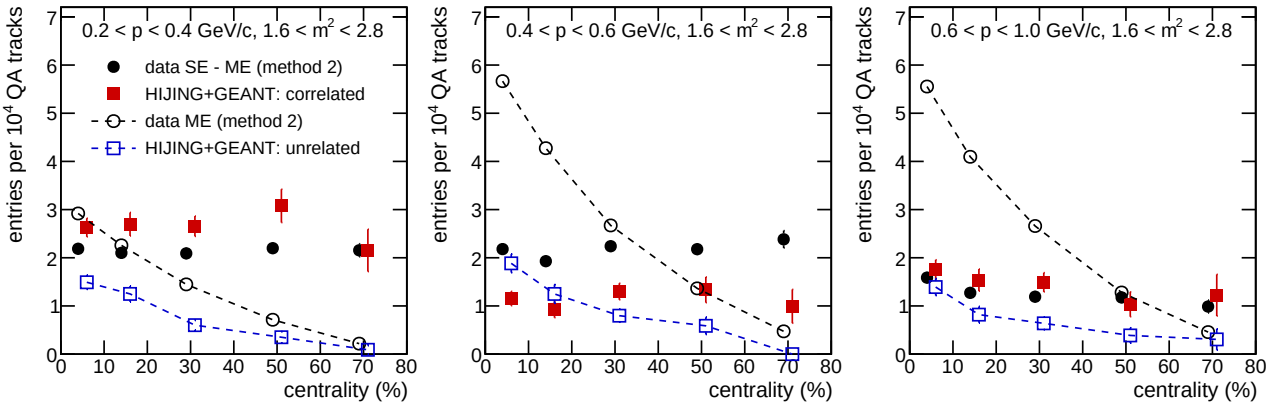
(a) Left: radius of the vertex where the branch leading to the hit leaves the track's particle, for decays (green) and secondaries (red), $0.2 < p < 1\,\text{GeV}/c$, log scale. Right: particle that made the hit, in % of each category.

Figure 42: Simulation: origin of correlated hits (`scripts/make_sim.C`).

7.10.2 Comparison with the data

In the p–d window ($1.6 < |q\,m^2| < 2.8\,\text{GeV}^2/c^4$) at 0.2–0.4 GeV/$c$ the simulation has 596 correlated entries; 96% of them are secondaries, 24 decays, and only 12 entries are own hits. Fig. 43a and Table 22 compare, per 10^4 QA tracks and versus centrality, the data SE−ME (method 2) with the simulated correlated matches, and the data ME with the simulated unrelated matches.

- 0.2–0.4 GeV/c: the simulated correlated rate is 2.7 per 10^4 tracks (all centralities) and, like the data, flat in centrality. The data/simulation ratio is 0.71–1.00 across the five classes, with at most $2.6\,\sigma$ difference.
- 0.4–0.6 GeV/c: the simulation (1.2) is below the data in every class; data/simulation 1.63–2.40, up to $5.9\,\sigma$.
- 0.6–1 GeV/c: the simulation (1.5) agrees with the data, at most $1.4\,\sigma$ apart.
- The simulated unrelated rate rises with centrality like the data ME but stays below it (Table 22): the simulation has fewer BTOF hits per event (no noise, no pile-up, untuned multiplicity).



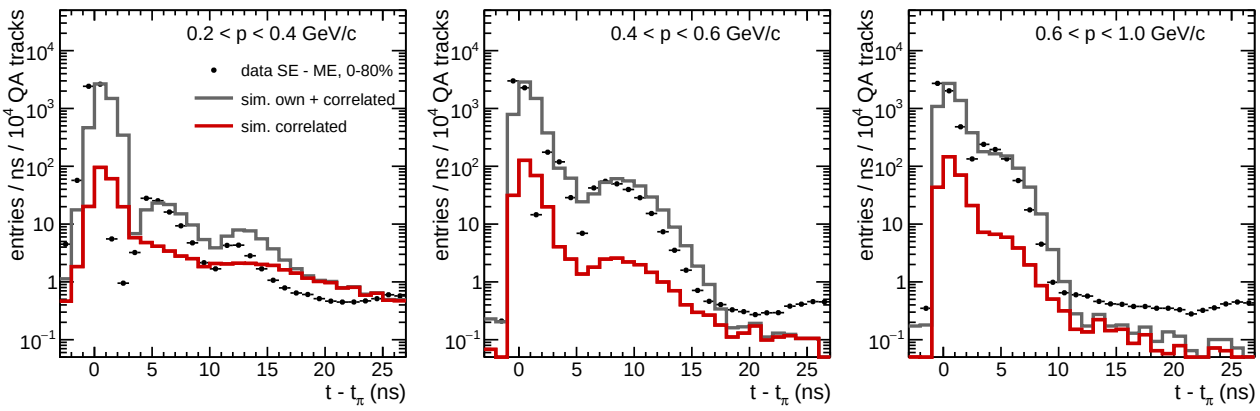
(a) Entries in $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$ per 10^4 QA tracks versus centrality, for 0.2–0.4, 0.4–0.6 and 0.6–1 GeV/c. Black full: data SE−ME (method 2); red squares: simulated correlated matches (decay + secondary + relative); black open, dashed: data ME; blue open, dashed: simulated unrelated matches. Simulation centrality from the refMult percentile (approximate). Statistical errors.

Figure 43: Simulation versus data: the track-correlated rate in the p–d window (`scripts/make_sim.C`).

Table 22: Entries in $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$ per 10^4 QA tracks. Data (method 2): SE−ME and ME. Simulation: correlated (decay + secondary + relative) and unrelated matches; simulation centrality from the refMult percentile of the simulated events (approximate). Δ/σ: (SE−ME − sim. correlated) over the combined statistical error.

p (GeV/c)	centrality	data SE−ME	sim. corr.	Δ/σ	data ME	sim. unrel.
0.2–0.4	0–10%	2.19 ± 0.06	2.63 ± 0.19	-2.2	2.92	1.49 ± 0.15
	10–20%	2.10 ± 0.07	2.70 ± 0.24	-2.4	2.26	1.25 ± 0.17
	20–40%	2.09 ± 0.06	2.65 ± 0.21	-2.6	1.45	0.60 ± 0.10
	40–60%	2.20 ± 0.08	3.08 ± 0.35	-2.5	0.71	0.35 ± 0.12
	60–80%	2.16 ± 0.14	2.15 ± 0.44	+0.0	0.22	0.09 ± 0.09
0.4–0.6	0–10%	2.18 ± 0.09	1.15 ± 0.15	+5.9	5.67	1.88 ± 0.19
	10–20%	1.93 ± 0.09	0.92 ± 0.17	+5.3	4.27	1.25 ± 0.19
	20–40%	2.24 ± 0.08	1.30 ± 0.17	+5.0	2.67	0.80 ± 0.13
	40–60%	2.18 ± 0.11	1.33 ± 0.27	+2.9	1.36	0.59 ± 0.18
	60–80%	2.38 ± 0.18	0.99 ± 0.35	+3.5	0.47	0.00 ± 0.00
0.6–1.0	0–10%	1.59 ± 0.08	1.76 ± 0.20	-0.8	5.55	1.39 ± 0.17
	10–20%	1.27 ± 0.09	1.54 ± 0.23	-1.1	4.10	0.82 ± 0.17
	20–40%	1.19 ± 0.07	1.49 ± 0.20	-1.4	2.66	0.64 ± 0.13
	40–60%	1.18 ± 0.09	1.03 ± 0.26	+0.5	1.28	0.39 ± 0.16
	60–80%	0.99 ± 0.13	1.22 ± 0.43	-0.5	0.46	0.31 ± 0.22

Time structure. Fig. 44a compares $t - t_\pi$ per QA track. In the late region 20–27 ns, beyond the K and p peaks, the data SE−ME is 0.50, 0.37 and 0.37 per ns per 10^4 tracks at 0.2–0.4, 0.4–0.6 and 0.6–1 GeV/c; the simulated correlated matches give 0.68, 0.11 and 0.05 (data/simulation 0.7, 3.4, 6.9). At the lowest momentum the simulation has the late plateau at a similar level, but not its momentum independence: in data the plateau stays at the same level up to 1 GeV/c, in the simulation it falls with momentum.



(a) $t - t_\pi$ per ns per 10^4 QA tracks. Black points: data SE−ME (method 2, 0–80%); gray: simulation, own + correlated matches (everything except unrelated); red: simulated correlated matches. Simulation with the true event time; the peak widths and the K/p yields of HIJING are not tuned to the data. Log scale.

Figure 44: Simulation versus data: time structure (`scripts/make_sim.C`).

Charge. Table 23 compares the negative/positive ratio. The data excess is positive-dominated at 0.2–0.3 GeV/c (0.40) and below 1 elsewhere; the simulated correlated matches are charge-symmetric at 0.2–0.3 GeV/c (0.91) and negative-dominated at 0.3–0.6 GeV/c (2.51, 1.43). The positive-only data component at the lowest momentum (Sec. 7.9.3) is therefore not reproduced (HIJING has no light nuclei, and the proton energy-loss tail is small in the simulation), and the negative excess of the simulation at 0.3–0.6 GeV/c is not seen in data.

Table 23: Negative over positive tracks in $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$: data SE−ME (method 2, 0–80%) and simulated correlated matches (all centralities), with statistical errors.

p (GeV/c)	data SE−ME neg/pos	sim. correlated neg/pos
0.2–0.3	0.40 ± 0.03	0.91 ± 0.09
0.3–0.4	0.85 ± 0.03	2.51 ± 0.44
0.4–0.6	0.80 ± 0.03	1.43 ± 0.21
0.6–1.0	0.69 ± 0.04	0.32 ± 0.05

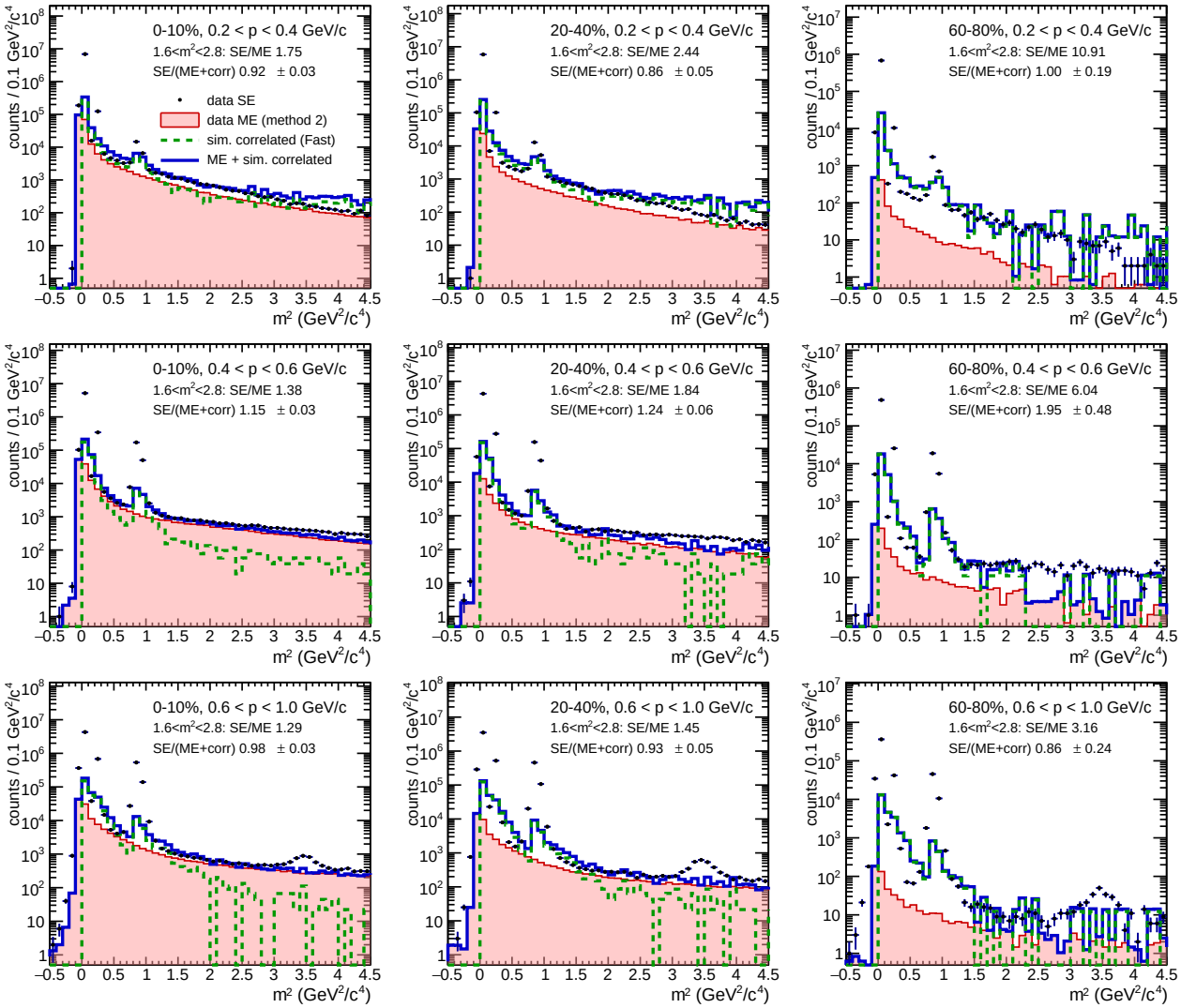
7.10.3 Data mass spectrum with the simulated correlated component

The simulated correlated matches (decay + secondary + relative), normalized per QA track to the data, are added to the data ME and compared with the data SE m^2 spectrum (Fig. 45, `scripts/make_simtmpl.C`). The simulated m^2 is $|q\,m^2|$ and its centrality is the refMult percentile of the simulated events.

- Between the p and d peaks (1.6–2.8 GeV$^2/c^4$) at 0.2–0.4 GeV/c, SE/ME is 1.75, 2.44 and 10.91 in 0–10, 20–40 and 60–80%; SE/(ME + sim. correlated) is 0.92, 0.86 and 1.00. At 0.6–1 GeV/c: 0.98, 0.93, 0.86. At 0.4–0.6 GeV/c an excess remains: 1.15, 1.24, 1.95.
- In the K–p valley (0.35–0.6 GeV$^2/c^4$) and above the deuteron (3.0–4.5 GeV$^2/c^4$) at 0.2–0.4 GeV/c the simulation predicts far more correlated matches than the data allow: SE/(ME + sim.) is 0.62–0.40 and 0.51–0.32 (0–10% and 60–80%), i.e. the simulated correlated component alone is larger than the data SE there (Table 24).

Table 24: Data SE over (ME + simulated correlated matches) in three m^2 windows (GeV$^2/c^4$), 0.2–0.4 GeV/c, for the 50k sample (no ToT) and the three BTOF responses of the 5k sample (`scripts/make_simtmpl.C`). A value of 1 means ME + simulation describes the data; the error on the p–d window is statistical.

	0–10%			20–40%			60–80%		
sample	0.35–0.6	1.6–2.8	3.0–4.5	0.35–0.6	1.6–2.8	3.0–4.5	0.35–0.6	1.6–2.8	3.0–4.5
50k, no ToT	0.62	0.92 ± 0.03	0.51	0.47	0.86 ± 0.05	0.36	0.40	1.00 ± 0.19	0.32
5k, no ToT	0.64	0.87 ± 0.09	0.48	0.50	0.97 ± 0.16	0.38	0.34	0.61 ± 0.29	0.19
5k, ToT(dE)	0.58	0.82 ± 0.09	0.65	0.42	0.92 ± 0.15	0.43	0.27	0.80 ± 0.43	0.19
5k, slow	0.57	0.86 ± 0.09	0.62	0.43	0.78 ± 0.13	0.61	0.31	0.41 ± 0.16	0.14



(a) Data SE m^2 (points) compared with the data ME (method 2, red filled), the simulated correlated matches (HIJING+GEANT, 50k events, no ToT; green dashed) scaled to the number of data QA tracks, and their sum (blue). Columns: 0–10, 20–40, 60–80%; rows: 0.2–0.4, 0.4–0.6, 0.6–1 GeV/c. 0.1 GeV$^2/c^4$ bins, log scale. The ratios in the p–d window are printed in each panel.

Figure 45: Data mass spectra with the simulated correlated component (`scripts/make_simtmpl.C`).

7.10.4 BTOF response: ToT and the slow simulator

The SL24y BTOF simulator writes the trailing edge equal to the leading edge in both its modes, so every simulated hit has ToT = 0 and the production rule "largest ToT wins" between several candidate hits never acts (all candidates tie and the closest-y rule decides); the slow mode has no setter. The patched `StBTofSimMaker` adds, through environment variables: (a) a ToT from the cell energy deposit in fast mode, $\mathrm{ToT} = 16.0 + 1.05\ln(dE/dE_{\mathrm{MIP}})$ ns with $dE_{\mathrm{MIP}} = 0.047$ keV (the median GEANT deposit of minimum-ionizing pions), and (b) the slow (avalanche) mode with its threshold, with the ToT from the summed avalanche charge in the same form ($Q_{\mathrm{MIP}} = 0.0095$ pC). The constants are taken from the data: in Run 19 the ToT of identified pions is 16.0 ns (16–84%: 15.1–17.1 ns) at minimum ionization and rises to 17.9 ns for protons at $\beta\gamma \approx 0.45$ (`macros/totData.C`, one MuDst file). The same 5k GEANT events (`sim/sim_job3.sh`) were reconstructed with the three responses (Table 25).

Table 25: The same 5k GEANT events with three BTOF responses. Entries per 10^4 QA tracks (all centralities) of correlated (decay + secondary + relative) and unrelated matches in three $|q\,m^2|$ windows (GeV$^2/c^4$), and the TOF-matching efficiency of QA tracks. Data: method 2, 0–80%, SE−ME and ME, same windows (signed m^2).

p (GeV/c)		match eff.	0.35–0.6		1.6–2.8		3.0–4.5	
			corr.	unrel.	corr.	unrel.	corr.	unrel.
0.2–0.4	no ToT	52.1%	7.7 ± 0.6	1.2 ± 0.2	2.7 ± 0.3	0.9 ± 0.2	2.2 ± 0.3	1.3 ± 0.2
	fast + ToT(dE)	54.0%	9.1 ± 0.6	2.9 ± 0.4	3.0 ± 0.4	1.5 ± 0.3	1.8 ± 0.3	1.7 ± 0.3
	slow + ToT(Q)	55.4%	9.0 ± 0.6	2.7 ± 0.3	3.1 ± 0.4	1.3 ± 0.2	1.6 ± 0.3	1.1 ± 0.2
	data SE−ME / ME		2.5	3.7	2.1	2.0	0.4	0.8
0.4–0.6	no ToT	60.7%	3.0 ± 0.4	1.2 ± 0.3	0.6 ± 0.2	0.9 ± 0.2	0.5 ± 0.2	0.5 ± 0.2
	fast + ToT(dE)	61.5%	5.7 ± 0.6	4.1 ± 0.5	1.4 ± 0.3	2.9 ± 0.4	0.6 ± 0.2	1.8 ± 0.3
	slow + ToT(Q)	61.8%	5.4 ± 0.6	3.8 ± 0.5	1.5 ± 0.3	2.4 ± 0.4	0.5 ± 0.2	1.0 ± 0.3
	data SE−ME / ME		1.5	4.6	2.1	3.9	2.2	2.3

- With a ToT the matching efficiency of QA tracks rises by 2–3 points (52.1% → 54.0%, 55.4% at 0.2–0.4 GeV/c), and the unrelated matches roughly double to triple: when several hits compete, the ToT rule picks a random hit more often than the closest-y rule. At 0.4–0.6 GeV/c the unrelated rate in the p–d window becomes 2.9 (fast + ToT) and 2.4 (slow) per 10^4 tracks, against 0.9 without ToT; the data ME is 3.9.
- The correlated rate in the p–d window at 0.4–0.6 GeV/c grows from 0.6 to 1.4 and 1.5 per 10^4 tracks (data SE−ME 2.1), closing part of the gap seen in Table 22. At 0.2–0.4 GeV/c it is unchanged within errors (2.7, 3.0, 3.1; data 2.1).
- The over-prediction in the K–p valley and above the deuteron at 0.2–0.4 GeV/c is not removed: 7.7, 9.1, 9.0 correlated per 10^4 tracks against a data SE−ME of 2.5 in the valley, and 2.2, 1.8, 1.6 against 0.4 at 3.0–4.5 GeV$^2/c^4$. The slow simulator's threshold does not remove these secondaries either.

Caveat

The 5k sample has 1–70 entries per cell of Table 25; differences below about 2σ are not significant. The ToT scale of the simulation is not tuned: own hits at 0.4–1 GeV/c have a median ToT of about 18 ns (data about 16 ns), which matters only through ToT differences between candidates. In slow mode the avalanche charge is saturated and does not depend on $\beta\gamma$, so the slow ToT does not favour heavily ionizing hits.

Choice

Conclusion: the simulation confirms that most of the low-momentum excess is track-correlated and not combinatorial. It is dominated by secondaries produced by the track's own particle in the BTOF trays ($R \approx 210$–$220\,\mathrm{cm}$), not by decays in flight, which stay under the pion peak. At 0.2–0.4 GeV/c the simulated rate is flat in centrality and within 0.71–1.00 (data/simulation) of the data.

Caveat

Not reproduced: the data rate at 0.4–0.6 GeV/c (data/simulation 1.63–2.40 without ToT; a ToT closes part of it, Sec. 7.10.4), the momentum-independent late plateau, the charge pattern, and the size of the correlated component outside the p–d window: at 0.2–0.4 GeV/c the simulation predicts about 3–4 times more correlated matches than the data allow in the K–p valley and above the deuteron, with or without ToT and with the slow simulator. The simulated secondaries in the trays are therefore not a reliable absolute template; the likely causes are the GEANT3 treatment of low-energy secondaries in the tray material and the absence of any electronics dead time or noise in the simulation (not tested).

Decision needed

Proposed: use the simulation as evidence for the origin of the low-momentum p–d excess, not as a correction: a simulated template would fix the p–d window at 0.2–0.4 GeV/c but over-subtract the K–p valley and the high-m^2 region by a factor of 2–3. The data-driven per-track excess (Sec. 7.9.3) remains the estimate of this component.

8 Cuts adopted and inputs for the mixed-event step

Table 26: Selection used in this QA (defaults in `macros/runTofMEQA.C` and the constructor of `StTofMEQAMaker`).

Level	Cut	Status
Event	trigger $\in \{640001, 640011, 640021, 640031, 640041, 640051\}$	applied
Event	$\lvert V_z\rvert < 70\,\mathrm{cm}$	applied
Event	$V_r < 2\,\mathrm{cm}$	applied
Event	$\lvert V_z - V_z^{\mathrm{VPD}}\rvert$	not applied
Event	pile-up / bad-run rejection	not implemented
Track	primary, $p_T > 0.15\,\mathrm{GeV}/c$	applied
Track	nHitsFit ≥ 15, nHitsFit/nHitsMax > 0.52	applied
Track	gDCA $< 3\,\mathrm{cm}$	applied
Track	$\lvert\eta\rvert < 1$	applied
TOF	btofMatchFlag > 0	applied
TOF	$\beta > 0$	applied (TOF-matched sample)
TOF	$\lvert y_{\mathrm{local}}\rvert$, $\lvert z_{\mathrm{local}}\rvert$	none beyond production ($\lvert z_{\mathrm{local}}\rvert < 3.05\,\mathrm{cm}$)

What this QA implies for the mixed-event matching:

- With 2.94 stored hits per TOF-matched QA track and a matching window of one pad in y and $\pm 3.05\,\mathrm{cm}$ in z, the random-match probability will depend strongly on event multiplicity; events must be mixed within multiplicity classes (refMult or number of BTOF hits) and V_z classes.

- The picoDst stores only the cell ID of each BTOF hit, so matching is done at the cell level with the projection of Sec. 7 (implemented).
- Dead trays/modules are identical between mixed events because mixing is restricted to one run (implemented).
- The y_{local} side-bands ($|y_{\mathrm{local}}| > 2.5\,\mathrm{cm}$) provide an independent same-event estimate of the random-match level that the mixed-event result can be compared to.

Decision needed

(1) Is the minimum-bias trigger list correct for P24iy, or should the analysis use 640051 only? (2) Should a $|V_z - V_z^{\mathrm{VPD}}|$ cut be used, given the VPD vertex is available for only 92.9% of events? (3) Mixing classes: refMult (used now) or number of BTOF hits, and is a 10 cm V_z class adequate? (4) Which matching rule should the final random-match correction use: production-like cell±1 (default proposal) or cell only? (5) Low-p_T projection: accept the DCA-helix projection, or add an energy-loss correction / use the production cell for same-event matches?

9 Audit

9.1 Checks that passed

- **MuDst production re-run:** 16451 of 16451 production matches reproduced in the same cell, 0/0/0 different/lost/new; time of flight median -14 ps (`StTofClosureMaker`, 300 events, `scripts/make_probe.C`, `logs/probe_numbers.txt`).
- **MuDst emulation:** the emulated production choice picks the production hit for 98.3% of the production matches (`scripts/make_mu.C`, 21116 events).
- **Overlay bookkeeping:** matched fraction of QA tracks 56.4% in the same event, 56.9% in the overlays without and 54.6% with partner tracks; of the same-event matches 95.2% keep their hit, 1.2% switch to a partner hit and 3.7% are lost with partner tracks (`scripts/make_ov.C`, 21160 events).
- **Two independent MuDst estimates agree:** overlay with partner tracks 2.62%/1.57% vs emulation 2.7%/1.6% (0–5%/0–80%; `scripts/make_mudst_table.py`, `tables/mudst_cmp.tex`).
- **Code identity:** md5 sums of `StTofMEQAMaker.{h,cxx}`, `runTofMEQA.C` and `drawTofMEQA.C` on the node equal those of the local copies (`logs/provenance.txt`, `logs/md5_local.txt`). The library was rebuilt (`cons`) before the run that produced `output/qa_test_10k.root`.
- **Event count:** 10000 events processed out of 17041 in the file, 1 file read (run log, `logs/runinfo.txt`); the first bin of `hEvtCut` equals the entries of `hVzAll` (10 000; PASS).
- **Cut-flow consistency:** entries of `hVz` (selected events) equal the last applied cut-flow bin (5 501; PASS); track counts in Table 4 decrease monotonically (PASS) (`scripts/make_numbers.C`, full test sample).
- **Cell decoding:** no matched track points to a module without hits (0 entries), consistent with the same `btofCellId/id()` encoding for traits and hits.
- **PID scale:** fitted m^2 peaks of π, K, p agree with PDG within $0.005\,\mathrm{GeV}^2/c^4$ (`scripts/make_numbers.C`, $0.4 < p < 1.0\,\mathrm{GeV}/c$).

- z_{local} **window:** the edges at ± 3.05 cm match `mZLocalCut=3.05` in `StRoot/StBTofMatchMaker/StBTofMatchMaker.cxx` of SL24y (source inspection).
- **API:** picoDst getters and compression factors were read from the SL24y `StRoot/StPicoEvent` headers (source inspection).
- **Geometry fit:** mean fit residual 0.0029 cm over 3699 modules; rotated-copy closure 0.297 cm mean (`macros/makeGeom.C`, `logs/makeGeom_test.log`).
- **Production neighbour rule:** the cell± 1 acceptance and `mWidthPad`=3.45 cm were read in `StBTofMatchMaker.cxx` (SL24y); the fitted pitch agrees.
- **Same input sample:** the mixed-event test selects the same 5 501 events as the QA. It has 960 080 QA tracks, 77 fewer than the QA maker (960 157), because the test build compared nHitsFit/nHitsMax with the double 0.52 instead of the float 0.52 used in the QA maker (tracks with ratio exactly 13/25 were dropped). Fixed before the 1M-event productions.
- **1M merge:** all 15 job logs end normally and contain the maker summary; the merged file has 1 011 835 events read (`scripts/merge_v3.sh`, `logs/hadd_v3.log`); the job file lists contain no duplicate file (checked on the node).
- **Path length and m^2:** for tracks whose production hit is chosen, L agrees with $\beta_{\mathrm{prod}} c\, t_{\mathrm{prod}}$ (mean 0.001 cm, RMS 0.26 cm) and m^2 with production (RMS 0.0007 GeV$^2/c^4$) (`hDLVsP`, `hDm2VsP`, `scripts/make_m2.C`).
- **Production hit choice:** largest ToT among candidate cells, then closest y, read in `StBTofMatchMaker.cxx` (SL24y, lines 782–833).
- **Background diagnostics:** every v4–v8 job log ends normally; all passes read the same 1 011 835 events (v6: 25 000 per run) (`scripts/merge_v4.sh`–`merge_v8.sh`).
- **Overlay ME does not exceed SE:** largest ME/SE over m^2 bins is 1.07 in 0–5%, $0.4 < p < 1.5$ GeV/c, against 3.8–5.5 without overlay (`scripts/make_m2.C`). It reaches 1.21 at $0.2 < p < 0.4$ GeV/c (Table 8).
- P_{ME} **= occupancy:** slope of P_{ME} vs N_{hits} is 0.99 times 1/23040 (`scripts/make_me.C`, test sample).
- Simulation: fast TPC simulator vs full TPC response on the same 1000 GEANT events: category fractions agree within $1.6\,\sigma$ (`scripts/make_sim.C`, Table 20). Every TOF-matched simulated track has a truth class after the `StBTofSimMaker` truth-id patch: 0 matches without truth in the 50k sample (`logs/sim_numbers.txt`).

9.2 Not verified (assumptions)

- MuDst: the database time stamp of the re-run is not pinned to the production; the residual ps-level differences are attributed to later calibration updates (not checked).
- MuDst: the restore of the event hit and track arrays after each overlay is not tested directly; the overlay matched fractions are consistent with it.
- MuDst: the partner tracks are selected with $n_{\mathrm{HitsFit}} \geq 15$ and $0 <$ flag < 1000, a superset of the match maker's valid tracks (the maker applies its own cuts); their effect on the event's own matches (3.7% lost) is not corrected.

- MuDst: 21160 (overlay) and 21116 (emulation) events are used; the difference comes from the start-time requirement (VPD start vs own T_0), not checked event by event.
- MuDst: the statement that the overlay ME follows the SE shape and that the pion-side spike disappears is read by eye (Fig. 25a); `tables/m2_v9.tex` is a stored output of `scripts/make_m2.C`, not regenerated by `build.sh`.
- Minimum-bias trigger IDs other than 640051 (not present in the test file).
- That the picoDst keeps all BTOF hits of an event (only inferred from equal mean/RMS of hit count and `btofTrayMultiplicity`).
- Meaning of trigger IDs 6, 35, 36, 37 and the 6406xx/6408xx IDs.
- Representativeness of run 20093036 for the dataset; no run-by-run QA.
- Catalog file/event counts are copied from the catalog query output in `logs/catalog_P24iy.txt`.
- The availability of NFS replicas was checked only for the latest runs (first 3000 catalog entries ordered by run number), not for the whole list.
- The geometry is fitted and the ME test is run on the same events (test sample). For the 1M sample the refitted geometry agrees with it to 0.0005 cm (`scripts/make_me_runs.C`), so this is checked for the runs used.
- The time distribution of BTOF hits without a stored time (43.2% of the chosen partner hits have one) is the same as for timed hits (all-hits normalization of Sec. 7.5).
- StRefMultCorr centrality and pile-up definitions for Run19 19.6 GeV were used as provided (not re-derived).
- ToT of a hit is independent of its origin (equal-chance emulation, Sec. 7.6); earliest time as a ToT proxy.
- The growth of the BTOF hit multiplicity between 70–80% and 0–5% is taken from refMult, not computed per class.
- Shape statements on the m^2 overlays (position of the random peak, where ME follows SE) are read by eye.
- Mixed-event independence of random hits on the track (no correlation of noise or afterpulses with real hits).
- Visual statements (shape of y_{local} tails, ϕ dip near -0.6, η dip of the efficiency at $\eta \approx 0$) are read from the figures, not fitted.
- Simulation: GEANT3 hadronic model and cuts for slow secondaries in the BTOF trays; ToT = 0 of the fast BTOF simulator; centrality from the simulated refMult percentile; shapes in Fig. 41a and the decay radius range are read by eye.

10 Open issues

Ordered by importance for the next step.

decision Matching rule for the final correction (cell±1 vs cell) and mixing classes.

1. Improve the low-p_T projection (energy loss, or outer helix not available in picoDst).
2. Translate P_{ME} into a random-match correction of the TOF-matched yield and m^2 background (needs the time of the random hit, not stored in picoDst).
3. Line shapes: no fit model is statistically acceptable below 2 GeV/c on the large sample (Sec. 7.8.3); try finer momentum bins or a resolution-scaled variable, and understand the continuum left at 0.2–0.4 GeV/c.
4. Track-correlated late hits (Sec. 7.9.3): a charge-symmetric SE excess of about 2.1–2.3 per 10^4 tracks in $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$ at 0.2–0.4 GeV/c, independent of centrality, plus a positive-only part at 0.2–0.3 GeV/c. The full simulation attributes it mainly to secondaries from the BTOF trays (Sec. 7.10); the positive-only part is not reproduced.
5. Simulation (Sec. 7.10): ToT and the slow BTOF simulator were tested (Sec. 7.10.4); a ToT raises the 0.4–0.6 GeV/c correlated and unrelated rates toward the data but none of the responses removes the 3–4× over-prediction of correlated matches in the K–p valley and above the deuteron at 0.2–0.4 GeV/c. Open: GEANT3 settings for low-energy secondaries in the trays (time and energy cuts, hadronic package), electronics dead time, multiplicity tuning. [decision] simulation as evidence only, not as a correction (Sec. 7.10.3).
6. Method 2 loses 3.6% of the own matches in the overlay; quantify the effect on the random fraction.
7. MuDst statistics: 98 files of 14 runs restored (`filelists/mudst/restore_100.txt`, 319 GB); one job per file. Run 20088015 needs a workaround for the trigger-data decoding errors.
8. Understand the slight p_T dependence of P_{ME} seen in the 1M sample.
9. Process the three removed files (030, 033, 040) if the full 57-file sample is needed.

decision Confirm the trigger list and decide on VPD-vertex and pile-up cuts.

10. Scale beyond 1M events if needed (one HTCondor job per run, about 70k events in 35–40 min; avoid the qgp017 server).
11. Understand the reduced TPC track density near $\phi \approx -0.6$, $\eta > 0$.

11 Glossary

BTOF Barrel Time-of-Flight: 120 trays × 32 MRPC modules × 6 pads (cells), $|\eta| \lesssim 0.9$.

VPD Vertex Position Detector; start time and V_z^{VPD}.

picoDst STAR compact analysis format (`StRoot/StPicoEvent`).

refMult / gRefMult Reference multiplicity from primary / global tracks in $|\eta| < 0.5$.

nBTOFMatch Number of primary tracks matched to BTOF (stored per event).

btofTrayMultiplicity Number of BTOF hits in the event (stored per event).

btofMatchFlag 0 no match, 1 one-to-one, 2 one hit shared by several track candidates.

y_{local}, z_{local} Track projection relative to the centre of the matched pad.

gDCA Distance of closest approach of the global track to the primary vertex.

$\varepsilon_{\text{match}}$ TOF-matched ($\beta > 0$) QA tracks divided by all QA tracks.

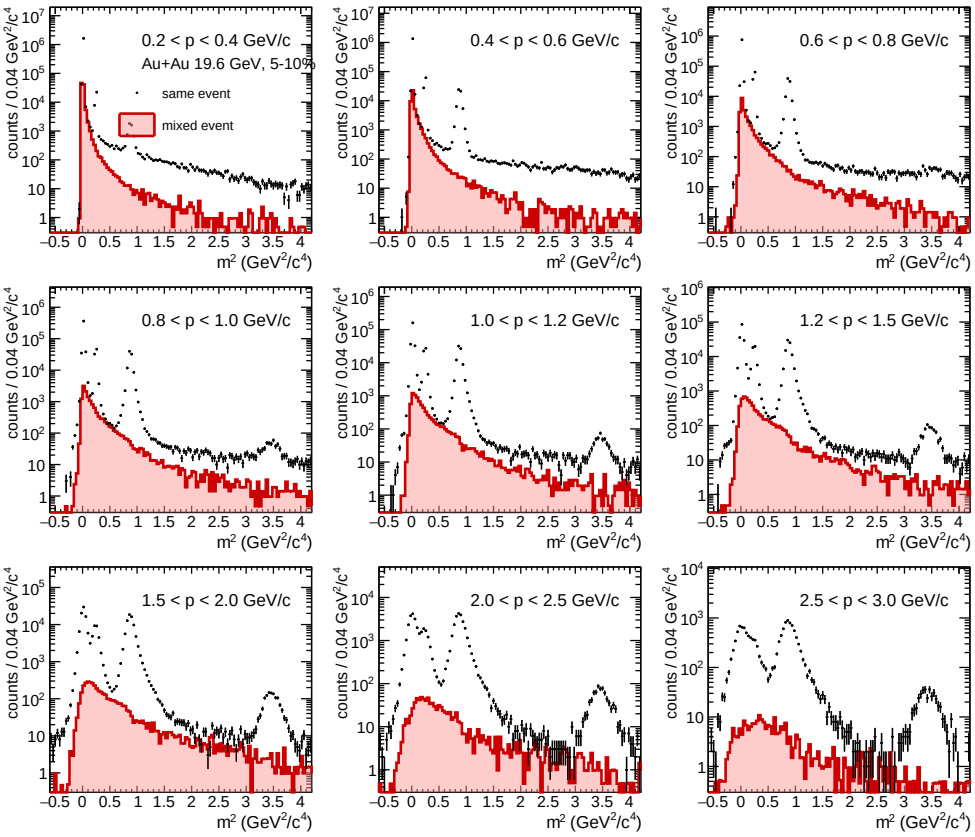
P_{SE}, P_{ME} Probability of a hit in the projected cell (or cell±1) in the same / a mixed event.

f_{rand} $P_{\text{ME}}/P_{\text{SE}}$, estimated random fraction of same-event matches.

Mixed event Pairing tracks of one event with BTOF hits of another event of similar class.

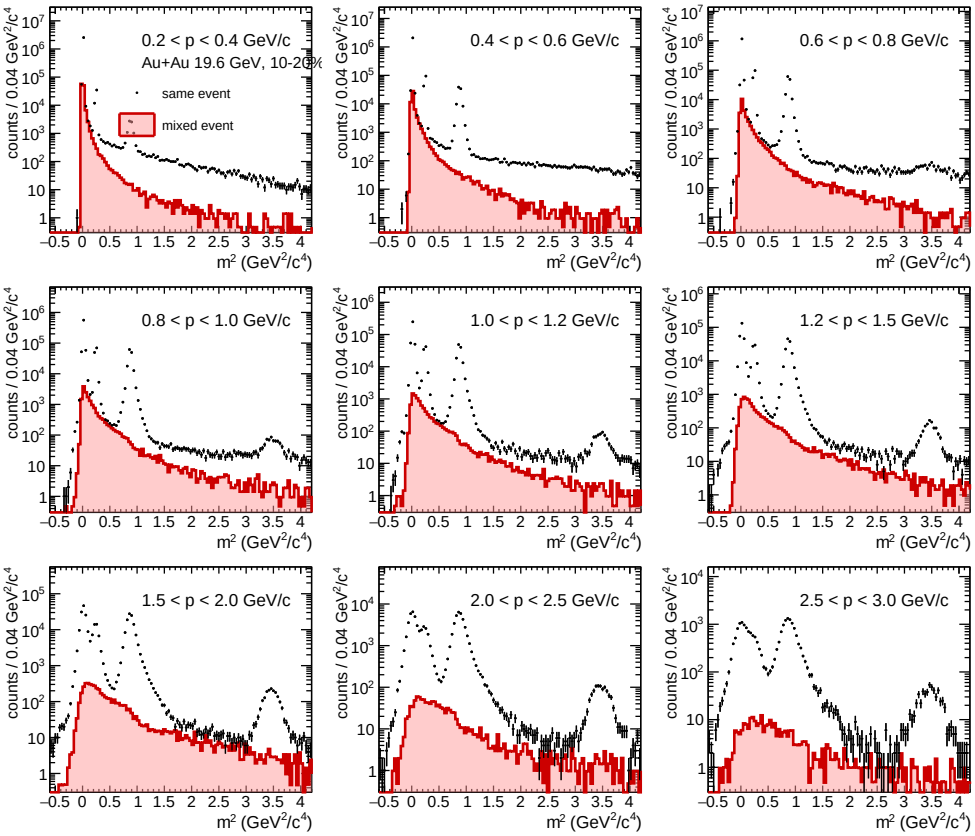
A SE vs ME m^2 for the other centrality classes

Same content as Figs. 15a–17a (`scripts/make_m2.C`, 1M sample).



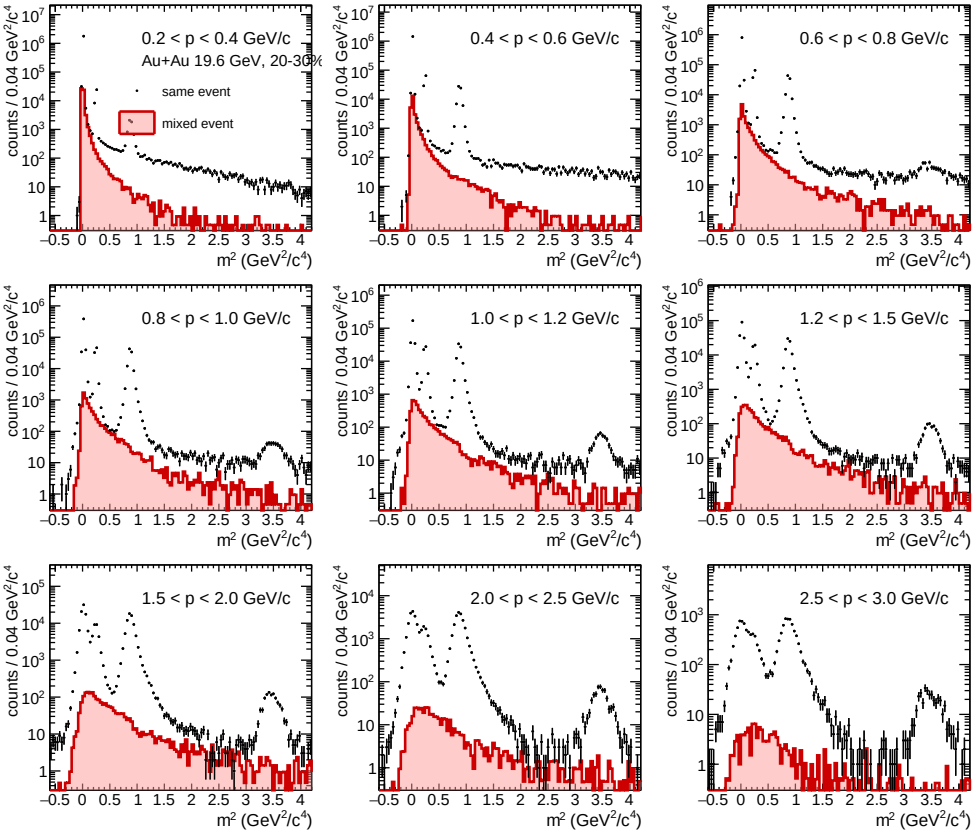
(a) m^2 of projected QA tracks, 5–10%, nine momentum bins (log scale, $0.04\,\text{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 46: SE vs ME m^2, 5–10%.



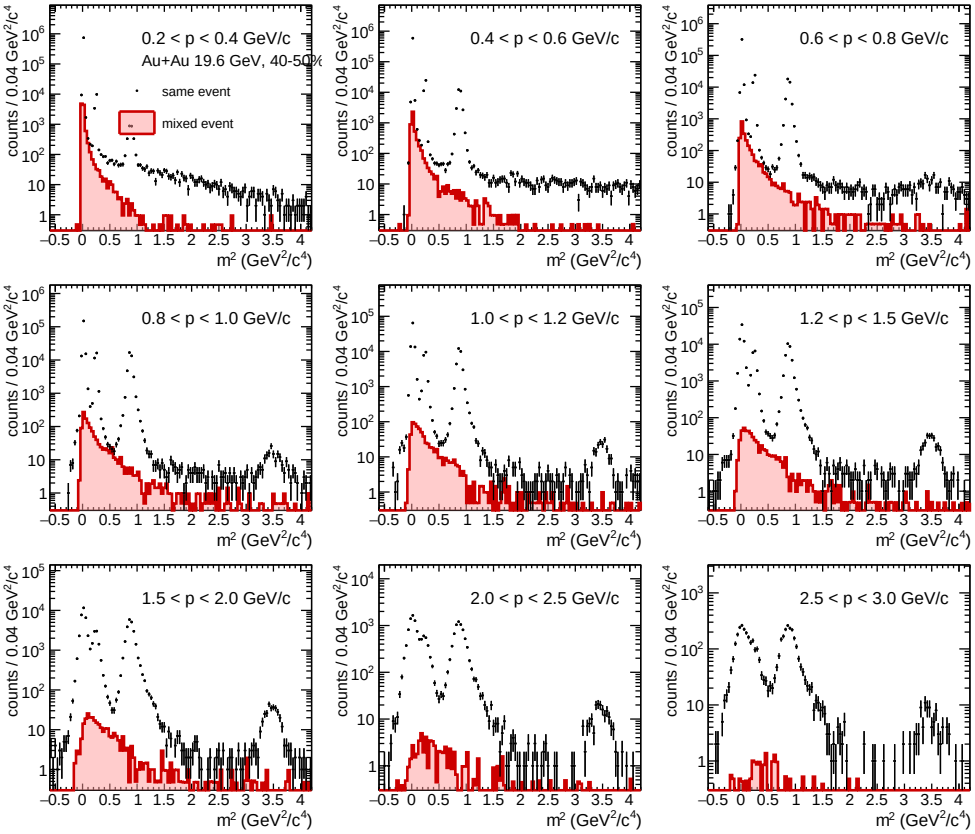
(a) m^2 of projected QA tracks, 10–20%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell$\pm$1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 47: SE vs ME m^2, 10–20%.



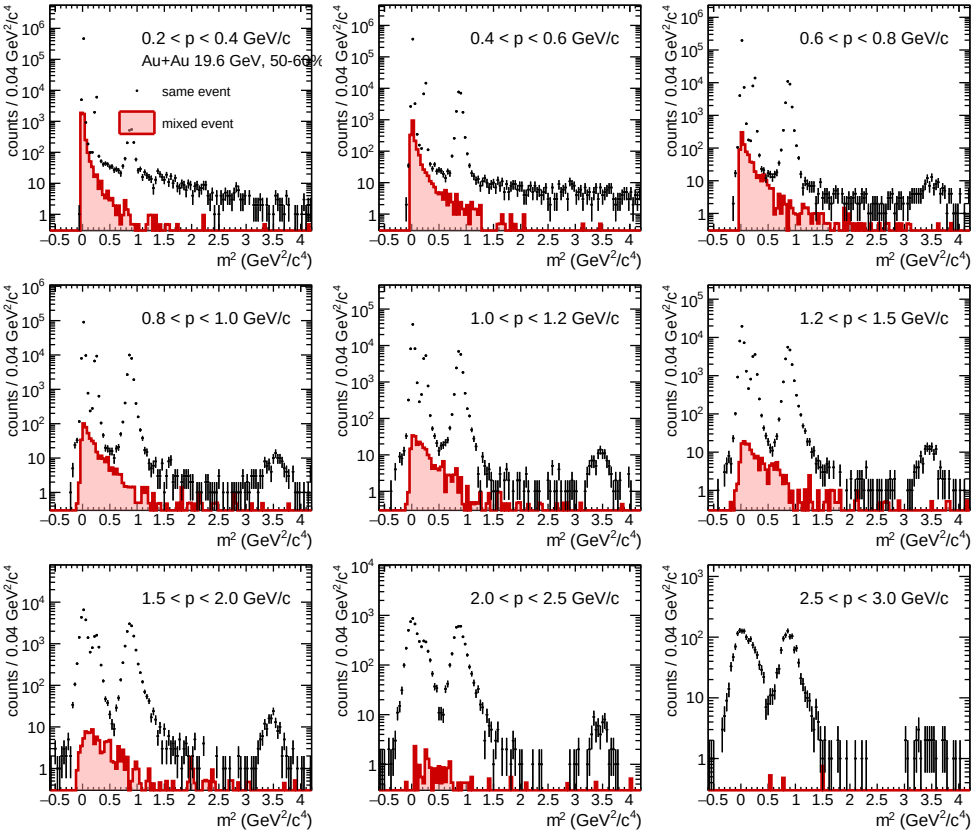
(a) m^2 of projected QA tracks, 20–30%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell$\pm$1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 48: SE vs ME m^2, 20–30%.



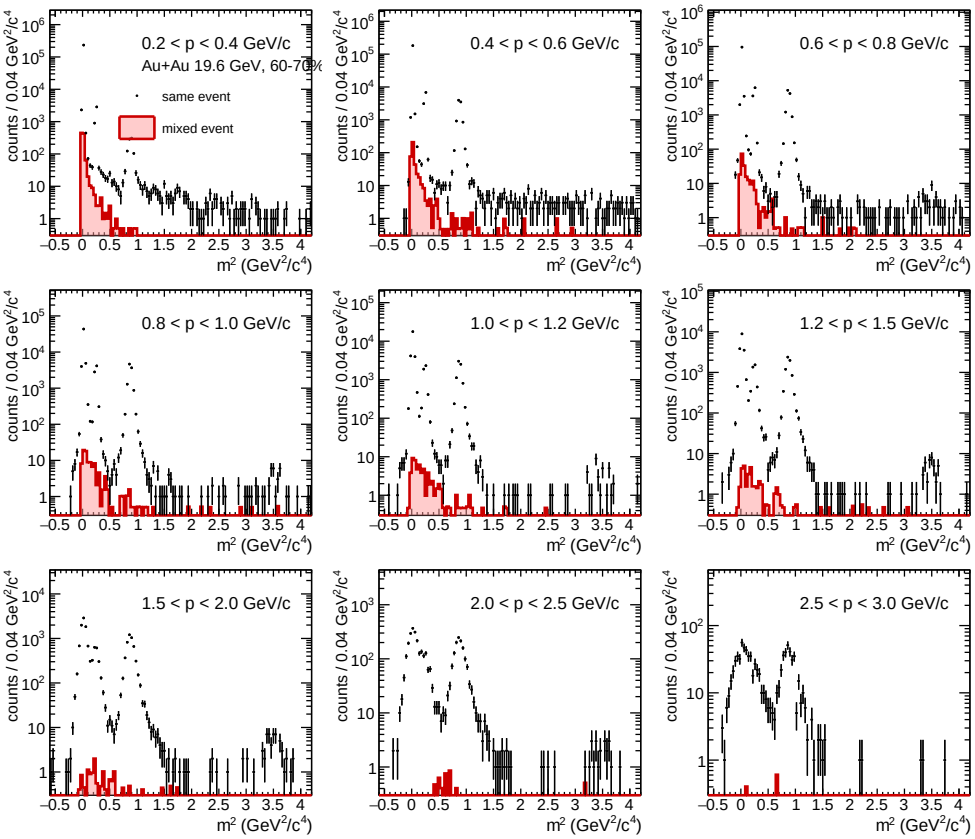
(a) m^2 of projected QA tracks, 40–50%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 49: SE vs ME m^2, 40–50%.



(a) m^2 of projected QA tracks, 50–60%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 50: SE vs ME m^2, 50–60%.



(a) m^2 of projected QA tracks, 60–70%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 51: SE vs ME m^2, 60–70%.